
On Expressive Power of Floating-Point Transformers

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Abstract

Research on the expressive power of transformers shows that transformers are equivariant to permutations and can approximate all permutation-equivariant continuous functions on a compact domain. However, these results assume real parameters and exact operations, whereas real-world implementations on computers can only use a finite set of numbers and inexact machine operations with round-off errors. In this work, we investigate the representability of floating-point transformers that use floating-point parameters and floating-point operations. Unlike existing results under exact arithmetic, we first show that floating-point transformers can represent non-permutation-equivariant functions even without positional encoding. Furthermore, we prove that floating-point transformers can represent all permutation-equivariant functions when the sequence length is bounded, but they cannot when the sequence length is sufficiently large. We also identify the minimal equivariance property in floating-point transformers, and show that all non-trivial additive positional encoding can harm the representability of floating-point transformers.

1. Introduction

Self-attention-based transformers are among the most popular neural network architectures for sequence-to-sequence learning. Starting with natural language processing (Vaswani et al., 2017), they have achieved state-of-the-art performance in various fields, including computer vision (Dosovitskiy et al., 2021), graph representation learning (Ying et al., 2021), and time-series forecasting (Zhou et al., 2021). Due to the architectural symmetry, it is well-known

that transformers are permutation-equivariant: if input tokens are permuted, then the outputs are also permuted correspondingly. While such a symmetry has been appreciated when the output is independent of input order (e.g., Lee et al. (2019)), it is often intentionally broken by using positional encoding if the input order matters.

Several works have investigated the expressive power of transformers. For example, it is known that transformers are dense in the L^p space of permutation-equivariant sequence-to-sequence functions on a compact domain, where the permutation-equivariance constraint can be removed by introducing positional encoding (Yun et al., 2020a). Similar results have also been shown for the space of continuous functions equipped with the uniform norm (Alberti et al., 2023), for shallow architectures and probabilistic setups (Furuya et al., 2025), for sparse architectures (Yun et al., 2020b), and for approximating shift-equivariant functions (Takakura & Suzuki, 2023). Furthermore, transformers can fit a finite number of input–output sequence pairs (Kim et al., 2023; Mahdavi et al., 2024; Dana et al., 2025).

However, existing theoretical results cannot be directly applied to real-world transformers implemented on computers since they assume real parameters and exact arithmetic. In practice, transformers use a finite set of numbers and inexact machine operations such as floating-point arithmetic. Only a few recent works have studied the expressive power of neural networks under floating-point arithmetic. Park et al. (2024) showed that fully-connected floating-point networks using ReLU or the binary step function can represent all floating-point functions from floating-point vectors to floating-point numbers. This result was later extended to a general class of activation functions (Hwang et al., 2025b) and interval arithmetic (Hwang et al., 2025a). A notable observation is that floating-point networks may not inherit important properties of neural networks analyzed under exact arithmetic, e.g., floating-point networks using the identity activation function can represent all floating-point functions. Floating-point transformers and state-space models have also been studied from a computational-complexity viewpoint: constant-precision transformers lie in $AC^0 \subsetneq TC^0$ (Li et al., 2024) and log-precision state-space models are confined to TC^0 (Merrill et al., 2024).

Prior works mainly consider sequential reduction orders

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in floating-point summations. In practice, however, GPU reductions may use runtime-dependent parallel reduction orders, and some hardware performs high-precision accumulation before converting the result back to the target precision. To the best of our knowledge, the only prior work considering arbitrary reduction orders is [Park et al. \(2026\)](#), which extended [Hwang et al. \(2025b\)](#). In addition, none of the prior works considered high-precision accumulation.

1.1. Contributions

In this work, we study the expressive power of floating-point transformers that use floating-point parameters and floating-point operations (e.g., addition and multiplication). Unlike prior works that mainly focus on sequential reduction orders, we additionally investigate arbitrary deterministic reduction orders and high-precision accumulation. Specifically, we study whether floating-point transformers inherit the well-known properties of transformers under exact arithmetic by addressing the following questions:

- Are floating-point transformers permutation-equivariant?
- Can floating-point transformers represent all permutation-equivariant sequence-to-sequence functions?

Surprisingly, we prove that the answers to both questions are *no*: floating-point transformers may not satisfy natural properties of transformers analyzed under exact arithmetic.

We first show that floating-point transformers can express non-permutation-equivariant functions. By exploiting the non-associativity of floating-point addition, we show that an attention layer can identify partial input-order information even without positional encoding. Using this construction, we prove that floating-point transformers can represent a class of non-permutation-equivariant functions (Theorem 2). This result holds under arbitrary deterministic reduction orders and high-precision accumulation, including deterministic parallel reductions in multi-thread CPUs and GPUs. We next show that floating-point transformers cannot represent some permutation-equivariant sequence-to-sequence functions when the sequence length n is sufficiently large (Theorem 3), under the left-to-right summation order, corresponding to deterministic single-thread CPU computations without parallel reduction.

On the other hand, when the sequence length n is properly bounded, we prove that floating-point transformers can represent all permutation-equivariant sequence-to-sequence functions. In particular, we first prove this result under the left-to-right summation order without high-precision accumulation (Theorem 4). We then prove an analogous result for arbitrary deterministic reduction orders and high-precision accumulation (Theorem 5). These results contrast with exact-arithmetic universal approximation results, where approximation is possible regardless of the sequence

length ([Yun et al., 2020a](#)). We also investigate various properties of floating-point transformers and clarify the limitations of our computation model in Section 4.

2. Preliminaries

2.1. Notations

We often use lowercase letters ($a, b, c, \dots$) to denote scalar values, bold lowercase letters ($\mathbf{a}, \mathbf{b}, \mathbf{c}, \dots$) to denote vectors, uppercase letters ($A, B, C, \dots$) to denote matrices, and calligraphic uppercase letters ($\mathcal{A}, \mathcal{B}, \mathcal{C}, \dots$) to denote sets. We use $\mathbb{N}$ and $\mathbb{R}$ to denote the set of positive integers and the set of real numbers, respectively. For $n \in \mathbb{N}$, we use $[n] := \{1, 2, \dots, n\}$. For $a, b \in \mathbb{R} \cup \{\pm\infty\}$ and a set $\mathcal{C}$, $(a, b) := \{x \in \mathbb{R} : a < x < b\}$, $[a, b] := \{x \in \mathbb{R} : a \leq x \leq b\}$, $(a, b)_{\mathcal{C}} := (a, b) \cap \mathcal{C}$, and $[a, b]_{\mathcal{C}} := [a, b] \cap \mathcal{C}$. We similarly define $[a, b)$, $(a, b]$, $[a, b)_{\mathcal{C}}$, and $(a, b]_{\mathcal{C}}$.

For $n \in \mathbb{N}$ and $x_1, \dots, x_n \in \mathbb{R}$, we use $(x_1, \dots, x_n) \in \mathbb{R}^n$ to denote the vector whose i -th coordinate is x_i . Throughout this paper, we treat all vectors as column vectors (i.e., $\mathbb{R}^n = \mathbb{R}^{n \times 1}$). For $n, m \in \mathbb{N}$ and $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^n$, we use $[\mathbf{x}_1, \dots, \mathbf{x}_m] \in \mathbb{R}^{n \times m}$ to denote the $n \times m$ matrix whose i -th column is $\mathbf{x}_i$. For $n, m \in \mathbb{N}$, $\mathbf{x} \in \mathbb{R}^n$, and $M \in \mathbb{R}^{n \times m}$, we use x_i to denote the i -th coordinate of $\mathbf{x}$, M_i to denote the i -th row of M , and M_{ij} to denote the element at the i -th row and the j -th column of M . We also use $M_{i:j}$ to denote the submatrix of M induced by $M_i, M_{i+1}, \dots, M_j$. When $i = 1$ or $j = n$, we use $M_{:j}$ or M_i to denote $M_{i:j}$, respectively. We use $\mathbf{0}_n$ and $\mathbf{1}_n$ to represent $(0, \dots, 0) \in \mathbb{R}^n$ and $(1, \dots, 1) \in \mathbb{R}^n$, respectively. Likewise, we use $\mathbf{0}_{n \times m}$ and $\mathbf{1}_{n \times m}$ to denote the $n \times m$ matrix consisting of zeros and the $n \times m$ matrix consisting of ones, respectively.

For a set $\mathcal{X}$, $\rho : \mathcal{X} \rightarrow \mathcal{X}$, $\sigma : \mathcal{X}^n \rightarrow \mathcal{X}^n$, we often apply ρ and σ to a matrix $M = [\mathbf{m}_1, \dots, \mathbf{m}_m] \in \mathcal{X}^{n \times m}$ as follows: $\rho(M), \sigma(M) \in \mathcal{X}^{n \times m}$ are matrices satisfying

$$\rho(M)_{ij} = \rho(M_{ij}), \quad \sigma(M) = [\sigma(\mathbf{m}_1), \dots, \sigma(\mathbf{m}_m)]. \quad (1)$$

$\mathcal{S}_n$ denotes the permutation group on $[n]$, i.e., $\mathcal{S}_n = \{\pi : [n] \rightarrow [n] : \pi([n]) = [n]\}$. For any set $\mathcal{X}$, we define a right group action of $\mathcal{S}_n$ on $\mathcal{X}^{m \times n}$ as follows: for $\pi \in \mathcal{S}_n$ and $X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathcal{X}^{m \times n}$, $\pi X := [\mathbf{x}_{\pi(1)}, \dots, \mathbf{x}_{\pi(n)}]$. For $m, m', n \in \mathbb{N}$, $\mathcal{T} \subset \mathcal{S}_n$, sets $\mathcal{X}, \mathcal{Y}$, a subset $\mathcal{D} \subset \mathcal{X}^{m \times n}$ closed under the action of $\mathcal{T}$, and $f : \mathcal{D} \rightarrow \mathcal{Y}^{m' \times n}$, we say “ f is $\mathcal{T}$ -equivariant” if $f(\pi X) = \pi f(X)$ for all $X \in \mathcal{D}$ and $\pi \in \mathcal{T}$. We say f is “permutation-equivariant” if f is $\mathcal{S}_n$ -equivariant. For a singleton set $\{\pi\}$, we use “ π -equivariant” instead of “ $\{\pi\}$ -equivariant”.

2.2. Floating-point arithmetic

Floating-point numbers. Throughout this paper, we consider the IEEE-754 standard for floating-point arithmetic

(IEEE, 2019). For $p, q \in \mathbb{N}$, we define $\mathbb{F}_{p,q}$ as the set of finite floating-point numbers as follows:

$$\mathbb{F}_{p,q} := \{\pm(1.m_1 \cdots m_p) \times 2^e, \pm(0.m_1 \cdots m_p) \times 2^{\epsilon_{\min}} : m_1, \dots, m_p \in \{0, 1\}, e \in [\epsilon_{\min}, \epsilon_{\max}]_{\mathbb{Z}}\} \quad (2)$$

where $1.m_1 \cdots m_p$ and $0.m_1 \cdots m_p$ are written in the binary representation, $\epsilon_{\min} := -2^{q-1} + 2$, and $\epsilon_{\max} := 2^{q-1} - 1$. Each floating-point number is represented by the sign s , mantissa $m_1, \dots, m_p$, and exponent e , i.e., $p + q + 1$ bits suffice to represent all elements in $\mathbb{F}_{p,q}$.

There are three additional (non-finite) floating-point numbers: *positive infinity* ∞ , *negative infinity* $-\infty$, and *not-a-number* NaN. In floating-point operations, ∞ and $-\infty$ may appear when the output exceeds the range of $\mathbb{F}_{p,q}$. NaN may appear when the output cannot be approximated by an element of $\mathbb{F}_{p,q} \cup \{-\infty, \infty\}$, e.g., when we add ∞ to $-\infty$. We assume that $-\infty, \infty$ follow the conventional order, i.e., $-\infty < x < \infty$ for all $x \in \mathbb{R}$. We denote the set of all floating-point numbers by $\overline{\mathbb{F}}_{p,q} := \mathbb{F}_{p,q} \cup \{-\infty, \infty, \text{NaN}\}$. Here, $\overline{\mathbb{F}}_{p,q}$ can also be represented by using $p + q + 1$ bits since we are not using all 2^q representations for the exponent in $\mathbb{F}_{p,q}$. We use $\omega := 2^{\epsilon_{\min}-p}$ and $\Omega := (2 - 2^{-p}) \times 2^{\epsilon_{\max}}$ to denote the smallest and largest positive floating-point numbers, respectively. For $x \in \mathbb{F}_{p,q}$, we use x^- to denote the largest element of $\mathbb{F}_{p,q} \cup \{-\infty, \infty\}$ that is strictly smaller than x , and we use x^+ to denote the smallest element of $\mathbb{F}_{p,q} \cup \{-\infty, \infty\}$ that is strictly larger than x (e.g., $\Omega^+ = \infty$, $(-\Omega)^- = -\infty$). We often use $\mathbb{F}$ (and $\overline{\mathbb{F}}$) to represent $\mathbb{F}_{p,q}$ (and $\overline{\mathbb{F}}_{p,q}$) when p, q are clear from the context.

Rounding to floating-point numbers. We define floating-point rounding $\lceil \cdot \rceil_{\mathbb{F}} : \mathbb{R} \cup \{-\infty, \infty, \text{NaN}\} \rightarrow \overline{\mathbb{F}}$:

$$\lceil x \rceil_{\mathbb{F}} := \begin{cases} \arg \min_{y \in \mathbb{F}} |x - y| & \text{if } |x| < \Omega + 2^{\epsilon_{\max}-p-1}, \\ \infty & \text{if } x \geq \Omega + 2^{\epsilon_{\max}-p-1}, \\ -\infty & \text{if } x \leq -\Omega - 2^{\epsilon_{\max}-p-1}, \\ \text{NaN} & \text{if } x = \text{NaN}. \end{cases}$$

When there are two (finite) floating-point numbers that are equidistant to $x \in \mathbb{R}$, we break ties using the *ties-to-even* rule (IEEE, 2019): $\lceil x \rceil_{\mathbb{F}}$ is the unique float whose last mantissa bit m_p is zero (see Eq. (2)). When $\mathbb{F}$ is clear from the context, we use $\lceil \cdot \rceil$ to denote $\lceil \cdot \rceil_{\mathbb{F}}$.

We define the *correctly rounded version* of ReLU and exp: for $\rho \in \{\text{ReLU}, \text{exp}\}$ and $x \in \overline{\mathbb{F}}$, $\lceil \rho \rceil_{\mathbb{F}}(x)$ is $\lceil \rho(x) \rceil_{\mathbb{F}}$ if $x \in \mathbb{F}$, 0 if $x = -\infty$, ∞ if $x = \infty$, and NaN if $x = \text{NaN}$. We note that the function values at $\pm\infty$ follow the convention induced by $\lim_{x \rightarrow -\infty} \rho(x) = 0$ and $\lim_{x \rightarrow \infty} \rho(x) = \infty$.

Floating-point operations. We now define floating-point addition, subtraction, multiplication, and division. For $x, y \in \mathbb{F}$, $x \oplus_{\mathbb{F}} y := \lceil x + y \rceil$, $x \ominus_{\mathbb{F}} y := \lceil x - y \rceil$, and $x \otimes_{\mathbb{F}} y := \lceil x \times y \rceil$. For $x, y \in \mathbb{F}$ with $y \neq 0$, $x \oslash_{\mathbb{F}} y :=$

$\lceil x/y \rceil$. We also define these operations for $x, y \in \overline{\mathbb{F}}$ in Appendix A.1, following IEEE (2019). We note that, unlike conventional addition and multiplication, floating-point addition and multiplication are *non-associative*: for $\odot \in \{\oplus_{\mathbb{F}}, \otimes_{\mathbb{F}}\}$, $(x \odot y) \odot z \neq x \odot (y \odot z)$ in general. Hence, we need to be careful about the ordering of operations. As in the conventional operations, we first perform floating-point multiplications and divisions (first priority), and we next perform floating-point additions (second priority). For operations with the same priority, we perform operations in the *left-associative* way, i.e., from left to right.

Order of floating-point summation. When we add n floating-point numbers, the order of summation can be specified by a (rooted) full binary tree $\mathcal{T}_n$ of n leaves indexed by $[n]$. Given $x_1, \dots, x_n$, we define $\bigoplus_{i \in [n]; \mathbb{F}}^{\mathcal{T}_n} x_i$ as the value of the root node of $\mathcal{T}_n$ computed by the following procedure:

- Each leaf node i has a value x_i .
- Each internal node's value is the floating-point sum $\oplus_{\mathbb{F}}$ of its children's values.

By choosing a proper $\mathcal{T}_n$ and leaf indices, $\bigoplus_{i \in [n]; \mathbb{F}}^{\mathcal{T}_n} x_i$ can represent an arbitrary summation order of n values. We note that there can be multiple trees corresponding to a single summation order. When we introduce $\mathcal{T}_n$ in this paper, we implicitly assume that its leaves are indexed by $[n]$ as well.

Given $\mathcal{T}_n$, we also define a group $\mathcal{G}(\mathcal{T}_n)$ generated by the following procedure, starting from $\Pi = \emptyset$:

- Iterate over all internal nodes v such that $\mathcal{T}_{v_L}$ and $\mathcal{T}_{v_R}$ are isomorphic. Here, v_L and v_R denote the children of v and $\mathcal{T}_u$ denotes the subtree of $\mathcal{T}_n$ with the root u .
 - For the sets $\mathcal{L}(\mathcal{T}_{v_L})$ and $\mathcal{L}(\mathcal{T}_{v_R})$ of leaf nodes of $\mathcal{T}_{v_L}$ and $\mathcal{T}_{v_R}$, let $g : \mathcal{L}(\mathcal{T}_{v_L}) \rightarrow \mathcal{L}(\mathcal{T}_{v_R})$ be a bijection induced by an isomorphism between them.
 - Add $\pi_v \in \mathcal{S}_n$ to Π where $\pi_v(i) = i$ if $i \notin \mathcal{L}(\mathcal{T}_{v_L}) \cup \mathcal{L}(\mathcal{T}_{v_R})$, $\pi_v(i) = g(i)$ if $i \in \mathcal{L}(\mathcal{T}_{v_L})$, and $\pi_v(i) = g^{-1}(i)$ if $i \in \mathcal{L}(\mathcal{T}_{v_R})$.
- Set $\mathcal{G}(\mathcal{T}_n)$ as the group generated by Π .

Then, the summation of n inputs whose order is specified by $\mathcal{T}_n$ is $\mathcal{G}(\mathcal{T}_n)$ -invariant as stated in the following lemma. The proof of Lemma 1 is in Appendix C.1.

Lemma 1. *Let $\mathcal{T}_n$ be a full binary tree with n leaves indexed by $[n]$. Then, for any $g \in \mathcal{G}(\mathcal{T}_n)$ and for any $x_1, \dots, x_n \in \overline{\mathbb{F}}_{p,q}$, $\bigoplus_{i \in [n]; \mathbb{F}_{p,q}}^{\mathcal{T}_n} x_i = \bigoplus_{i \in [n]; \mathbb{F}_{p,q}}^{\mathcal{T}_n} x_{g(i)}$.*

For $n \in \mathbb{N}$ and $x_1, \dots, x_n \in \overline{\mathbb{F}}$, we also define

$$\bigoplus_{i=1; \mathbb{F}}^n x_i := (\cdots ((x_1 \oplus_{\mathbb{F}} x_2) \oplus_{\mathbb{F}} x_3) \oplus_{\mathbb{F}} \cdots) \oplus_{\mathbb{F}} x_n \quad (3)$$

and call this summation order the *left-to-right order*. We also set $\bigoplus_{i=1;\mathbb{F}}^0 x_i := 0$, and likewise for the empty tree sum. We also use $\oplus$ and $\bigoplus$ in the left-associative manner:

$$\bigoplus_{i=1;\mathbb{F}}^n x_i \oplus \bigoplus_{j=1;\mathbb{F}}^m y_j = x_1 \oplus_{\mathbb{F}} \cdots \oplus_{\mathbb{F}} x_n \oplus_{\mathbb{F}} y_1 \oplus_{\mathbb{F}} \cdots \oplus_{\mathbb{F}} y_m.$$

Floating-point matrix operations. For floating-point matrices $M, N \in \overline{\mathbb{F}}^{m \times n}$, we define $(M \oplus_{\mathbb{F}} N)_{ij} = M_{ij} \oplus_{\mathbb{F}} N_{ij}$. We also define matrix product under $\mathbb{F}$. For $M \in \overline{\mathbb{F}}^{m \times n}$ and $N \in \overline{\mathbb{F}}^{n \times l}$, $M \otimes_{\mathbb{F}} N \in \overline{\mathbb{F}}^{m \times l}$ satisfies $(M \otimes_{\mathbb{F}} N)_{ij} = \bigoplus_{k \in [n]; \mathbb{F}}^{\tau_n} (M_{ik} \otimes_{\mathbb{F}} N_{kj})$. Here, to consider a more realistic setup, we consider the *high-precision accumulation* in GPUs (Wang et al., 2018). Given the target precision p, q and a higher precision $p' \geq p, q' \geq q$ used to accumulate intermediate values, we define $\hat{\otimes}_{\mathbb{F}_{p,q}, \mathbb{F}_{p',q'}}$ as follows:

$$(M \hat{\otimes}_{\mathbb{F}_{p,q}, \mathbb{F}_{p',q'}} N)_{ij} = \left[\bigoplus_{k \in [n]; \mathbb{F}_{p',q'}}^{\tau_n} (M_{ik} \otimes_{\mathbb{F}_{p',q'}} N_{kj}) \right]_{\mathbb{F}_{p,q}}.$$

Throughout this paper, we use $\mathbb{F}$ and $\mathbb{F}'$ to denote $\mathbb{F}_{p,q}$ and $\mathbb{F}_{p',q'}$, respectively. In addition, we use $\oplus, \bigoplus, \otimes, \ominus$, and $\oslash$ to denote $\oplus_{\mathbb{F}}, \bigoplus_{\mathbb{F}}, \otimes_{\mathbb{F}}, \ominus_{\mathbb{F}}$, and $\oslash_{\mathbb{F}}$; we explicitly write the subscript $\mathbb{F}'$ when the computation is under $\mathbb{F}'$. Likewise, we use $\hat{\otimes}$ instead of $\otimes_{\mathbb{F}_{p,q}, \mathbb{F}_{p',q'}}$. The priority of $\hat{\otimes}$ is identical to that of $\otimes$, and that of $\ominus$ is identical to that of $\oplus$.

Condition on $\mathbb{F}$ and $\mathbb{F}'$. In this paper, we assume Condition 1. Note that most practical floating-point formats satisfy this condition, including double/single/half-precision formats (IEEE, 2019), bfloat16 (Google), and FP8 (E5M2, E4M3, Micikevicius et al. (2022)); see Table 1.

Condition 1. $2 \leq p \leq 2^{q-1} - 3$ and $2 \leq p' \leq 2^{q'-1} - 3$.

2.3. Floating-point transformer

As in transformers under exact arithmetic (Vaswani et al., 2017), we consider a floating-point transformer consisting of feed-forward networks and multi-head self-attention layers with target precision p, q and higher precision parameters p', q' for $\hat{\otimes}$ and the softmax σ . Specifically, given the hidden dimensions d, r , and the input sequence length n , the *feed-forward networks* have the following form: for $\rho = \lceil \text{ReLU} \rceil$, $W_1 \in \mathbb{F}^{r \times d}$, $W_2 \in \mathbb{F}^{d \times r}$, $\mathbf{b}_1 \in \mathbb{F}^r$, $\mathbf{b}_2 \in \mathbb{F}^d$, and $X \in \overline{\mathbb{F}}^{d \times n}$,

$$\text{FF}_{\theta}^{r,d,n}(X) := X \oplus (W_2 \hat{\otimes} \rho(W_1 \hat{\otimes} X \oplus (\mathbf{b}_1 \mathbf{1}_n^{\top})) \oplus (\mathbf{b}_2 \mathbf{1}_n^{\top}))$$

where $\theta \in \mathbb{F}^{2dr+r+d}$ is the vectorization of $W_1, W_2, \mathbf{b}_1, \mathbf{b}_2$ and ρ is applied element-wise (see Eq. (1)). We note that the multiplications in $\mathbf{b}_1 \mathbf{1}_n^{\top}$ and $\mathbf{b}_2 \mathbf{1}_n^{\top}$ are exact. To define the self-attention layer, we first define the (floating-point) softmax function $\sigma : \overline{\mathbb{F}}^n \rightarrow \overline{\mathbb{F}}^n$ as follows: for

$$\mathbf{x} = (x_1, \dots, x_n) \in \overline{\mathbb{F}}^n \text{ and } x_* = \max_{i \in [n]} x_i,¹$$

$$\sigma(\mathbf{x})_i := \left[\lceil \exp \rceil (x_i \ominus x_*) \oslash \left(\bigoplus_{j \in [n]} \lceil \exp \rceil (x_j \ominus x_*) \right) \right]_{\mathbb{F}}. \quad (4)$$

Here, all rounding, $\ominus$, $\oslash$, and $\bigoplus$ inside $\lceil \cdot \rceil_{\mathbb{F}}$ are under $\mathbb{F}'$. We subtract x_* in the exponent following standard implementations of the softmax function in PyTorch (Paszke et al., 2019) and TensorFlow (Abadi et al., 2016); they subtract x_* for better numerical stability.

Given the number of heads $h \in \mathbb{N}$ and the key/query/value dimension $m \in \mathbb{N}$, the *attention layer* has the following form: for $W_i^K, W_i^Q, W_i^V \in \mathbb{F}^{m \times d}$ and $W_i^O \in \mathbb{F}^{d \times m}$ for all $i \in [h]$, for $X \in \overline{\mathbb{F}}^{d \times n}$,

$$\text{AT}_{\phi}^{h,m,d,n}(X) := X \oplus \left(\bigoplus_{i=1}^h W_i^O \hat{\otimes} (V_i \hat{\otimes} \sigma(K_i^{\top} \hat{\otimes} Q_i)) \right) \quad (5)$$

where $\phi \in \mathbb{F}^{4hdm}$ is the vectorization of all W_i^K, W_i^Q, W_i^V , and $W_i^O, K_i = W_i^K \hat{\otimes} X, Q_i = W_i^Q \hat{\otimes} X, V_i = W_i^V \hat{\otimes} X$, and σ is applied column-wise (see Eq. (1)).

Each *transformer block* consists of a composition of an attention layer and a feed-forward network. Given $h, m, r, d, n \in \mathbb{N}$, we use $\mathcal{B}^{h,m,r,d,n}$ to denote all possible compositions of transformer blocks defined as follows:

$$\mathcal{B}^{h,m,r,d,n} := \{ \text{FF}_{\theta_L}^{r,d,n} \circ \text{AT}_{\phi_L}^{h,m,d,n} \circ \cdots \circ \text{FF}_{\theta_1}^{r,d,n} \circ \text{AT}_{\phi_1}^{h,m,d,n} : L \in \mathbb{N}, \theta_l \in \mathbb{F}^{2dr+r+d}, \phi_l \in \mathbb{F}^{4hdm}, \forall l \in [L] \}.$$

Given the input dimension d_{in} , the output dimension d_{out} , a function $f : \mathbb{F}^{d_{\text{in}} \times n} \rightarrow \overline{\mathbb{F}}^{d_{\text{out}} \times n}$ is a *floating-point transformer* if there exist $d, h, m, r \in \mathbb{N}$, $W_{\text{in}} \in \mathbb{F}^{d \times d_{\text{in}}}$, $\mathbf{b}_{\text{in}} \in \mathbb{F}^d$, $W_{\text{out}} \in \mathbb{F}^{d_{\text{out}} \times d}$, $\mathbf{b}_{\text{out}} \in \mathbb{F}^{d_{\text{out}}}$, and $g \in \mathcal{B}^{h,m,r,d,n}$ such that

$$f(X) = W_{\text{out}} \hat{\otimes} g(W_{\text{in}} \hat{\otimes} X \oplus (\mathbf{b}_{\text{in}} \mathbf{1}_n^{\top})) \oplus (\mathbf{b}_{\text{out}} \mathbf{1}_n^{\top}). \quad (6)$$

We use “transformers under exact arithmetic” to denote functions in Eq. (6) where all parameters are real, and all operations are replaced by their exact counterparts.

3. Main results

We investigate whether floating-point transformers inherit the properties of transformers under exact arithmetic. Specifically, we study the following questions: for any input/output dimensions $d_{\text{in}}, d_{\text{out}}$ and sequence length n ,

- Are floating-point transformers permutation-equivariant?
- Can floating-point transformers represent all floating-point permutation-equivariant functions?

¹ $x_* = \text{NaN}$ if $x_i = \text{NaN}$ for some i .

As introduced in Section 1.1, the answers to both questions are no. In Section 3.1, we answer the first question by proving that floating-point transformers can represent a class of non-permutation-equivariant functions. In Section 3.2, we answer the second question by showing that there exists a permutation-equivariant function that cannot be represented by floating-point transformers when the sequence length n is large under the left-to-right summation order. Furthermore, we show that floating-point transformers can represent all permutation-equivariant functions when n is properly bounded. Unless otherwise noted, results in this section are stated under summation orders specified by $\mathcal{T}_1, \mathcal{T}_2, \dots$ and high-precision accumulation.

3.1. Floating-point transformers are not $\mathcal{S}_n$ -equivariant

We first show that floating-point transformers can represent a class of non-permutation-equivariant functions.

Theorem 2. *Suppose $2^q - 4 \geq p' - p$ and $q' \leq 2^{q-1}$. Let $d_{in}, d_{out}, n \in \mathbb{N}$ such that $n \geq 4$, and*

$$\Delta_n := \{[\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{F}^{d_{in} \times n} : \mathbf{x}_i \neq \mathbf{x}_j \forall i \neq j\}.$$

Then, for any $\mathcal{G}(\mathcal{T}_n)$ -equivariant $f^ : \Delta_n \rightarrow \mathbb{F}^{d_{out} \times n}$, there exists a floating-point transformer $f : \mathbb{F}^{d_{in} \times n} \rightarrow \mathbb{F}^{d_{out} \times n}$ such that $f = f^*$ on Δ_n .*

Theorem 2 implies that floating-point transformers can represent all $\mathcal{G}(\mathcal{T}_n)$ -equivariant functions f^* from Δ_n to $\mathbb{F}^{d_{out} \times n}$. Note that f^* may not be permutation-equivariant since for any non-trivial permutation $\pi \notin \mathcal{G}(\mathcal{T}_n)$, it holds that $\pi f^*(X) \neq f^*(\pi X)$ in general. In addition, by augmenting each token with positional information (e.g., $1, \dots, n$), floating-point transformers can represent arbitrary functions from $\mathbb{F}^{d_{in} \times n}$ to $\mathbb{F}^{d_{out} \times n}$ when $[n] \subset \mathbb{F}$ (see Section 4.2). Furthermore, if the input is padded with distinct dummy tokens so that no two sibling leaves of $\mathcal{T}_m$ both carry real tokens, then the identity is the only element of $\mathcal{G}(\mathcal{T}_m)$ fixing all dummy positions. Hence, any $f^* : \Delta_n \rightarrow \mathbb{F}^{d_{out} \times n}$ extends $\mathcal{G}(\mathcal{T}_m)$ -equivariantly to Δ_m , and the transformer from Theorem 2 outputs $f^*(X)$ at the real positions.

To prove Theorem 2, we exploit the non-associativity of floating-point addition. For example, consider two matrices $V \in \mathbb{F}^{m \times n}$, $\Sigma \in \mathbb{F}^{n \times n}$ and a non-trivial permutation $\pi \in \mathcal{S}_n$. Let $Z = V \times \Sigma$ and $\hat{Z} = V \hat{\otimes} \Sigma$, i.e., Z is computed by using exact arithmetic, whereas $\hat{Z}$ is computed using floating-point matrix multiplication. Let $\Sigma^\pi = \pi(\pi(\Sigma^\top))^\top$ be the row- and column-permuted version of Σ , i.e., $\Sigma_{ij}^\pi = \Sigma_{\pi(i)\pi(j)}$. Then, we have $\pi Z = (\pi V) \times \Sigma^\pi$. However, this does not hold for $\hat{Z}$ since

$$(\pi \hat{Z})_{ij} = \left[\bigoplus_{k \in [n]; \mathbb{F}'} V_{ik} \otimes_{\mathbb{F}'} \Sigma_{k\pi(j)} \right]_{\mathbb{F}}$$

$$\neq \left[\bigoplus_{k \in [n]; \mathbb{F}'} V_{i\pi(k)} \otimes_{\mathbb{F}'} \Sigma_{\pi(k)\pi(j)} \right]_{\mathbb{F}} = ((\pi V) \hat{\otimes} \Sigma^\pi)_{ij}$$

in general, due to the non-associativity of $\oplus_{\mathbb{F}'}$.

The proof of Theorem 2 is based on this observation. By exploiting the $V_i \hat{\otimes} \sigma(K_i^\top \hat{\otimes} Q_i)$ part in Eq. (5) (consider $V \leftarrow V_i$ and $\Sigma \leftarrow \sigma(K_i^\top \hat{\otimes} Q_i)$), we first construct non-permutation-equivariant floating-point attention layers that extract partial information about the input order induced by the reduction tree. We then use this information to identify the input up to the action of $\mathcal{G}(\mathcal{T}_n)$. Finally, we adapt the universal approximator construction of Hwang et al. (2025b) to obtain a token-wise universal approximator, which allows us to generate the target output $f^*(X)$ for any $\mathcal{G}(\mathcal{T}_n)$ -equivariant function $f^* : \Delta_n \rightarrow \mathbb{F}^{d_{out} \times n}$. Here, we note that $\mathcal{G}(\mathcal{T}_n)$ -equivariance in our construction is necessary; see Section 4.1 for more discussions. We present the full proof of Theorem 2 in Section 5.1 and Appendix D.

3.2. Floating-point transformers cannot represent all $\mathcal{S}_n$ -equivariant functions

In Section 3.1, we observed that floating-point transformers can represent a class of non-permutation-equivariant functions, i.e., they are not permutation-equivariant in general. In this section, we first show that even some permutation-equivariant floating-point functions cannot be represented by floating-point transformers when the sequence length is large under the left-to-right summation order. This is in contrast to the setting of transformers under exact arithmetic, which can universally approximate permutation-equivariant L^s functions, regardless of the sequence length (Yun et al., 2020a). We then show that floating-point transformers can represent all permutation-equivariant floating-point functions when the sequence length is properly bounded.

To describe our results, we first define (α, β) -similarity.

Definition 1. *Let $d, n, \alpha, \beta \in \mathbb{N}$ such that $\alpha < \beta \leq n$. We say that $X = [\mathbf{x}_1, \dots, \mathbf{x}_n], Y = [\mathbf{y}_1, \dots, \mathbf{y}_n] \in \mathbb{F}^{d \times n}$ are “ (α, β) -similar” if there exist $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{F}^d$ such that*

$$\mathbf{x}_i = \begin{cases} \mathbf{z}_1 & \text{if } 1 \leq i \leq \alpha, \\ \mathbf{z}_2 & \text{if } \alpha < i \leq \beta, \end{cases} \quad \mathbf{y}_i = \begin{cases} \mathbf{z}_1 & \text{if } 1 \leq i < \alpha, \\ \mathbf{z}_2 & \text{if } \alpha \leq i \leq \beta, \end{cases}$$

and $\mathbf{x}_j = \mathbf{y}_j$ for all $j \in \{\beta + 1, \dots, n\}$.

Note that if X and Y are (α, β) -similar, then they only differ at the α -th column.

We now introduce the following theorem, which shows the existence of permutation-equivariant floating-point functions that cannot be represented by floating-point transformers, when the left-to-right summation order is assumed (see Eq. (3)). The proof of Theorem 3 is presented in Section 5.3.

Theorem 3. Consider the left-to-right summation order. Let $d_{in}, d_{out}, n, \alpha, \beta \in \mathbb{N}$ such that $\alpha \geq 3 \times 2^{p'}$, $\beta - \alpha \geq 6 \times 2^{p'}$, and $n \geq \beta$. Then, for any (α, β) -similar $X, Y \in \mathbb{F}^{d_{in} \times n}$ and for any floating-point transformer $f : \mathbb{F}^{d_{in} \times n} \rightarrow \mathbb{F}^{d_{out} \times n}$, $f(X)$ and $f(Y)$ are also (α, β) -similar.

Theorem 3 implies that if inputs X, Y are (α, β) -similar, then the corresponding outputs $f(X), f(Y)$ are also (α, β) -similar for all floating-point transformers f under the left-to-right summation order when $\alpha \geq 3 \times 2^{p'}$, $\beta - \alpha \geq 6 \times 2^{p'}$, and $n \geq \beta$. That is, floating-point transformers cannot represent all permutation-equivariant floating-point functions when $n \geq 9 \times 2^{p'}$ since they cannot represent a non-empty class of permutation-equivariant floating-point functions f^* such that $f^*(X)$ and $f^*(Y)$ are not (α, β) -similar for some (α, β) -similar inputs X, Y . This also implies that to represent all permutation-equivariant functions, $p' = \Omega(\log n)$ is necessary. We note that without the high-precision accumulation (i.e., $p' = p$ and $q' = q$), 9×2^p can be small, especially for low-precision formats, e.g., E5M2 and E4M3.

We prove Theorem 3 using the following observation (Lemmas 16 and 17): for any $a, b \in \mathbb{F}'$, $\bigoplus_{i=1;\mathbb{F}'}^\alpha a \oplus \bigoplus_{i=\alpha+1;\mathbb{F}'}^\beta b = \bigoplus_{i=1;\mathbb{F}'}^{\alpha-1} a \oplus \bigoplus_{i=\alpha;\mathbb{F}'}^\beta b$ when α, β are large. That is, for any $a, b \in \mathbb{F}'$, even if we sequentially add different counts of a and b under floating-point arithmetic, the resulting outputs can be identical due to round-off errors. This property arises from the round-off error in floating-point addition: if we repeatedly add the same number, it can saturate to some finite number. Using this property, we show that if two inputs to the attention layers are (α, β) -similar, then the corresponding outputs should also be (α, β) -similar. Hence, Theorem 3 naturally follows.

The result in Theorem 3 only holds when n and α, β are large. By contrast, under the left-to-right summation order, when $n \leq 6 \times 2^p - 2$, $p = p'$, and $q = q'$ (i.e., no high-precision accumulation), floating-point transformers can represent all permutation-equivariant floating-point functions. We note that while one can hard-code all permutation-equivariant functions when n is bounded, it is not obvious whether they can be represented by floating-point transformers. The proof of Theorem 4 is in Section 5.2.

Theorem 4. Consider the left-to-right summation order, $p = p'$, and $q = q'$. Let $d_{in}, d_{out}, n \in \mathbb{N}$ such that $n \leq 6 \times 2^p - 2$. Then, for any permutation-equivariant function $f^* : \mathbb{F}^{d_{in} \times n} \rightarrow \mathbb{F}^{d_{out} \times n}$, there exists a floating-point transformer $f : \mathbb{F}^{d_{in} \times n} \rightarrow \mathbb{F}^{d_{out} \times n}$ such that $f = f^*$ on $\mathbb{F}^{d_{in} \times n}$.

We lastly show that floating-point transformers can also represent all permutation-equivariant functions under high-precision accumulation and general reduction orders if $p' \geq 2p$ and $n \leq 2^{p+1} - 2$. The condition $p' \geq 2p$ is satisfied for popular floating-point formats, when the high-precision

format uses twice as many bits as the low-precision format (see Table 1). The proof of Theorem 5 is in Appendix C.2.

Theorem 5. Suppose that $p' \geq 2p$ and let $d_{in}, d_{out}, n \in \mathbb{N}$ such that $n \leq 2^{p+1} - 2$. Then, for any permutation-equivariant function $f^* : \mathbb{F}^{d_{in} \times n} \rightarrow \mathbb{F}^{d_{out} \times n}$, there exists a floating-point transformer $f : \mathbb{F}^{d_{in} \times n} \rightarrow \mathbb{F}^{d_{out} \times n}$ such that $f = f^*$ on $\mathbb{F}^{d_{in} \times n}$.

4. Discussions

4.1. Properties of floating-point transformers

In this section, we discuss additional properties of floating-point transformers that arise from the definition. We first show that floating-point transformers must be $\mathcal{G}(\mathcal{T}_n)$ -equivariant. The proof of Theorem 6 is presented in Appendix C.3, which uses Lemma 1 to show the $\mathcal{G}(\mathcal{T}_n)$ -equivariance of attention layers.

Theorem 6. Let $d_{in}, d_{out}, n \in \mathbb{N}$ such that $n \geq 2$. Then, for any floating-point transformer $f : \mathbb{F}^{d_{in} \times n} \rightarrow \mathbb{F}^{d_{out} \times n}$, f is $\mathcal{G}(\mathcal{T}_n)$ -equivariant.

We note that $\mathcal{G}(\mathcal{T}_n)$ -equivariance is the minimal equivariance property in floating-point transformers. Specifically, for any $n \geq 4$ and any permutation $\pi \in \mathcal{S}_n \setminus \mathcal{G}(\mathcal{T}_n)$, there exists a floating-point transformer that is not π -equivariant. The existence of such a floating-point transformer follows from Theorem 2, under its hypotheses.

We next show that if some input columns are identical, then the corresponding output columns should also be the same for floating-point transformers. We note that such a property also exists in transformers under exact arithmetic. The proof of Theorem 7 is in Appendix C.4.

Definition 2. Let $d_{in}, d_{out}, n \in \mathbb{N}$, $\mathcal{X}, \mathcal{Y}$ be sets, and $f : \mathcal{X}^{d_{in} \times n} \rightarrow \mathcal{Y}^{d_{out} \times n}$. We say “ f preserves equality” if for any $X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathcal{X}^{d_{in} \times n}$ and for $f(X) = [\mathbf{y}_1, \dots, \mathbf{y}_n]$, $\mathbf{y}_i = \mathbf{y}_j$ whenever $\mathbf{x}_i = \mathbf{x}_j$ for all $i, j \in [n]$.

Theorem 7. Floating-point transformers preserve equality.

4.2. Effects of positional encoding

In this section, we discuss the effect of positional encoding on the expressive power of floating-point transformers. Given the sequence length n , let $\mathbf{p}_1, \dots, \mathbf{p}_n \in \mathbb{F}^{d_{in}}$ be the positional encoding vectors. A typical strategy to encode positional information in the input X is to add the position vectors to the corresponding position as $X \oplus [\mathbf{p}_1, \dots, \mathbf{p}_n]$ (Vaswani et al., 2017). However, for any non-trivial choice of $\mathbf{p}_1, \dots, \mathbf{p}_n \in \mathbb{F}^{d_{in}}$, some information in X will be lost during encoding as shown in the following lemma. We present the proof of Lemma 8 in Appendix C.5.

Lemma 8. For any $z \in \mathbb{F} \setminus \{0\}$, $x \mapsto x \oplus z$ is not injective on $\mathbb{F}$.

By Lemma 8, one can observe that $X \mapsto X \oplus [\mathbf{p}_1, \dots, \mathbf{p}_n]$ is not injective unless $\mathbf{p}_i = (0, \dots, 0)$ for all $i \in [n]$. In other words, floating-point transformers cannot distinguish $[\mathbf{x}_1, \dots, \mathbf{x}_n]$ and $[\mathbf{y}_1, \dots, \mathbf{y}_n]$ if $\mathbf{x}_i \oplus \mathbf{p}_i = \mathbf{y}_i \oplus \mathbf{p}_i$ for all $i \in [n]$. This contrasts with transformers under exact arithmetic, where positional encoding can increase the expressive power (Yun et al., 2020a). Here, let $x \in \mathbb{F}$ be an input value and $v \in \mathbb{F}$ be the corresponding positional-encoding value. Such a collapse can occur for a wide range of inputs unless $|v| \ll |x|$ so that $x \oplus v = x$, i.e., v may not deliver positional information. For example, consider $v = 1^+$, $x_1 = 2$, and $x_2 = 2^-$. Then, $x_1 \oplus v = 3 = x_2 \oplus v$. Likewise, for $|x_1|, |x_2| \ll |v|$, $x_1 \oplus v = v = x_2 \oplus v$.

We can also encode positional information by introducing an additional dimension to an input X , e.g., consider the positional encoding vector $\mathbf{p}^{\text{pos}} = (1, \dots, n) \in \mathbb{F}^{n \times 1}$ and a new input $[X^\top, \mathbf{p}^{\text{pos}}]^\top = [\mathbf{x}'_1, \dots, \mathbf{x}'_n]$. In this case, $\mathbf{x}'_1, \dots, \mathbf{x}'_n$ are always distinct when $[n] \subset \mathbb{F}$. Hence, floating-point transformers can represent all functions of $X \in \mathbb{F}^{d_{\text{in}} \times n}$ on this augmented input, with values in $\mathbb{F}^{d_{\text{out}} \times n}$, by Theorem 2, under its hypotheses.

4.3. On GPU computations

Our analysis assumes deterministic reduction orders specified by full binary trees $\mathcal{T}_1, \mathcal{T}_2, \dots$. This setting is most closely related to deterministic reduction implementations, such as fixed-order CPU reductions and deterministic GPU execution modes. In contrast, practical GPU implementations often employ runtime-determined reduction orders that may vary across executions due to parallel execution, thread scheduling, memory access patterns, and hardware-level optimization strategies. Consequently, the same floating-point computation may produce slightly different outputs across executions, even when the inputs and model are identical.

5. Proofs

In this section, we present the proofs of Theorems 3 and 4 and of Theorem 2 for $p' \geq p + 2$ (the case $p' \leq p + 1$ is in Appendix D). For clarity, we prove Theorems 2 and 4 in Sections 5.1 and 5.2, and prove Theorem 3 in Section 5.3.

5.1. Proof of Theorem 2 when $p' \geq p + 2$

In this proof, we explicitly construct a floating-point transformer f that represents a target $\mathcal{G}(\mathcal{T}_n)$ -equivariant function $f^* : \Delta_n \rightarrow \mathbb{F}^{d_{\text{out}} \times n}$ when $p' \geq p + 2$. In particular, our floating-point transformer f can be represented as

$$f(X) = W_{\text{out}} \hat{\otimes} (\psi \circ \phi(W_{\text{in}} \hat{\otimes} X)) \quad (7)$$

for $X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{F}^{d_{\text{in}} \times n}$ and some transformer blocks ϕ, ψ . To this end, for some $d > d_{\text{in}}, d_{\text{out}}$, we choose $W_{\text{in}} \in \mathbb{F}^{d \times d_{\text{in}}}$ such that $(W_{\text{in}})_{ii} = 1$ for all $i \in [d_{\text{in}}]$ and all

other entries are zero. That is, $W_{\text{in}} \hat{\otimes} X$ preserves the input information in the first d_{in} dimensions and pads zeros for the remaining $d - d_{\text{in}}$ dimensions. We will specify d at the end of this proof. Here, the i -th column of the output matrix of ϕ will hold the corresponding input $\mathbf{x}_i$ together with information identifying the orbit $\mathcal{G}(\mathcal{T}_n)X$ of the input. Given this, as stated in the following lemma, the information in each column is sufficient to generate the i -th column of the target output. The proof of Lemma 9 is in Appendix C.6.

Lemma 9. *Let $f^* : \Delta_n \rightarrow \mathbb{F}^{d_{\text{out}} \times n}$ be a $\mathcal{G}(\mathcal{T}_n)$ -equivariant function. Then, there exists $\tilde{f} : \mathbb{F}^{d_{\text{in}}} \times \mathbb{F}^{d_{\text{in}} \times n} \rightarrow \mathbb{F}^{d_{\text{out}}}$ such that for each $X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \Delta_n$ and $\pi \in \mathcal{G}(\mathcal{T}_n)$,*

$$\begin{aligned} f^*(X) &= [\tilde{f}(\mathbf{x}_1, X), \dots, \tilde{f}(\mathbf{x}_n, X)] \\ &= [\tilde{f}(\mathbf{x}_1, \pi X), \dots, \tilde{f}(\mathbf{x}_n, \pi X)]. \end{aligned}$$

In ψ , we use the information in each column of $\phi(Z)$ to produce the corresponding column of the target output; we store it in the first d_{out} dimensions of $\psi(Z)$. Lastly, we choose $W_{\text{out}} \in \mathbb{F}^{d_{\text{out}} \times d}$ such that $(W_{\text{out}})_{ii} = 1$ for all $i \in [d_{\text{out}}]$ and all other entries are zero so that $W_{\text{out}} \hat{\otimes} Z = Z_{:d_{\text{out}}}$.

To recover the input sequence up to the action of $\mathcal{G}(\mathcal{T}_n)$, we encode how different groups of input positions are combined during the floating-point reduction specified by $\mathcal{T}_n$. Since floating-point addition is not associative, the output of a floating-point reduction may depend on the reduction order. The following construction assigns a value to each quadruple of positions that reflects how the corresponding leaves interact in the reduction tree.

For each quadruple of distinct indices $\mathcal{I} = (\iota_1, \iota_2, \iota_3, \iota_4)$ in $[n]$ and each pair of distinct elements (j, k) in $[4]$, we define

$$\tau_{\mathcal{I},(j,k)} = \left[\bigoplus_{i \in [n]; \mathbb{F}'} \delta_{\mathcal{I},(j,k),i} \right]_{\mathbb{F}} \text{ where}$$

$$\delta_{\mathcal{I},(j,k),i} = \begin{cases} 2^{e^*} & \text{if } i = \iota_j, \\ 2^{e^* - p - 1} & \text{if } i = \iota_k, \\ 2^{e^* - p' - 1} & \text{if } i \in \{\iota_1, \iota_2, \iota_3, \iota_4\} \setminus \{\iota_j, \iota_k\}, \\ 0 & \text{otherwise,} \end{cases}$$

and $e^* = 2^{q-1} - 3$. Intuitively, the value of $\tau_{\mathcal{I},(j,k)}$ depends on how the four nonzero values are grouped during the reduction specified by $\mathcal{T}_n$. Thus, $\tau_{\mathcal{I},(j,k)}$ encodes local reduction-order information among the positions $\iota_1, \iota_2, \iota_3, \iota_4$. See Lemma 28 in Appendix for more details.

We then define $\mathbf{q}(\pi) = (\tau_{(\pi(\iota_1), \dots, \pi(\iota_4)), (j,k)})_{\mathcal{I},(j,k)}$, for $\pi \in \mathcal{S}_4$ where $\mathcal{I} = (\iota_1, \iota_2, \iota_3, \iota_4)$ ranges over all quadruples of distinct indices in $[n]$ and (j, k) ranges over all ordered pairs of distinct elements in $[4]$.

The following lemma shows that $\mathbf{q}(\pi)$ uniquely determines the permutation π up to $\mathcal{G}(\mathcal{T}_n)\pi$, i.e., the reduction-order information encoded by $\mathbf{q}(\pi)$ is sufficient to identify $\mathcal{G}(\mathcal{T}_n)\pi$. The proof of Lemma 10 is in Appendix C.7.

Lemma 10. Let $p' \geq p + 2$, $n \geq 4$, and $\mathcal{T}_n$ be a full binary tree of n leaves indexed by $[n]$. Then, for each $\pi \in \mathcal{S}_n$, $q^{-1}(q(\pi)) = \mathcal{G}(\mathcal{T}_n)\pi$.

Lemma 10 implies that given $q(\pi)$, we can recover the permutation π up to its $\mathcal{G}(\mathcal{T}_n)$ -equivalent counterpart. Suppose that an input $X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \Delta_n$ to a floating-point transformer is given. If we identify the set of input tokens $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ together with the relative reduction-order information among every quadruple of distinct tokens in $\mathcal{X}$ given by q , then we can recover the input sequence up to the action of $\mathcal{G}(\mathcal{T}_n)$. By Lemma 9, this information is sufficient to generate the target $\mathcal{G}(\mathcal{T}_n)$ -equivariant output.

Based on this, we design an attention layer in the following lemma that collects the input information and saves it in each column. In Lemma 11, we use $\gamma = \binom{|\mathbb{F}|^{d_{in}}}{4}$, $\mathcal{Z}_1 = \{\mathbf{z}_{1,1}, \mathbf{z}_{1,2}, \mathbf{z}_{1,3}, \mathbf{z}_{1,4}, \dots, \mathbf{z}_{\gamma,1}, \mathbf{z}_{\gamma,2}, \mathbf{z}_{\gamma,3}, \mathbf{z}_{\gamma,4}\}$ to denote all quadruples of distinct elements in $\mathbb{F}^{d_{in}}$, and $\{(\alpha_1, \beta_1), \dots, (\alpha_{12}, \beta_{12})\} = \{(i, j) : i, j \in [4], i \neq j\}$. The proof of Lemma 11 is in Appendix C.9.

Lemma 11. Suppose $p' \geq p + 2$, $2^q - 4 \geq p' - p$, and $q' \leq 2^{q-1}$. Let $d_{in}, n \in \mathbb{N}$ with $n \geq 4$, and $e = d_{in} + 12\gamma$. There exist $h, t_d, t_m, t_r \in \mathbb{N}$ such that for any $d \geq t_d$, $m \geq t_m$, and $r \geq t_r$, there exists $\phi \in \mathcal{B}^{h,m,r,d,n}$ satisfying the following. For each $Z \in \mathbb{F}^{d \times n}$ with $Z_{:d_{in}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \Delta_n$ and $Z_{d_{in}+1:} = \mathbf{0}_{(d-d_{in}) \times n}$, $j \in [\gamma]$, and $k \in [12]$,

- $\phi(Z)_{:d_{in}} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ and $\phi(Z)_{e+1:} = \mathbf{0}_{(d-e) \times n}$,
 - $\phi(Z)_{d_{in}+12(j-1)+k} = \left[\bigoplus_{i \in [n]; \mathbb{F}'} \varepsilon_j \otimes_{\mathbb{F}'} \zeta_{i,j,k} \right]_{\mathbb{F}} \times \mathbf{1}_n^\top$,
- $$\varepsilon_j := \begin{cases} \left[1 \otimes_{\mathbb{F}'} \left(\bigoplus_{i \in [n]; \mathbb{F}'} 1 \right) \right]_{\mathbb{F}} & \text{if } \mathbf{x}_i \notin \mathcal{Z}_j \forall i, \\ \left[1 \otimes_{\mathbb{F}'} |\{i \in [n] : \mathbf{x}_i \in \mathcal{Z}_j\}| \right]_{\mathbb{F}} & \text{otherwise,} \end{cases}$$
- $$\zeta_{i,j,k} := \begin{cases} 2^{e^*+2} & \text{if } \mathbf{x}_i = \mathbf{z}_{j,\alpha_k}, \\ 2^{e^*-p+1} & \text{if } \mathbf{x}_i = \mathbf{z}_{j,\beta_k}, \\ 2^{e^*-p'+1} & \text{if } \mathbf{x}_i \in \mathcal{Z}_j \setminus \{\mathbf{z}_{j,\alpha_k}, \mathbf{z}_{j,\beta_k}\}, \\ 0 & \text{otherwise.} \end{cases}$$

For $d \geq t_d$, $m \geq t_m$, and $r \geq t_r$, let $\phi_{d,m,r} \in \mathcal{B}^{h,m,r,d,n}$ be the transformer block in Lemma 11. Then, the i -th column of the output of $\phi_{d,m,r}$ contains $\mathbf{x}_i$ in the first d_{in} coordinates. For the next $e - d_{in}$ coordinates, the output stores the information about the set $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ and the relative reduction-order information among quadruples of distinct input tokens encoded through Lemma 10.

By Lemmas 9 and 10, it follows that there exists a column-wise function $\tilde{\psi} : \mathbb{F}^e \rightarrow \mathbb{F}^{d_{out}}$ such that $\tilde{\psi}(\phi_{d,m,r}(W_{in} \hat{\otimes} X)_{:e}) = f^*(X)$. Note that the value $\phi_{d,m,r}(W_{in} \hat{\otimes} X)_{:e}$ is independent of d, m, r . We now construct the target output using the following lemma. We present the proof of Lemma 12 in Appendix C.8.

Lemma 12. Let $d_1, d_2, m, h, n \in \mathbb{N}$. There exist $t'_d, t'_r \in \mathbb{N}$ such that for any $d \geq t'_d$, $r \geq t'_r$, and $\psi^* : \mathbb{F}^{d_1} \rightarrow \mathbb{F}^{d_2}$, there exists $\psi \in \mathcal{B}^{h,m,r,d,n}$ satisfying the following properties. For each $Z \in \mathbb{F}^{d \times n}$ with $Z_{:d_1} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ and $Z_{d_1+1:} = \mathbf{0}_{(d-d_1) \times n}$, $\psi(Z)_{:d_2} = [\psi^*(\mathbf{x}_1), \dots, \psi^*(\mathbf{x}_n)]$ and $\psi(Z)_{d_2+1:} = \mathbf{0}_{(d-d_2) \times n}$.

By Lemma 12, for any large enough d, r and any $m \geq t_m$, there exists a transformer block $\psi_{d,m,r} \in \mathcal{B}^{h,m,r,d,n}$ such that $W_{out} \hat{\otimes} (\psi_{d,m,r} \circ \phi_{d,m,r}(W_{in} \hat{\otimes} X)) = f^*(X)$. Choose $d \geq \max\{t_d, t'_d\}$, $m \geq t_m$, and $r \geq \max\{t_r, t'_r\}$ where t_d, t_m, t_r are from Lemma 11 and t'_d, t'_r are from Lemma 12, and choose $\phi = \phi_{d,m,r}$ and $\psi = \psi_{d,m,r}$. As we mentioned at the beginning of this proof, we construct f as in Eq. (7). This completes the proof.

5.2. Proof of Theorem 4

We construct a floating-point transformer f that implements a permutation-equivariant function $f^* : \mathbb{F}^{d_{in} \times n} \rightarrow \mathbb{F}^{d_{out} \times n}$ when $n \leq 6 \times 2^p - 2$. Our construction is given by

$$f(X) = W_{out} \hat{\otimes} (\psi \circ \phi(W_{in} \hat{\otimes} X)) \quad (8)$$

for some transformer blocks ϕ and ψ . As in the proof of Theorem 2, we also choose $W_{in} \in \mathbb{F}^{d \times d_{in}}$ and $W_{out} \in \mathbb{F}^{d_{out} \times d}$ so that $W_{in} \hat{\otimes} (x_1, \dots, x_{d_{in}}) = (x_1, \dots, x_{d_{in}}, 0, \dots, 0)$ and $W_{out} \hat{\otimes} (x_1, \dots, x_d) = (x_1, \dots, x_{d_{out}})$ where d will be specified at the end of this proof. To describe the functions of ϕ and ψ , we introduce the following lemma where $\mathbf{z}_1, \dots, \mathbf{z}_{|\mathbb{F}|^{d_{in}}}$ denote all vectors in $\mathbb{F}^{d_{in}}$. The proof of Lemma 13 is in Appendix C.10.

Lemma 13. For any permutation-equivariant function $f^* : \mathbb{F}^{d_{in} \times n} \rightarrow \mathbb{F}^{d_{out} \times n}$, there exists $\tilde{f} : \mathbb{F}^{d_{in}} \times (\mathbb{N} \cup \{0\})^{|\mathbb{F}|^{d_{in}}} \rightarrow \mathbb{F}^{d_{out}}$ such that for each $X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{F}^{d_{in} \times n}$,

$$f^*(X) = [\tilde{f}(\mathbf{x}_1, \mathbf{c}_X), \dots, \tilde{f}(\mathbf{x}_n, \mathbf{c}_X)]$$

where $\mathbf{c}_X \in (\mathbb{N} \cup \{0\})^{|\mathbb{F}|^{d_{in}}}$ is a vector whose i -th coordinate is the number of columns of X equal to $\mathbf{z}_i$.

Lemma 13 states that for any permutation-equivariant function f^* , if we know the i -th input token and the input distribution, then we can generate the i -th column of the output. Using the following lemmas, we design a transformer block ϕ in which the i -th column of the output of ϕ contains the distribution of inputs and the i -th input token. This information will be used in ψ to generate the target output. The proofs of Lemmas 14 and 15 are in Appendices C.11 and C.12, respectively.

Lemma 14. Assume the left-to-right summation order, $p = p'$, and $q = q'$. Let $d_{in}, n \in \mathbb{N}$, $h = 1$, and $e = d_{in} + |\mathbb{F}|^{d_{in}}$. Then, there exist $t_d, t_m, t_r \in \mathbb{N}$ such that for any $d \geq t_d$, $m \geq t_m$, and $r \geq t_r$, there exists $\phi \in \mathcal{B}^{h,m,r,d,n}$ satisfying the following. For each $Z \in \mathbb{F}^{d \times n}$ satisfying $Z_{:d_{in}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{F}^{d_{in} \times n}$ and $Z_{d_{in}+1:} = \mathbf{0}_{(d-d_{in}) \times n}$,

- $\phi(Z)_{:d_{in}} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ and $\phi(Z)_{e+1:} = \mathbf{0}_{(d-e) \times n}$,
- for each $j \in [e] \setminus [d_{in}]$, $\phi(Z)_j = (\bigoplus_{i=1}^{k_j} 1^+) \times \mathbf{1}_n^\top$ where k_j denotes the number of $i \in [n]$ such that $\mathbf{x}_i = \mathbf{z}_{j-d_{in}}$.

Lemma 15. *The values $\bigoplus_{i=1}^{k_j} 1^+$ for all $k \in [3 \times 2^p - 1] \cup \{0\}$ are distinct and $\bigoplus_{i=1}^{3 \times 2^p - 1 + n} 1^+ = 2^{p+2}$ for all $n \in \mathbb{N} \cup \{0\}$.*

Lemmas 14 and 15 imply that for each $d \geq t_d, m \geq t_m$, and $r \geq t_r$, there exists $\phi_{d,m,r} \in \mathcal{B}^{1,m,r,d,n}$ such that each column of the output of $\phi_{d,m,r}$ encodes the input distribution and the corresponding input token. Here, the input distribution can be recovered from $\bigoplus_{i=1}^{k_j} 1^+$ for all j by using Lemma 15. If $\bigoplus_{i=1}^{k_j} 1^+ < 2^{p+2}$ for all j , then each output column of $\phi_{d,m,r}$ contains the exact distribution of inputs. If $\bigoplus_{i=1}^{k_j} 1^+ = 2^{p+2}$ for two indices j (say j_1, j_2), then we can conclude that $X = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ contains exactly $3 \times 2^p - 1$ columns of $\mathbf{z}_{j_1}$ and $3 \times 2^p - 1$ columns of $\mathbf{z}_{j_2}$. If $\bigoplus_{i=1}^{k_j} 1^+ = 2^{p+2}$ for a single j (say j_3), then we can recover the number of columns in X that are equal to $\mathbf{z}_{j_3}$ since we have the exact count of columns in X that are equal to $\mathbf{z}_j$ for all $j \neq j_3$. Hence, there exists $\tilde{\psi} : \mathbb{F}^e \rightarrow \mathbb{F}^{d_{out}}$ such that $\tilde{\psi}(\phi_{d,m,r}(W_{in} \hat{\otimes} X)_{:e}) = f^*(X)$. Here, the value of $\phi_{d,m,r}(W_{in} \hat{\otimes} X)_{:e}$ is independent of d, m, r .

We then apply Lemma 12 with $\psi^* \leftarrow \tilde{\psi}$. This implies that for any $d \geq \max\{t_d, t'_d\}, r \geq \max\{t_r, t'_r\}, m \geq t_m$ (t'_d, t'_r are from Lemma 12 and t_d, t_m, t_r are from Lemma 14), there exists $\psi_{d,m,r}$ such that $W_{out} \hat{\otimes} (\psi_{d,m,r} \circ \phi_{d,m,r}(W_{in} \hat{\otimes} X)) = f^*(X)$. Then, choosing $\phi = \phi_{d,m,r}$ and $\psi = \psi_{d,m,r}$ for $d \geq \max\{t_d, t'_d\}, r \geq \max\{t_r, t'_r\}, m \geq t_m$ completes the proof.

5.3. Proof of Theorem 3

For simplicity, we write the proof for $p' = p$ and $q' = q$ so that $\otimes_{\mathbb{F}'}, \oplus_{\mathbb{F}'}$ coincide with $\otimes, \oplus$, and $\hat{\otimes}$. For $p' > p$ and/or $q' > q$, replace $\otimes, \oplus$ by $\otimes_{\mathbb{F}'}, \oplus_{\mathbb{F}'}$ inside $\hat{\otimes}$ and σ , and apply Lemmas 16 and 17 to $\mathbb{F}'$ (see the remark after Lemma 17).

Let $\alpha, \beta \in \mathbb{N}$ such that $\alpha \geq 3 \times 2^{p'}, \beta - \alpha \geq 6 \times 2^{p'}$, and $n \geq \beta$. Since the feed-forward networks in a floating-point transformer are applied column-wise, they preserve the (α, β) -similarity of inputs: if their inputs are (α, β) -similar, then the corresponding outputs are also (α, β) -similar. Likewise, for any ψ' that preserves the (α, β) -similarity of the inputs and for any floating-point matrix W , functions $Z \mapsto W \otimes Z$ and $Z \mapsto Z \oplus \psi'(Z)$ also preserve the (α, β) -similarity. Hence, to prove Theorem 3, it suffices to show that for any $W^V, W^K, W^Q \in \mathbb{F}^{m \times d}$, the function $g : Z \mapsto (W^V \otimes Z) \otimes \sigma((W^K \otimes Z)^\top \otimes (W^Q \otimes Z))$ preserves the (α, β) -similarity of the inputs.

For (α, β) -similar $Z, Z' \in \mathbb{F}^{d \times n}$, let $S = [\mathbf{s}_1, \dots, \mathbf{s}_n] = \sigma((W^K \otimes Z)^\top \otimes (W^Q \otimes Z))$ and $V = [\mathbf{v}_1, \dots, \mathbf{v}_n] =$

$W^V \otimes Z$, i.e., $g(Z) = V \otimes S$. We also define $S', \mathbf{s}'_j, V', \mathbf{v}'_j$ analogously. By the definition of the floating-point softmax σ (Eq. (4)), one can observe that $\mathbf{s}_j^\top, (\mathbf{s}'_j)^\top$ are also (α, β) -similar for all $j \in [n] \setminus \{\alpha\}$. Furthermore, we have

$$g(Z)_i = \left[\bigoplus_{j=1}^n (V_{ij} \otimes s_{1,j}), \dots, \bigoplus_{j=1}^n (V_{ij} \otimes s_{n,j}) \right]$$

where $(s_{k,1}, \dots, s_{k,n}) = \mathbf{s}_k$. A similar expression holds for $g(Z')_i$. Since V, V' are (α, β) -similar and $\mathbf{s}_j^\top, (\mathbf{s}'_j)^\top$ are (α, β) -similar for all $j \neq \alpha$, each coordinate $j \neq \alpha$ of $g(Z)_i$ and $g(Z')_i$ can be written as

$$\begin{aligned} g(Z)_{ij} &= \bigoplus_{k=1}^{\alpha} a_{i,j} \oplus \bigoplus_{k=\alpha+1}^{\beta} b_{i,j} \oplus \bigoplus_{k=\beta+1}^n (V_{ik} \otimes s_{j,k}), \\ g(Z')_{ij} &= \bigoplus_{k=1}^{\alpha-1} a_{i,j} \oplus \bigoplus_{k=\alpha}^{\beta} b_{i,j} \oplus \bigoplus_{k=\beta+1}^n (V_{ik} \otimes s_{j,k}), \end{aligned} \quad (9)$$

where $a_{i,j} = V_{ik} \otimes s_{j,k} = V'_{ik'} \otimes s'_{j,k'}$ for all $k \in [\alpha], k' \in [\alpha - 1]$ and $b_{i,j} = V_{ik} \otimes s_{j,k} = V'_{ik'} \otimes s'_{j,k'}$ for all $k \in [\beta] \setminus [\alpha], k' \in [\beta] \setminus [\alpha - 1]$.

If all operations are exact (i.e., not based on floating-point arithmetic), then $g(Z)_i$ and $g(Z')_i$ are not (α, β) -similar in general since the counts of $a_{i,j}$ and $b_{i,j}$ are different in Eq. (9) for $g(Z)_{ij}$ and $g(Z')_{ij}$. However, this does not hold for floating-point addition, because repeated addition of the same number will saturate due to the round-off error. Hence, for large enough α, β , we have $\bigoplus_{k=1}^{\alpha} a_{i,j} \oplus \bigoplus_{k=\alpha+1}^{\beta} b_{i,j} = \bigoplus_{k=1}^{\alpha-1} a_{i,j} \oplus \bigoplus_{k=\alpha}^{\beta} b_{i,j}$. We formally prove this intuition in the following lemmas. The proofs of Lemmas 16 and 17 are in Appendices C.13 and C.14.

Lemma 16. *For any $x \in \mathbb{F}$ and $n, m \in \mathbb{N} \cup \{0\}$, it holds that $\bigoplus_{i=1}^{3 \times 2^p - 1 + n} x = \bigoplus_{i=1}^{3 \times 2^p - 1 + m} x$. Furthermore, $\bigoplus_{i=1}^{3 \times 2^p - 1} x \in \{0, \pm\infty, \text{NaN}\} \cup (\{\pm 2^e : e \in \mathbb{Z}\} \cap \mathbb{F})$.*

Lemma 17. *For any $x \in \mathbb{F}, z \in \{0, \pm\infty, \text{NaN}\} \cup (\{\pm 2^e : e \in \mathbb{Z}\} \cap \mathbb{F})$, and $n, m \in \mathbb{N} \cup \{0\}$, it holds that $z \oplus \bigoplus_{i=1}^{6 \times 2^p + n} x = z \oplus \bigoplus_{i=1}^{6 \times 2^p + m} x$.*

We note that Lemmas 16 and 17, and Lemmas 15 and 18 and Lemma 1 used in their proofs, refer to a single floating-point format and rely only on Condition 1. Hence, they also hold for numbers and operations $\mathbb{F}'$.

By Lemmas 16 and 17 and our choices of α, β , the two expressions in Eq. (9) coincide, i.e., $g(Z)_j = g(Z')_j$ for all $j \neq \alpha$. We now consider the remaining column $j = \alpha$: if two columns of Z coincide, then so do the corresponding columns of $W^Q \otimes Z$, hence those of S , and of $g(Z) = V \otimes S$, and likewise for Z' . Thus, $g(Z)$ is constant on the columns $1, \dots, \alpha$ and on the columns $\alpha + 1, \dots, \beta$, while $g(Z')$ is constant on the columns $1, \dots, \alpha - 1$ and on the columns $\alpha, \dots, \beta$, i.e., $g(Z)$ and $g(Z')$ are (α, β) -similar. This completes the proof.

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Impact Statement

This paper analyzes the representation power of transformers under floating-point arithmetic. We could not find notable potential societal consequences of our work.

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A. Additional preliminaries

For a logical proposition P , we define $\mathbb{1}_P = 1$ if P is true and $\mathbb{1}_P = 0$ if P is false. We use ρ to denote $\lceil \text{ReLU} \rceil$. We use $\mathbb{F}_-$ and $\mathbb{F}_+$ to denote the set of negative finite floats and the set of positive finite floats, respectively.

Definition 3. Let $d, d' \in \mathbb{N}$ and $f : \mathbb{F}^d \rightarrow \mathbb{F}^{d'}$. We say f is a “floating-point affine transformation” if there exist $W \in \mathbb{F}^{d' \times d}$ and $\mathbf{b} \in \mathbb{F}^{d'}$ such that $f(\mathbf{x}) = (W \otimes \mathbf{x}) \oplus \mathbf{b}$ for all $\mathbf{x} \in \mathbb{F}^d$.

A.1. Floating-point operations

In this section, we extend floating-point addition, subtraction, multiplication, and division to $\mathbb{F} \times \mathbb{F}$. For $\odot$, we specify the extension only for $0 \leq x \leq y < \infty$ with $y > 0$, for $0 \leq x < \infty$ with $y = \infty$, or $y = \text{NaN}$, since the other cases do not appear in this paper (on the remaining pairs of $\{(x, y) \in \mathbb{F}^2 : y \neq 0\}$, $x \odot y = \lceil x/y \rceil$ as in Section 2.2). This is because the division only arises during the computation of the floating-point softmax function. Recall the definition of the floating-point softmax function $\sigma : \mathbb{F}^n \rightarrow \mathbb{F}^n$ in Eq. (4): for $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}^n$ and $x_* = \max_{i \in [n]} x_i$,

$$\sigma(\mathbf{x})_i := \left[\lceil \exp \rceil_{\mathbb{F}'}(x_i \ominus_{\mathbb{F}'} x_*) \odot_{\mathbb{F}'} \left(\bigoplus_{j \in [n]; \mathbb{F}'}^{\mathcal{T}_n} \lceil \exp \rceil_{\mathbb{F}'}(x_j \ominus_{\mathbb{F}'} x_*) \right) \right]_{\mathbb{F}}.$$

Here, if $x_i \in \{\text{NaN}, \infty\}$ for some $i \in [n]$, then $x_i - x_* = \text{NaN}$, i.e., division by NaN occurs. Likewise, if $x_i = -\infty$ for all $i \in [n]$, then the division by NaN occurs. If $x_i \in \mathbb{F} \cup \{-\infty\}$ for all $i \in [n]$ and $x_j \neq -\infty$ for some $j \in [n]$, then the denominator in the computation of the floating-point softmax is non-zero, and the numerator is not larger than the denominator. Indeed, each summand $\lceil \exp \rceil_{\mathbb{F}'}(x_k \ominus_{\mathbb{F}'} x_*)$ lies in $[0, 1]$ (as $x_k \ominus_{\mathbb{F}'} x_* \leq 0$) and at least one of them equals 1; since $a \oplus_{\mathbb{F}'} b \geq \max\{a, b\}$ for all $a, b \geq 0$ by the monotonicity of rounding, the tree sum is at least 1 and at least each summand. For large n the tree sum may be ∞ , in which case $\sigma(\mathbf{x})$ is the all-zero vector by the definition of $\odot$ below. Under this observation, we now define floating-point addition, subtraction, and multiplication on $\mathbb{F} \times \mathbb{F}$, and division on the pairs listed below, as follows:

$$\begin{aligned} x \oplus y &= y \oplus x := \begin{cases} \lceil x + y \rceil & \text{if } (x, y) \in \mathbb{F}^2, \\ \infty & \text{if } (x, y) \in (\mathbb{F} \cup \{\infty\}) \times \{\infty\}, \\ -\infty & \text{if } (x, y) \in (\mathbb{F} \cup \{-\infty\}) \times \{-\infty\}, \\ \text{NaN} & \text{otherwise,} \end{cases} \\ x \otimes y &= y \otimes x := \begin{cases} \lceil x \times y \rceil & \text{if } (x, y) \in \mathbb{F}^2, \\ \infty & \text{if } (x, y) \in ((\mathbb{F}_- \cup \{-\infty\}) \times \{-\infty\}) \cup ((\mathbb{F}_+ \cup \{\infty\}) \times \{\infty\}), \\ -\infty & \text{if } (x, y) \in ((\mathbb{F}_- \cup \{-\infty\}) \times \{\infty\}) \cup ((\mathbb{F}_+ \cup \{\infty\}) \times \{-\infty\}), \\ \text{NaN} & \text{otherwise,} \end{cases} \\ x \ominus y &:= x \oplus (-1 \otimes y), \\ x \odot y &:= \begin{cases} \lceil x/y \rceil & \text{if } 0 \leq x \leq y < \infty \text{ and } y > 0, \\ 0 & \text{if } 0 \leq x < \infty \text{ and } y = \infty, \\ \text{NaN} & \text{if } y = \text{NaN}. \end{cases} \end{aligned}$$

A.2. Values of p and q in practical floating-point formats

Table 1. Values of p and q in practical floating-point formats.

Floating-point format	(p, q)
64-bit double-precision (IEEE, 2019)	(52, 11)
32-bit single-precision (IEEE, 2019)	(23, 8)
16-bit half-precision (IEEE, 2019)	(10, 5)
bfloat16 (Google)	(7, 8)
8-bit E5M2 (Micikevicius et al., 2022)	(2, 5)
8-bit E4M3 (Micikevicius et al., 2022)	(3, 4)

B. Technical lemmas

Lemma 18. For any $x \in \mathbb{F}$ and $z \in \mathbb{F} \cup \{\infty\}$ with $x, z \geq 0$, and for any $n, m \in \mathbb{N} \cup \{0\}$, it holds that $z \oplus \bigoplus_{i=1}^{3 \times 2^p - 1 + n} x = z \oplus \bigoplus_{i=1}^{3 \times 2^p - 1 + m} x$. Furthermore, $z \oplus \bigoplus_{i=1}^{3 \times 2^p - 1} x \in \{0, z, z^+, \pm\infty, \text{NaN}\} \cup (\{2^e : e \in \mathbb{Z}\} \cap \mathbb{F})$ and $\bigoplus_{i=1}^{3 \times 2^p - 1} x \in \{0, \pm\infty, \text{NaN}\} \cup (\{2^e : e \in \mathbb{Z}\} \cap \mathbb{F})$.

Proof. Without loss of generality, suppose that $z < \infty$ and $x \neq 0$. Choose $e \in \mathbb{Z}$ such that $2^e \leq x < 2^{e+1}$. We consider all possible cases.

Case $e \geq \epsilon_{\max} - p - 1$ and $x > 2^{\epsilon_{\max} - p - 1}$.

In this case, it holds that

$$z \oplus \bigoplus_{i=1}^{3 \times 2^p - 1} x \geq \bigoplus_{i=1}^{3 \times 2^p - 1} x \geq \bigoplus_{i=1}^{3 \times 2^p - 1} (2^{\epsilon_{\max} - p - 1})^+ = \infty,$$

where the last equality is from Lemma 15. This implies that $z \oplus \bigoplus_{i=1}^{3 \times 2^p - 1 + n} x = \infty$ for all $n \in \mathbb{N} \cup \{0\}$ and the result follows.

Case $e \geq \epsilon_{\max} - p - 1$ and $x = 2^{\epsilon_{\max} - p - 1}$.

Since

$$z \oplus \bigoplus_{i=1}^{2^{p+1}} x \geq \bigoplus_{i=1}^{2^{p+1}} x = 2^{\epsilon_{\max}},$$

and $(2^{\epsilon_{\max}})^+ - 2^{\epsilon_{\max}} = 2^{\epsilon_{\max} - p} = 2x$, one can observe that $z \oplus \bigoplus_{i=1}^{2^{p+1} + n} x$ is either z, z^+ (when $z \geq 2^{\epsilon_{\max}}$) or $2^{\epsilon_{\max}}$ (when $z < 2^{\epsilon_{\max}}$) for all $n \in \mathbb{N} \cup \{0\}$. Hence, the result follows.

Case $e \leq \epsilon_{\max} - p - 2$ and $z \geq 2^{e+p+2}$.

In this case, since $2^e \leq x < 2^{e+1}$ and $z^+ - z \geq (2^{e+p+2})^+ - 2^{e+p+2} = 2^{e+2}$, one can observe that $z \oplus \bigoplus_{i=1}^k x = z$ for all $k \in \mathbb{N}$. Hence, the result follows.

Case $e \leq \epsilon_{\max} - p - 2, z < 2^{e+p+2}$, and $x = 2^e$.

If $z \geq 2^{e+p+1}$, then $z \oplus \bigoplus_{i=1}^k x = z$ (or z^+) for all $k \in \mathbb{N}$ and the result follows. For $z < 2^{e+p+1}$, it holds that

$$z \oplus \bigoplus_{i=1}^{2^{p+1}} x \geq \bigoplus_{i=1}^{2^{p+1}} x = \bigoplus_{i=1}^{2^{p+1}} 2^e = 2^{e+p+1} \quad \text{and} \quad z \oplus \bigoplus_{i=1}^k x \leq 2^{e+p+1}$$

for all $k \in \mathbb{N}$. Hence, one can observe that $z \oplus \bigoplus_{i=1}^{2^{p+1} + n} x = 2^{e+p+1}$ for all $n \in \mathbb{N} \cup \{0\}$ and the result follows.

Case $e \leq \epsilon_{\max} - p - 2, z < 2^{e+p+2}$, and $x > 2^e$.

Since $2^e < x < 2^{e+1}$ and $(2^{e+p+2})^+ - 2^{e+p+2} = 2^{e+2}$, one can observe that

$$z \oplus \bigoplus_{i=1}^k x \leq 2^{e+p+2} \tag{10}$$

for all $k \in \mathbb{N}$. Furthermore, for $e \geq \epsilon_{\min}$, by Lemma 15,

$$z \oplus \bigoplus_{i=1}^{3 \times 2^p - 1} x \geq \bigoplus_{i=1}^{3 \times 2^p - 1} x \geq \bigoplus_{i=1}^{3 \times 2^p - 1} (2^e)^+ = 2^{e+p+2}. \quad (11)$$

Here, the last equality of Eq. (11) follows from Lemma 15 rescaled by 2^e . From Eqs. (10) and (11), the result follows.

For $e < \epsilon_{\min}$, x is subnormal and $2^e < (2^e)^+ < 2^{e+1}$ since we excluded the case that $x = 2^{\epsilon_{\min}-p}$. Write $(2^e)^+ = 2^e + 2^{e-k}$, i.e., $2^{e-k} = \omega$ and $k \in \{2, \dots, p-1\}$, and let $S_j = \bigoplus_{i=1}^j (2^e)^+$. Since every integer multiple of ω smaller than $2^{\epsilon_{\min}+1}$ is in $\mathbb{F}$ and $j \times (2^e + 2^{e-k}) \leq 2^{p-k} \times 2^e + 2^{p-k} \times 2^{e-k} < 2^{\epsilon_{\min}+1}$ for all $j \leq 2^{p-k}$, the first 2^{p-k} additions are exact. In particular, $S_{2^{p-k}} = 2^{p-k} \times 2^e + 2^{p-k} \times 2^{e-k} > 2^{p-k} \times 2^e$. We now show $S_j > j \times 2^e$ for all $j \in [2^{p-k}, 2^p]_{\mathbb{Z}}$ by induction, the above being the base case. Note that $j \times 2^e \in \mathbb{F}$ for all $j \leq 2^p$, as it is a multiple of 2^e smaller than 2^{e+p+1} . Hence, $S_j > j \times 2^e$ implies $S_j \geq (j \times 2^e)^+$, so that the exact value of $S_j + (2^e)^+$ exceeds $(j+1) \times 2^e$ by more than half the spacing of $\mathbb{F}$ at $(j+1) \times 2^e$, and therefore $S_{j+1} = \lceil S_j + (2^e)^+ \rceil > (j+1) \times 2^e$. In particular, $S_{2^p} > 2^{e+p}$. Furthermore, for every $z \in [2^{e+p}, 2^{e+p+2})_{\mathbb{F}}$, we have $z \oplus (2^e)^+ \geq z^+$, since the exact value of $z + (2^e)^+$ is larger than z^+ . As $[2^{e+p}, 2^{e+p+2})_{\mathbb{F}}$ consists of 2^{p+1} elements and $S_{2^p} > 2^{e+p}$, at most $2^{p+1} - 1$ further additions are required, i.e., $S_{3 \times 2^p - 1} \geq 2^{e+p+2}$ since $2^p + 2^{p+1} - 1 = 3 \times 2^p - 1$. Consequently,

$$z \oplus \bigoplus_{i=1}^{3 \times 2^p - 1} x \geq \bigoplus_{i=1}^{3 \times 2^p - 1} x \geq \bigoplus_{i=1}^{3 \times 2^p - 1} (2^e)^+ \geq 2^{e+p+2}. \quad (12)$$

From Eqs. (10) and (12), the result follows. This completes the proof. $\square$

Lemma 19. For any $x \in (1, 2]_{\mathbb{F}}$, there exist $y \in (2^{-1}, 1]_{\mathbb{F}}$ and $z \in (2^{-1}, 1^+)_{\mathbb{F}}$ such that $x \otimes y = 1^+$ and $x \otimes z = 1^{++}$.

Proof. We first show the existence of y . We assume $x \in (1^+, 2)_{\mathbb{F}}$ since we can choose $y = 1$ for $x = 1^+$ and $y = (2^{-1})^+$ for $x = 2$. For $x \in (1^+, 2)_{\mathbb{F}}$, let α_x be the smallest number in $[2^{-1}, 1]_{\mathbb{F}}$ such that $x \times \alpha_x > 1$ (the exact multiplication here). Then, $\alpha_x \in (2^{-1}, 1)_{\mathbb{F}}$ for all $x \in (1^+, 2)_{\mathbb{F}}$ since $2^- \times 2^{-1} < 1$ and $1^- \times 1^{++} > 1$. Furthermore, since $x \in (1^+, 2)_{\mathbb{F}}$, $\alpha_x - \alpha_x^- = 2^{-p-1}$, and $x \times \alpha_x^- \leq 1$ (note that $\alpha_x > 2^{-1}$ but α_x is chosen from $[2^{-1}, 1]_{\mathbb{F}}$), it holds that

$$x \times \alpha_x = x \times \alpha_x^- + x \times (\alpha_x - \alpha_x^-) = x \times \alpha_x^- + x \times 2^{-p-1} \in [1, 1^+],$$

i.e., $x \otimes \alpha_x \in \{1, 1^+\}$.

If $x \otimes \alpha_x = 1^+$, then choose $y = \alpha_x$. Hence, we now assume $x \otimes \alpha_x = 1$, i.e., $x \times \alpha_x \in (1, 1 + 2^{-p-1}]$. In this case, since $\alpha_x < 1$ and $x \in (1^+, 2)_{\mathbb{F}}$, we have

$$x \times \alpha_x^+ = x \times \alpha_x + x \times (\alpha_x^+ - \alpha_x) = x \times \alpha_x + x \times 2^{-p-1} \in (1 + 2^{-p-1}, 1 + 3 \times 2^{-p-1}).$$

This implies $x \otimes \alpha_x^+ = 1^+$, and choosing $y = \alpha_x^+$ shows the existence of the desired y for all $x \in (1, 2]_{\mathbb{F}}$.

We now show the existence of z . Again, without loss of generality, we assume $x \in (1^{++}, 2]_{\mathbb{F}}$; we can choose $z = 1^+$ for $x = 1^+$ and $z = 1$ for $x = 1^{++}$. For each $x \in (1^{++}, 2]_{\mathbb{F}}$, choose $y_x \in (2^{-1}, 1]_{\mathbb{F}}$ such that $x \otimes y_x = 1^+$. Here, the existence of such y_x is given by the first part of the proof. Then, one can observe that $y_x \in (2^{-1}, 1^-)_{\mathbb{F}}$ since $1^{+++} \otimes 1^- = 1^{++}$. Furthermore, since $x \otimes y_x = 1^+$, it holds that $x \times y_x \in (1 + 2^{-p-1}, 1 + 3 \times 2^{-p-1})$. Since $x \in (1^{++}, 2]_{\mathbb{F}}$ and $y_x^+ - y_x = y_x^{++} - y_x^+ = 2^{-p-1}$ by $y_x \in (2^{-1}, 1^-)_{\mathbb{F}}$, one can observe that

$$x \times y_x^+ \in [1 + 3 \times 2^{-p-1}, 1 + 5 \times 2^{-p-1}] \text{ or } x \times y_x^{++} \in [1 + 3 \times 2^{-p-1}, 1 + 5 \times 2^{-p-1}].$$

This implies that $x \otimes y_x^+ = 1^{++}$ or $x \otimes y_x^{++} = 1^{++}$. Since $y_x^+, y_x^{++} \in (2^{-1}, 1]_{\mathbb{F}}$, choosing z corresponding to y_x^+ or y_x^{++} completes the proof. $\square$

Lemma 20. Consider the left-to-right summation orders, $p = p'$, and $q = q'$, i.e., no high-precision accumulation. For any $z \in \mathbb{F}$ and $s \in \{-1, 1\}$, there exist $w, b \in \mathbb{F}$ with $w \in \{0, \pm 2^{-1}, \pm 1\}$ such that for all $x \in \mathbb{F}$,

$$\rho(w \otimes x \oplus b) = 0 \text{ if } s \times x \geq s \times z, \quad \rho(w \otimes x \oplus b) \in [\omega, \Omega]_{\mathbb{F}} \text{ otherwise.}$$

Proof. It suffices to treat $s = 1$; for $s = -1$ one replaces (w, b) by $(-w, -b)$. If $s \times z = -\Omega$, then $s \times x \geq s \times z$ for all $x \in \mathbb{F}$, so the second case is vacuous and $w = b = 0$ works. Otherwise, we choose

$$(w, b) = (-1, z) \text{ if } |z| < 1, \quad (w, b) = (-2^{-1}, z \otimes 2^{-1}) \text{ if } |z| \geq 1.$$

If $|z| \geq 1$, the exponent of z is at least 0, so $z/2 \in \mathbb{F}$ and $b = z/2$ exactly; in both cases $\lceil w \times z \rceil = -b$.

If $x \geq z$, then $w \times x \leq w \times z$ as $w < 0$, so $\lceil w \times x \rceil \leq \lceil w \times z \rceil = -b$ by the monotonicity of $\lceil \cdot \rceil$, which implies $w \otimes x \oplus b \leq 0$ and $\rho(w \otimes x \oplus b) = 0$. If $x < z$, then $x \leq z^-$ and, again by monotonicity, $w \otimes x \oplus b \geq \lceil \lceil w \times z^- \rceil + b \rceil$. For $|z| < 1$ this lower bound is $z - z^- \geq \omega$. For $|z| \geq 1$ the exponent of z^- is at least -1 , so $z^-/2 \in \mathbb{F}$ and the bound is $(z - z^-)/2 \geq 2^{-p-2} \geq \omega$, where the last inequality uses $\epsilon_{\min} \leq -2$. Finally, no overflow occurs: for $|z| < 1$ we have $|-x + b| < \Omega + 1 \leq \Omega + 2^{\epsilon_{\max}-p-1}$ since $p \leq \epsilon_{\max} - 1$ by Condition 1, and for $|z| \geq 1$ both $\lceil -x/2 \rceil$ and $|b|$ are at most $\Omega/2$. $\square$

Lemma 21. Consider the left-to-right summation orders, $p = p'$, and $q = q'$, i.e., no high-precision accumulation. Let $N = 2(p - \epsilon_{\min})$ and, for each $k \in [\epsilon_{\min} - p, -1]_{\mathbb{Z}}$ and $l = k - \epsilon_{\min} + p$, let

$$\alpha_{2l+1} = (2^{k+p+1})^+, \quad \alpha_{2l+2} = -(2^{k+p+1})^+.$$

Then, $\alpha_1, \dots, \alpha_N \in \mathbb{F}$, and for every $v \in [-2^{\epsilon_{\min}+1}, \omega]_{\mathbb{F}}$, all the values $v \oplus \bigoplus_{i=1}^j \alpha_i$, $j \in [N]$, are finite and

$$v \oplus \bigoplus_{i=1}^N \alpha_i = 1 \text{ if } v = \omega, \quad v \oplus \bigoplus_{i=1}^N \alpha_i \leq 0 \text{ if } v \leq 0.$$

Proof. Fix k and write $\alpha = (2^{k+p+1})^+ = 2^{k+p+1} + 2^{k+1}$, which lies in $\mathbb{F}$ as $k + p + 1 \leq p \leq \epsilon_{\max}$ by Condition 1. For $u = 2^k$, the exact value $u + \alpha = 2^{k+p+1}(1 + 2^{-p} + 2^{-p-1})$ lies exactly halfway between the consecutive floats $(2^{k+p+1})^+$ and $(2^{k+p+1})^{++} = 2^{k+p+1} + 2^{k+2}$, whose last mantissa bits are 1 and 0. Hence, $u \oplus \alpha = (2^{k+p+1})^{++}$ by the ties-to-even rule, and $(u \oplus \alpha) \oplus (-\alpha) = 2^{k+1}$ exactly. Applying this successively for $k = \epsilon_{\min} - p, \dots, -1$ carries $\omega = 2^{\epsilon_{\min}-p}$ to $2^0 = 1$, which proves the first claim.

For the second claim, let $v \leq 0$. By the monotonicity of $\oplus$, $v \oplus \alpha \leq 0 \oplus \alpha = \alpha$ and therefore $(v \oplus \alpha) \oplus (-\alpha) \leq \alpha \oplus (-\alpha) = 0$; thus each consecutive pair $(\alpha, -\alpha)$ maps a nonpositive value to a nonpositive value, and the claim follows by induction. For finiteness, monotonicity also gives $v \leq v \oplus \alpha \leq \alpha$ at every stage, and since $|v| \leq 2^{\epsilon_{\min}+1}$ and $\alpha \leq 2^{p+1}$, every intermediate value is bounded in absolute value by $2^{p+2} \leq 2^{\epsilon_{\max}} \leq \Omega$, again by Condition 1. $\square$

Lemma 22. Consider the left-to-right summation orders, $p = p'$, and $q = q'$, i.e., no high-precision accumulation. Let $d_0 \in \mathbb{N}$ and $\mathbf{z} \in \mathbb{F}^{d_0}$. Then, there exist $d_1, d_2 \in \mathbb{N}$ and floating-point affine transformations $\phi_1 : \mathbb{F}^{d_0} \rightarrow \mathbb{F}^{d_1}$, $\phi_2 : \mathbb{F}^{d_1} \rightarrow \mathbb{F}^{d_2}$, and $\phi_3 : \mathbb{F}^{d_2} \rightarrow \mathbb{F}$ such that for any $\mathbf{x} \in \mathbb{F}^{d_0}$,

$$\rho \circ \phi_3 \circ \rho \circ \phi_2 \circ \rho \circ \phi_1(\mathbf{x}) = \mathbb{1}_{\mathbf{x}=\mathbf{z}}.$$

Furthermore, all values that appear during the intermediate computation are finite, d_1, d_2 depend only on d_0 , and ϕ_1, ϕ_2, ϕ_3 can be chosen so that in their evaluation, every product $w \otimes v$ satisfies $w \in \{0, \pm 2^{-1}, \pm 1\}$.

Proof. For each $j \in [d_0]$ and $s \in \{-1, 1\}$, let $w_{j,s}, b_{j,s} \in \mathbb{F}$ be given by Lemma 20 applied to z_j and s , and let $\alpha_1, \dots, \alpha_N$ be as in Lemma 21. By Definition 3, a row of a floating-point affine transformation evaluates $\bigoplus_k w_k \otimes v_k$ over all input coordinates k in increasing order, followed by the addition of the bias. Since $0 \otimes v = 0$ for every finite v and $u \oplus 0 = u$, the coordinates carrying weight 0 contribute exact zeros and leave the running value unchanged. Hence such a row evaluates to the left-to-right sum of its nonzero-weight terms, taken in increasing coordinate order. We therefore lay out the coordinates below so that this order is the one displayed.

Set $d_1 = 2d_0 + 1$ and let ϕ_1 be the affine transformation with

$$(\phi_1(\mathbf{x}))_1 = \omega, \quad (\phi_1(\mathbf{x}))_{2j} = w_{j,1} \otimes x_j \oplus b_{j,1}, \quad (\phi_1(\mathbf{x}))_{2j+1} = w_{j,-1} \otimes x_j \oplus b_{j,-1}$$

for $j \in [d_0]$, i.e., the first row has all weights 0 and bias ω , and each of the remaining $2d_0$ rows has a single nonzero weight. Writing $\mathbf{a} = \rho(\phi_1(\mathbf{x}))$, Lemma 20 gives $a_1 = \omega$ and, for each $j \in [d_0]$, $a_{2j} = a_{2j+1} = 0$ if $x_j = z_j$, while otherwise exactly one of a_{2j}, a_{2j+1} is 0 and the other lies in $[\omega, \Omega]_{\mathbb{F}}$.

Set $d_2 = 2d_0 + N + 1$ and let ϕ_2 be given, for $j \in [d_0]$, by

$$(\phi_2(\mathbf{a}))_{2j} = a_1 \ominus a_{2j} \ominus a_{2j+1},$$

i.e., the row of index $2j$ has weight 1 at coordinate 1 and weight -1 at coordinates $2j$ and $2j + 1$; every other row of ϕ_2 has all weights 0, with bias ω for the rows of index $1, 3, \dots, 2d_0 + 1$ and bias $|\alpha_i|$ for the row of index $2d_0 + 1 + i$, $i \in [N]$. By the convention above, the reduction of the row $2j$ is evaluated as displayed, and since at most one of a_{2j}, a_{2j+1} is nonzero, each of its intermediate values lies in $[-\Omega, \omega]_{\mathbb{F}}$ and is finite. Hence $\mathbf{y} = \rho(\phi_2(\mathbf{a}))$ satisfies $y_{2j} = \omega$ if $x_j = z_j$ and $y_{2j} = 0$ otherwise, while $y_l = \omega$ for all $l \in \{1, 3, \dots, 2d_0 + 1\}$ and $y_{2d_0+1+i} = |\alpha_i|$ for all $i \in [N]$.

Finally, let ϕ_3 be the single row with weight 1 at coordinate 1, weight 1 at each coordinate $2j$, weight -1 at each coordinate $2j + 1$ for $j \in [d_0]$, weight 1 at coordinate $2d_0 + 1 + i$ for odd $i \in [N]$ and weight -1 for even $i \in [N]$, and bias 0, so that

$$\phi_3(\mathbf{y}) = y_1 \oplus y_2 \ominus y_3 \oplus \dots \oplus y_{2d_0} \ominus y_{2d_0+1} \oplus \bigoplus_{i=1}^N (\pm 1) \otimes y_{2d_0+1+i}.$$

Here, the i -th of the last N summands equals α_i , since $\alpha_i = |\alpha_i|$ for odd i and $\alpha_i = -|\alpha_i|$ for even i by Lemma 21, and multiplication by ± 1 is exact. Each pair $y_{2j} \ominus y_{2j+1}$ adds 0 if $x_j = z_j$ and $-\omega$ otherwise, so the value after the first $2d_0 + 1$ summands equals ω if $\mathbf{x} = \mathbf{z}$ and is nonpositive otherwise. That value is also at least $-2^{\epsilon_{\min}+1}$: adding $-\omega$ repeatedly to ω produces $\omega, 0, -\omega, \dots$ and saturates at $-2^{\epsilon_{\min}+1}$, because $-2^{\epsilon_{\min}+1} \oplus (-\omega) = -2^{\epsilon_{\min}+1}$ by the ties-to-even rule. Lemma 21 therefore applies and yields $\phi_3(\mathbf{y}) = 1$ if $\mathbf{x} = \mathbf{z}$ and $\phi_3(\mathbf{y}) \leq 0$ otherwise, so that $\rho(\phi_3(\mathbf{y})) = \mathbb{1}_{\mathbf{x}=\mathbf{z}}$, with all intermediate values finite.

The dimensions $d_1 = 2d_0 + 1$ and $d_2 = 2d_0 + N + 1$ depend only on d_0 , as N depends only on p, q . For the weight bound, the weights of ϕ_1 lie in $\{0, \pm 2^{-1}, \pm 1\}$ by Lemma 20, and all weights of ϕ_2 and ϕ_3 lie in $\{0, \pm 1\}$, since every quantity other than a weight is carried by a bias. This completes the proof. $\square$

Lemma 23. *Consider the left-to-right summation orders, $p = p'$, and $q = q'$, i.e., no high-precision accumulation. Let $d_0, d_4 \in \mathbb{N}$ and $f^* : \mathbb{F}^{d_0} \rightarrow \mathbb{F}^{d_4}$. Then, there exist $d_1, d_2, d_3 \in \mathbb{N}$ and floating-point affine transformations $\phi_i : \mathbb{F}^{d_{i-1}} \rightarrow \mathbb{F}^{d_i}$ for all $i \in [4]$ such that*

$$f(\mathbf{x}) = \phi_4 \circ \rho \circ \phi_3 \circ \rho \circ \phi_2 \circ \rho \circ \phi_1(\mathbf{x}) = f^*(\mathbf{x})$$

for all $\mathbf{x} \in \mathbb{F}^{d_0}$. Furthermore, all values that appear during the intermediate computation are finite, d_1, d_2, d_3 depend only on d_0 , and $\phi_1, \dots, \phi_4$ can be chosen so that every product $w \otimes z$ arising in the evaluation of f satisfies $w \in \{0, \pm 2^{-1}, \pm 1\}$ or $z \in \{0, 1\}$.

Proof. Let $d_3 = |\mathbb{F}|^{d_0}$ and $\mathbf{z}_1, \dots, \mathbf{z}_{d_3}$ be all elements in $\mathbb{F}^{d_0}$. Then, by Lemma 22 applied to each $\mathbf{z}_i$ separately, there exist $d_1, d_2 \in \mathbb{N}$ and floating-point affine transformations $\phi_j : \mathbb{F}^{d_{j-1}} \rightarrow \mathbb{F}^{d_j}$ for all $j \in [3]$ such that

$$(\rho \circ \phi_3 \circ \rho \circ \phi_2 \circ \rho \circ \phi_1(\mathbf{x}))_i = \mathbb{1}_{\mathbf{x}=\mathbf{z}_i},$$

for all $i \in [d_3]$. Here, ϕ_1, ϕ_2, ϕ_3 are obtained by stacking the d_3 three-layer networks of Lemma 22 on the common input $\mathbf{x}$, letting them share a single row of constant value 1 in ϕ_1 . Since stacking only copies rows of the individual weight matrices, padded with zeros outside the corresponding block, it preserves the weight bound of Lemma 22, and $d_1 = 2d_0d_3 + 1$ and $d_2 = (2d_0 + N + 1)d_3$ depend only on d_0 , where N is as in Lemma 21. Choose $\phi_4(\mathbf{y}) = W \otimes \mathbf{y}$ where $W = [\mathbf{w}_1, \dots, \mathbf{w}_{d_3}] \in \mathbb{F}^{d_4 \times d_3}$ satisfies $\mathbf{w}_i = f^*(\mathbf{z}_i)$ for all $i \in [d_3]$. Then, one can observe that

$$\phi_4(\rho \circ \phi_3 \circ \rho \circ \phi_2 \circ \rho \circ \phi_1(\mathbf{x})) = f^*(\mathbf{x})$$

for all $\mathbf{x} \in \mathbb{F}^{d_0}$. The last claim of the lemma follows since the weights of Lemma 22 lie in $\{0, \pm 2^{-1}, \pm 1\}$, and $\phi_4(\mathbf{y}) = W \otimes \mathbf{y}$ multiplies only the indicator values in $\{0, 1\}$. Furthermore, the dimensions d_1, d_2, d_3 produced by Lemma 22 depend only on d_0 . This completes the proof. $\square$

Lemma 24. *Suppose $p \geq 3$. Then, it holds that*

$$\begin{aligned} (1^+ \oplus 1^{++}) \oplus 1^{++} &= (1^{++} \oplus 1^+) \oplus 1^{++} = 3^{+++}, \\ (1^{++} \oplus 1^{++}) \oplus 1^+ &= 3^{++}. \end{aligned}$$

Proof. We have

$$\begin{aligned} (1^+ \oplus 1^{++}) \oplus 1^{++} &= \lceil 2 + 3 \times 2^{-p} \rceil_{\mathbb{F}} \oplus (1 + 2^{1-p}) = (2 + 2^{2-p}) \oplus (1 + 2^{1-p}) = 3 + 3 \times 2^{1-p} = 3^{+++}. \\ (1^{++} \oplus 1^{++}) \oplus 1^+ &= (2 + 2^{2-p}) \oplus (1 + 2^{-p}) = \lceil 3 + 5 \times 2^{-p} \rceil_{\mathbb{F}} = 3 + 2^{2-p} = 3^{++}. \end{aligned}$$

□

Lemma 25. Suppose $p = 2$. Then, it holds that

$$\begin{aligned} (1^+ \oplus 1) \oplus 1^+ &= (1 \oplus 1^+) \oplus 1^+ = 3, \\ (1^+ \oplus 1^+) \oplus 1 &= 3^+. \end{aligned}$$

Proof. We have

$$\begin{aligned} (1^+ \oplus 1) \oplus 1^+ &= (1 \oplus 1^+) \oplus 1^+ = \lceil 2 + 2^{-2} \rceil \oplus (1 + 2^{-2}) = 2 \oplus (1 + 2^{-2}) = 3, \\ (1^+ \oplus 1^+) \oplus 1 &= \lceil 2 + 2^{-1} \rceil \oplus 1 = (2 + 2^{-1}) \oplus 1 = 3^+. \end{aligned}$$

This completes the proof. □

Lemma 26. For any $x \in (1, 2]_{\mathbb{F}}$, there exists $y \in (2^{-1}, 1]_{\mathbb{F}}$ such that $1 \leq \lceil x \times y \rceil_{\mathbb{F}'} \leq 1 + 2^{-p}$.

Proof. Let $1^+ = 1 + 2^{-p}$. We assume $x \in (1^+, 2)_{\mathbb{F}}$ since we can choose $y = 1$ for $x = 1^+$ and $y = (2^{-1})^+$ for $x = 2$. For $x \in (1^+, 2)_{\mathbb{F}}$, let y be the smallest number in $[2^{-1}, 1]_{\mathbb{F}}$ such that $x \times y > 1$ (the exact multiplication here). Then, $y \in (2^{-1}, 1)_{\mathbb{F}}$ for all $x \in (1^+, 2)_{\mathbb{F}}$ since $2^- \times 2^{-1} < 1$ and $1^- \times 1^{++} > 1$. Furthermore, since $x \in (1^+, 2)_{\mathbb{F}}$, $y - y^- = 2^{-p-1}$, and $x \times y^- \leq 1$ (note that $y > 2^{-1}$ but y is chosen from $[2^{-1}, 1]_{\mathbb{F}}$), it holds that

$$x \times y = x \times y^- + x \times (y - y^-) = x \times y^- + x \times 2^{-p-1} \in [1, 1^+],$$

i.e., $\lceil x \times y \rceil_{\mathbb{F}'} \in [1, 1^+]$. □

Lemma 27. Let $p + 1 = p'$. For $a = 1$, $b = 2^{-p-1}$, and $c = 2^{-p'-1}$, it holds that

$$\lceil (a \oplus_{\mathbb{F}'} b) \oplus_{\mathbb{F}'} c \rceil_{\mathbb{F}} = 1^+, \quad \lceil (a \oplus_{\mathbb{F}'} c) \oplus_{\mathbb{F}'} b \rceil_{\mathbb{F}} = 1, \quad \lceil (b \oplus_{\mathbb{F}'} c) \oplus_{\mathbb{F}'} a \rceil_{\mathbb{F}} = 1^+.$$

Proof. Since $p' = p + 1$, we have $c = 2^{-p-2}$. For the first identity, $a \oplus_{\mathbb{F}'} b = 1 + 2^{-p-1}$ exactly, and adding c gives $1 + 3 \times 2^{-p-2}$, the midpoint of the adjacent values $1 + 2^{-p-1}$ and $1 + 2^{-p}$ of $\mathbb{F}'$; the tie is resolved to the even significand, i.e., to $1 + 2^{-p} = 1^+$, which lies in $\mathbb{F}$ and hence is unchanged by $\lceil \cdot \rceil_{\mathbb{F}}$. For the second identity, $a + c = 1 + 2^{-p-2}$ is the midpoint of the adjacent values 1 and $1 + 2^{-p-1}$ of $\mathbb{F}'$, so $a \oplus_{\mathbb{F}'} c = 1$ by the ties-to-even rule. Adding b then gives $1 + 2^{-p-1}$ exactly, which is the midpoint of the adjacent values 1 and 1^+ of $\mathbb{F}$ and hence rounds to 1 . For the third identity, $b \oplus_{\mathbb{F}'} c = 3 \times 2^{-p-2}$ exactly, and adding a gives $1 + 3 \times 2^{-p-2}$, which rounds to 1^+ as in the first identity. □

Lemma 28. Let $p' \geq p + 2$. For $a = 1$, $b = 2^{-p-1}$, and $c = d = 2^{-p'-1}$, it holds that

$$\begin{aligned} \lceil (a \oplus_{\mathbb{F}'} b) \oplus_{\mathbb{F}'} (c \oplus_{\mathbb{F}'} d) \rceil_{\mathbb{F}} &= 1^+, \quad \lceil (a \oplus_{\mathbb{F}'} c) \oplus_{\mathbb{F}'} (b \oplus_{\mathbb{F}'} d) \rceil_{\mathbb{F}} = 1, \quad \lceil (a \oplus_{\mathbb{F}'} d) \oplus_{\mathbb{F}'} (b \oplus_{\mathbb{F}'} c) \rceil_{\mathbb{F}} = 1, \\ \lceil ((a \oplus_{\mathbb{F}'} b) \oplus_{\mathbb{F}'} c) \oplus_{\mathbb{F}'} d \rceil_{\mathbb{F}} &= 1, \quad \lceil ((a \oplus_{\mathbb{F}'} b) \oplus_{\mathbb{F}'} d) \oplus_{\mathbb{F}'} c \rceil_{\mathbb{F}} = 1, \quad \lceil ((a \oplus_{\mathbb{F}'} c) \oplus_{\mathbb{F}'} b) \oplus_{\mathbb{F}'} d \rceil_{\mathbb{F}} = 1, \\ \lceil ((a \oplus_{\mathbb{F}'} c) \oplus_{\mathbb{F}'} d) \oplus_{\mathbb{F}'} b \rceil_{\mathbb{F}} &= 1, \quad \lceil ((a \oplus_{\mathbb{F}'} d) \oplus_{\mathbb{F}'} b) \oplus_{\mathbb{F}'} c \rceil_{\mathbb{F}} = 1, \quad \lceil ((a \oplus_{\mathbb{F}'} d) \oplus_{\mathbb{F}'} c) \oplus_{\mathbb{F}'} b \rceil_{\mathbb{F}} = 1, \\ \lceil ((b \oplus_{\mathbb{F}'} c) \oplus_{\mathbb{F}'} a) \oplus_{\mathbb{F}'} d \rceil_{\mathbb{F}} &= 1, \quad \lceil ((b \oplus_{\mathbb{F}'} c) \oplus_{\mathbb{F}'} d) \oplus_{\mathbb{F}'} a \rceil_{\mathbb{F}} = 1^+, \quad \lceil ((b \oplus_{\mathbb{F}'} d) \oplus_{\mathbb{F}'} a) \oplus_{\mathbb{F}'} c \rceil_{\mathbb{F}} = 1, \\ \lceil ((b \oplus_{\mathbb{F}'} d) \oplus_{\mathbb{F}'} c) \oplus_{\mathbb{F}'} a \rceil_{\mathbb{F}} &= 1^+, \quad \lceil ((c \oplus_{\mathbb{F}'} d) \oplus_{\mathbb{F}'} a) \oplus_{\mathbb{F}'} b \rceil_{\mathbb{F}} = 1^+, \quad \lceil ((c \oplus_{\mathbb{F}'} d) \oplus_{\mathbb{F}'} b) \oplus_{\mathbb{F}'} a \rceil_{\mathbb{F}} = 1^+. \end{aligned}$$

Proof. The statement follows from the direct computation. □

C. Proofs

C.1. Proof of Lemma 1

It suffices to prove the claim for each generator of $\mathcal{G}(\mathcal{T}_n)$, i.e., an element of Π in the definition of $\mathcal{G}(\mathcal{T}_n)$ in Section 2.2. Let v be an internal node such that $\mathcal{T}_{v_L}$ and $\mathcal{T}_{v_R}$ are isomorphic, and let π_v be the corresponding permutation swapping the leaves of the two subtrees via the induced bijection. Let $y_i = x_{\pi_v(i)}$. By construction, the restriction of π_v maps $\mathcal{T}_{v_L}$ to $\mathcal{T}_{v_R}$ along the tree isomorphism. Hence the value computed at the root of $\mathcal{T}_{v_L}$ under input y equals the value computed at the root of $\mathcal{T}_{v_R}$ under input x , and vice versa. Therefore, if the two child values at v under input x are a and b , then under input y they are b and a .

Since floating-point addition is commutative,

$$a \oplus b = b \oplus a,$$

the value at node v is unchanged. All other nodes either lie outside the subtree of v or depend only on the value at v , so the value at the root is unchanged. Thus,

$$\bigoplus_{i \in [n]}^{\mathcal{T}_n} x_i = \bigoplus_{i \in [n]}^{\mathcal{T}_n} x_{\pi_v(i)}.$$

Since $\mathcal{G}(\mathcal{T}_n)$ is generated by such permutations, the claim follows for all $g \in \mathcal{G}(\mathcal{T}_n)$.

C.2. Proof of Theorem 5

The proof of Theorem 5 follows that of Theorem 4, using Lemma 30 in place of Lemma 14, where the hypothesis $n \leq 2^{p+1}$ of Lemma 30 is satisfied here since $n \leq 2^{p+1} - 2$ by the assumption of Theorem 5. Here, the count-recovery step differs, which is given below. In Lemma 30, $\mathbf{z}_1, \dots, \mathbf{z}_{|\mathbb{F}^{d_{\text{in}}}|}$ denote all vectors in $\mathbb{F}^{d_{\text{in}}}$.

As in the proof of Theorem 4, we recover the input distribution by reading, for each token $\mathbf{z}_j$ present in the input, its count k_j from the aggregated value $\left[\bigoplus_{i \in [n]; \mathbb{F}'}^{\mathcal{T}_n} w_i \right]_{\mathbb{F}}$ (which we call the encoding), where $w_i = x$ at the k_j positions holding $\mathbf{z}_j$ and $w_i = 0$ otherwise, and $x \in [1, 1 + 2^{-p}]$ is the value produced by Lemma 30. Since 0 is an additive identity, this equals $\left[\bigoplus_{i \in [k_j]; \mathbb{F}'}^{\mathcal{T}} x \right]_{\mathbb{F}}$, the sum of k_j copies of x over the sub-tree $\mathcal{T}$ of $\mathcal{T}_n$ induced by the positions of $\mathbf{z}_j$. In particular, this encoding depends on those positions, not only on k_j , unlike the target-precision sums $\bigoplus 1^+$ of Lemma 15. Recovery instead uses Lemma 29 below: over any induced trees, the encodings of counts $0, \dots, 2^p - 1$ are distinct, and any cross-tree collision $\left[\bigoplus_{i \in [a]; \mathbb{F}'}^{\mathcal{T}} x \right]_{\mathbb{F}} = \left[\bigoplus_{i \in [b]; \mathbb{F}'}^{\mathcal{T}'} x \right]_{\mathbb{F}}$ ($a < b$) has $a + b \geq 2^{p+1} - 1 > n$ (as $n \leq 2^{p+1} - 2$) for all full binary trees $\mathcal{T}$ and $\mathcal{T}'$ with a and b leaves, respectively. This gives injectivity of the count-to-encoding map: if two length- n inputs had the same encoding but distinct count vectors $\mathbf{c} \neq \mathbf{c}'$, whose entries are the numbers of occurrences of the tokens $\mathbf{z} \in \mathbb{F}^{d_{\text{in}}}$, then on the set D of tokens where they differ each (c_j, c'_j) is such a colliding pair, so $c_j + c'_j > n$, while $\sum_{j \in D} c_j = \sum_{j \in D} c'_j \leq n$; hence $2n \geq \sum_{j \in D} (c_j + c'_j) > |D|n$, forcing $|D| < 2$, a contradiction. Thus the input distribution is recovered, and the target output is generated as in the proof of Theorem 4 via Lemmas 12 and 13.

Lemma 29. *Suppose $p' \geq 2p$ and $x \in [1, 1 + 2^{-p}]_{\mathbb{F}'}$. For a full binary tree $\mathcal{T}$ on k leaves, write $z_k^{\mathcal{T}} = \left[\bigoplus_{i \in [k]; \mathbb{F}'}^{\mathcal{T}} x \right]_{\mathbb{F}}$ ($z_0^{\mathcal{T}} = 0$). Then, for any counts $a < b$ with $b \leq 2^{p+1} - 2$ and any full binary trees $\mathcal{T}, \mathcal{T}'$ (with a, b leaves, respectively), a collision $z_a^{\mathcal{T}} = z_b^{\mathcal{T}'}$ implies $a \geq 2^p - 1$ and $a + b \geq 2^{p+1} - 1$. Consequently $z_a^{\mathcal{T}} \neq z_b^{\mathcal{T}'}$ for all distinct $a, b \in \{0, 1, \dots, 2^p - 1\}$ and all full binary trees $\mathcal{T}, \mathcal{T}'$.*

Proof. Fix $x \in [1, 1 + 2^{-p}]_{\mathbb{F}'}$ and a full binary tree $\mathcal{T}$ on k leaves. For an internal node v , let m_v be the number of leaves below v and $\text{val}(v)$ the value it computes; the root computes the $\mathbb{F}'$ tree-sum $y_k^{\mathcal{T}} := \bigoplus_{i \in [k]; \mathbb{F}'}^{\mathcal{T}} x$, whose rounding is the encoding $z_k^{\mathcal{T}} = \lceil y_k^{\mathcal{T}} \rceil_{\mathbb{F}}$. We first bound the accumulated rounding error for $k \leq 2^p$ (Steps 1–3), then prove distinctness (Step 4).

Step 1 (error bound when $k \leq 2^p$). Since each internal node performs one rounded addition in $\mathbb{F}'$, we have $y_k^{\mathcal{T}} = kx + \sum_v \delta_v$, summed over the $k - 1$ internal nodes, where $|\delta_v| \leq \frac{1}{2} \text{ulp}_{\mathbb{F}'}(\text{val}(v))$. Here, for $u \geq 2^{\epsilon_{\min}}$, $\text{ulp}_{\mathbb{F}'}(u) := 2^{\lfloor \log_2 u \rfloor - p'}$ and $\text{ulp}_{\mathbb{F}}(u) := 2^{\lfloor \log_2 u \rfloor - p}$ denote the unit in the last place of u , i.e. the spacing of adjacent representable values around u in $\mathbb{F}'$ and $\mathbb{F}$ respectively. Note that rounding incurs error at most $\frac{1}{2} \text{ulp}$. By monotonicity of floating-point addition in the

summand, $\text{val}(v)$ lies in $[m_v, m_v(1 + 2^{-p})]$ since $\bigoplus_{\mathbb{F}'} m_v$ copies of 1 is m_v and of $1 + 2^{-p}$ is $m_v(1 + 2^{-p})$ (each partial sum has at most $2p$ significant bits and $p' \geq 2p$). As $m_v \leq 2^p$, this interval is contained in $[m_v, m_v + 1]$, so $\lfloor \log_2 \text{val}(v) \rfloor = \lfloor \log_2 m_v \rfloor$ and $\frac{1}{2} \text{ulp}_{\mathbb{F}'}(\text{val}(v)) = 2^{\lfloor \log_2 m_v \rfloor - p' - 1}$. Hence

$$|y_k^T - k \times x| \leq 2^{-p'-1} \Phi(\mathcal{T}), \quad \Phi(\mathcal{T}) := \sum_{v \text{ internal}} 2^{\lfloor \log_2 m_v \rfloor}.$$

Step 2 (Φ is maximized by the left-to-right tree). Writing $\phi(k) := \max_{\mathcal{T}} \Phi(\mathcal{T})$ over trees on k leaves, the recursion $\Phi(\mathcal{T}) = 2^{\lfloor \log_2 k \rfloor} + \Phi(\mathcal{T}_L) + \Phi(\mathcal{T}_R)$ (with $\mathcal{T}_L, \mathcal{T}_R$ the subtrees of sizes $i, k-i$) gives $\phi(k) = 2^{\lfloor \log_2 k \rfloor} + \max_{1 \leq i < k} [\phi(i) + \phi(k-i)]$. We claim $\phi(k) = g(k) := \sum_{m=2}^k 2^{\lfloor \log_2 m \rfloor}$, the value of the left-to-right (caterpillar) tree, and that the maximum is attained at $i = 1$. Indeed, assuming $\phi(j) = g(j)$ for $j < k$, and taking $1 \leq i \leq k-i$, the two sums $g(i) + g(k-i)$ and $g(k-1) = g(i) + \sum_{m=i+1}^{k-1} 2^{\lfloor \log_2 m \rfloor}$ have the same number of terms beyond $g(i)$, namely $g(k-i) = \sum_{m=2}^{k-i} 2^{\lfloor \log_2 m \rfloor}$ versus $\sum_{m=i+1}^{k-1} 2^{\lfloor \log_2 m \rfloor}$; matching the j -th terms, $2^{\lfloor \log_2(1+j) \rfloor} \leq 2^{\lfloor \log_2(i+j) \rfloor}$ because $i \geq 1$ and $m \mapsto 2^{\lfloor \log_2 m \rfloor}$ is nondecreasing. Thus $g(i) + g(k-i) \leq g(k-1)$ and $\phi(k) = 2^{\lfloor \log_2 k \rfloor} + g(k-1) = g(k)$. For $k \leq 2^p$, summing $2^{\lfloor \log_2 m \rfloor}$ over $m = 2, \dots, 2^p$ in dyadic blocks gives $\Phi(\mathcal{T}) \leq g(2^p) = \frac{1}{3}(4^p - 4) + 2^p$.

Step 3 ($E < \frac{1}{4}$). Let $E := \max_{\mathcal{T}} |y_k^T - k \times x|$ over trees with $k \leq 2^p$ leaves. By Steps 1–2 and $p' \geq 2p$,

$$E \leq 2^{-2p-1} \left(\frac{1}{3}(4^p - 4) + 2^p \right) < \frac{1}{4} \iff (2^p - 2)(2^p - 4) > 0,$$

which holds for all $p \geq 3$. For $p = 2$ the strict bound $E \leq 2^{-p'-1} \Phi \leq 2^{-p'} < \frac{1}{4}$ already covers $p' > 4$, and at the boundary $p' = 2p = 4$ one can directly check that $E \leq 2^{-4} < \frac{1}{4}$ (the $2^{p'-p} + 1 = 5$ admissible values of $x, k \leq 4$). Thus $E < \frac{1}{4}$ in all cases.

Step 4 (distinctness). We use two bounds on the $\mathbb{F}'$ tree-sums: by monotonicity, any tree-sum of $k \leq 2^{p+1}$ copies satisfies $y_k^T \geq k$ (the sum of k ones equals the integer k , exact in $\mathbb{F}'$), and by Steps 1–3, any tree-sum of $k \leq 2^p$ copies lies in $(kx - E, kx + E)$. We also use the following consequence of round-to-nearest: since $\lceil y \rceil_{\mathbb{F}}$ differs from y by at most $\frac{1}{2} \text{ulp}_{\mathbb{F}}(y)$, if $u < w$ both lie in an interval on which $\text{ulp}_{\mathbb{F}}(v) \leq \delta$ for every v and $w - u > \delta$, then $\lceil u \rceil_{\mathbb{F}} < \lceil w \rceil_{\mathbb{F}}$.

For contradiction, suppose that $z_a^T = z_b^{T'}$ for some $a < b$ with $a \leq 2^p - 2$ and $b \leq 2^{p+1} - 2$, i.e., $\lceil y_a^T \rceil_{\mathbb{F}} = \lceil y_b^{T'} \rceil_{\mathbb{F}}$. We contradict this by showing $\lceil y_a^T \rceil_{\mathbb{F}} < \lceil y_b^{T'} \rceil_{\mathbb{F}}$.

Case $b > 2^p$. Since $a \leq 2^p - 2$, we have $y_a^T < ax + E < 2^p$, i.e., $\lceil y_a^T \rceil_{\mathbb{F}} \leq 2^p$. Furthermore, since $y_b^{T'} \geq b \geq 2^p + 1$, we have $\lceil y_b^{T'} \rceil_{\mathbb{F}} \geq 2^p + 1$ (the integers 2^p and $2^p + 1$ are exactly representable in $\mathbb{F}$). Hence $\lceil y_a^T \rceil_{\mathbb{F}} < \lceil y_b^{T'} \rceil_{\mathbb{F}}$.

Case $b \leq 2^p$. Since $y_a^T \in (ax - E, ax + E)$ and $y_b^{T'} \in (bx - E, bx + E)$,

$$y_b^{T'} - y_a^T > (bx - E) - (ax + E) = (b - a)x - 2E \geq x - 2E > \frac{1}{2},$$

using $x \geq 1$ and $E < \frac{1}{4}$. We bound the magnitudes of $y_a^T, y_b^{T'}$ and apply the rounding fact above.

- If $b = a + 1$, then by monotonicity $y_a^T, y_b^{T'} \leq b(1 + 2^{-p}) \leq (2^p - 1)(1 + 2^{-p}) = 2^p - 2^{-p} < 2^p$, i.e., $y_a^T, y_b^{T'} \in [0, 2^p]_{\mathbb{F}'}$. Since $y_b^{T'} - y_a^T > \frac{1}{2}$, we obtain $\lceil y_a^T \rceil_{\mathbb{F}} < \lceil y_b^{T'} \rceil_{\mathbb{F}}$.
- If $b \geq a + 2$, then $y_b^{T'} - y_a^T \geq 2x - 2E > \frac{3}{2}$, and $y_a^T, y_b^{T'} \leq b(1 + 2^{-p}) \leq 2^p(1 + 2^{-p}) = 2^p + 1 < 2^{p+1}$, i.e., $y_a^T, y_b^{T'} \in [0, 2^{p+1}]_{\mathbb{F}'}$. Since $y_b^{T'} - y_a^T > \frac{3}{2} > 1$, we obtain $\lceil y_a^T \rceil_{\mathbb{F}} < \lceil y_b^{T'} \rceil_{\mathbb{F}}$.

In every case $\lceil y_a^T \rceil_{\mathbb{F}} < \lceil y_b^{T'} \rceil_{\mathbb{F}}$, contradicting $z_a^T = z_b^{T'}$. Hence, every collision $z_a^T = z_b^{T'}$ ($a < b$) has $a \geq 2^p - 1$ and hence, since $b > a, a + b \geq (2^p - 1) + 2^p = 2^{p+1} - 1$. In particular, $z_a^T \neq z_b^{T'}$ for all distinct $a, b \in \{0, \dots, 2^p - 1\}$ and all trees. $\square$

Lemma 30. Let $d_{\text{in}}, n \in \mathbb{N}$. Let $h = 1$ and $e = d_{\text{in}} + \lceil \mathbb{F} \rceil^{d_{\text{in}}}$, and suppose $n \leq 2^{p+1}$. Then, there exist $t_d, t_m, t_r \in \mathbb{N}$ and $x \in [1, 1 + 2^{-p}]_{\mathbb{F}'}$ such that for any $d \geq t_d, m \geq t_m$, and $r \geq t_r$, there exists $\phi \in \mathcal{B}^{h,m,r,d,n}$ satisfying the following properties. For each $Z \in \mathbb{F}^{d \times n}$ satisfying $Z_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{F}^{d_{\text{in}} \times n}$ and $Z_{d_{\text{in}}+1:} = \mathbf{0}_{(d-d_{\text{in}}) \times n}$,

- $\phi(Z)_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ and $\phi(Z)_{e+1:} = \mathbf{0}_{(d-e) \times n}$,
- for each $j \in [e] \setminus [d_{\text{in}}]$, $\phi(Z)_j = \left[\bigoplus_{i \in [n]; \mathbb{F}'}^{\mathcal{T}_n} w_{ji} \right]_{\mathbb{F}} \times \mathbf{1}_n^\top$ where $w_{ji} = x$ if $\mathbf{x}_i = \mathbf{z}_{j-d_{\text{in}}}$ and $w_{ji} = 0$ otherwise.

Proof of Lemma 30. Let $\gamma = \lceil \mathbb{F} \rceil^{d_{\text{in}}}$ and $\alpha := \left[1 \odot_{\mathbb{F}'} \left(\bigoplus_{i \in [n]; \mathbb{F}'}^{\mathcal{T}_n} 1 \right) \right]_{\mathbb{F}}$, i.e., the common value of the entries of σ applied to an all-zero column (see Eq. (4)). Since $n \leq 2^{p+1} \leq 2^{p'+1}$, every partial sum of the denominator is an integer at most n and is therefore exact in $\mathbb{F}'$, so the denominator equals n under any reduction order and $\alpha = \lceil [1/n]_{\mathbb{F}'} \rceil_{\mathbb{F}} \geq 2^{-p-1}$ by the monotonicity of rounding. Here, Condition 1 supplies the two facts used: all integers up to $2^{p'+1}$ are exactly representable in $\mathbb{F}'$ (as $p' + 1 \leq 2^{q'-1} - 2$), and 2^{-p-1} is a normal float of $\mathbb{F}$ (as $-p - 1 \geq \epsilon_{\min}$), so that it is a valid lower bound after rounding and the scaling by powers of two used below is exact. Let $\beta \in \mathbb{F}$ be such that $\lceil \beta \times \alpha \rceil_{\mathbb{F}'} \in [1, 1 + 2^{-p}]$. Such β exists by Lemma 26: write $\alpha = a \times 2^{-s}$ with $a \in (1, 2]_{\mathbb{F}}$ and $s \in \{1, \dots, p+2\}$ (this scaling is exact since it is by a power of two, and $s \leq p+2$ since $\alpha \geq 2^{-p-1}$ and $a \leq 2$), take $y \in (2^{-1}, 1]_{\mathbb{F}}$ from Lemma 26 with $\lceil a \times y \rceil_{\mathbb{F}'} \in [1, 1 + 2^{-p}]$, and set $\beta := y \times 2^s$, which lies in $\mathbb{F}$ since $\beta \leq 2^{p+2} \leq \Omega$ by Condition 1. Then $\beta \times \alpha = a \times y$, and hence, $\lceil \beta \times \alpha \rceil_{\mathbb{F}'} = \lceil a \times y \rceil_{\mathbb{F}'}$. (The resulting value $x = \lceil \beta \times \alpha \rceil_{\mathbb{F}'} \in [1, 1 + 2^{-p}]$ is exactly the value to which Lemma 29 is applied in Appendix C.2.) Without loss of generality, we assume that $d = t_d, m = t_m$, and $r = t_r$; otherwise, we can add unused dimensions. We set $t_m = \gamma$, and we will choose the explicit values of $t_d \geq d_{\text{in}} + 2\gamma, t_r$ later. We construct ϕ as a composition of two transformer blocks $g_1, g_2 \in \mathcal{B}^{1,t_m,t_r,t_d,n}$. Here, g_1 maps $Z \in \mathbb{F}^{t_d \times n}$ with $Z_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{F}^{d_{\text{in}} \times n}$ and $Z_{d_{\text{in}}+1:} = \mathbf{0}_{(t_d-d_{\text{in}}) \times n}$ to $Y = [\mathbf{y}_1, \dots, \mathbf{y}_n] \in \mathbb{F}^{t_d \times n}$ such that

$$(\mathbf{y}_i)_l = \begin{cases} (\mathbf{x}_i)_l & \text{if } l \leq d_{\text{in}}, \\ \beta & \text{if } l = d_{\text{in}} + \gamma + j \text{ and } \mathbf{x}_i = \mathbf{z}_j \text{ for some } j \in [\gamma], \\ 0 & \text{otherwise,} \end{cases}$$

for all $i \in [n]$ and $l \in [t_d]$. Note that these row positions do not depend on t_d . By Lemma 12 applied with $d_1 = d_{\text{in}}, d_2 = d_{\text{in}} + 2\gamma$, and the present $m = t_m, h = 1$, and n , there exist $t'_d, t'_r \in \mathbb{N}$ (which do not depend on t_d or t_r) such that the desired $g_1 \in \mathcal{B}^{1,t_m,t_r,t_d,n}$ exists for all $d' \geq t'_d$ and $r' \geq t'_r$. Choose $t_d = \max\{t'_d, d_{\text{in}} + 2\gamma\}$, $t_r = \max\{t'_r, \gamma\}$, and $g_1 \in \mathcal{B}^{1,t_m,t_r,t_d,n}$.

We now construct g_2 . Define an attention layer $f_1 : \mathbb{F}^{t_d \times n} \rightarrow \overline{\mathbb{F}}^{t_d \times n}$ as

$$\begin{aligned} f_1(Z) &= Z \oplus (W^O \hat{\otimes} ((W^V \hat{\otimes} Z) \hat{\otimes} \sigma((\mathbf{0}_{t_m \times t_d} \hat{\otimes} Z)^\top \hat{\otimes} (\mathbf{0}_{t_m \times t_d} \hat{\otimes} Z)))) \\ &= Z \oplus (W^O \hat{\otimes} ((W^V \hat{\otimes} Z) \hat{\otimes} (\alpha \mathbf{1}_{n \times n}))) \end{aligned}$$

for some $W^V \in \mathbb{F}^{t_m \times t_d}$ and $W^O \in \mathbb{F}^{t_d \times t_m}$. Specifically, we choose W^V, W^O so that $W^V \hat{\otimes} Z = Z_{d_{\text{in}}+\gamma+1:d_{\text{in}}+2\gamma}$, $(W^O \hat{\otimes} Y)_{d_{\text{in}}+1:d_{\text{in}}+\gamma} = Y$, and all other rows of $W^O \hat{\otimes} Y$ are zero for all $Y \in \mathbb{F}^{t_m \times n}$. Then, by the definitions of g_1, f_1 , it holds that for each $Z \in \mathbb{F}^{t_d \times n}$ with $Z_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{F}^{d_{\text{in}} \times n}$ and $Z_{d_{\text{in}}+1:} = \mathbf{0}_{(t_d-d_{\text{in}}) \times n}$, for each $j \in [\gamma]$,

- $(f_1 \circ g_1(Z))_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$,
- $(f_1 \circ g_1(Z))_{d_{\text{in}}+j} = \left[\bigoplus_{i \in [n]; \mathbb{F}'}^{\mathcal{T}_n} w_{d_{\text{in}}+j,i} \right]_{\mathbb{F}} \times \mathbf{1}_n^\top$,
- $(f_1 \circ g_1(Z))_{d_{\text{in}}+2\gamma+1:} = \mathbf{0}_{(t_d-2\gamma-d_{\text{in}}) \times n}$, and the rows $d_{\text{in}} + \gamma + 1, \dots, d_{\text{in}} + 2\gamma$ of $f_1 \circ g_1(Z)$ coincide with the corresponding rows of $g_1(Z)$. In particular, they are non-negative.

Choose a feed-forward network $f_2 : \mathbb{F}^{t_d \times n} \rightarrow \mathbb{F}^{t_d \times n}$ as $f_2(Z) = Z \oplus (W_2 \hat{\otimes} \rho(W_1 \hat{\otimes} Z))$ where $W_1 \in \mathbb{F}^{t_r \times t_d}$ and $W_2 \in \mathbb{F}^{t_d \times t_r}$ are matrices satisfying that for $\mathbf{z} = (z_1, \dots, z_{t_d}) \in \mathbb{F}^{t_d}$ and $\mathbf{y} = (y_1, \dots, y_{t_r}) \in \mathbb{F}^{t_r}$,

$$W_1 \otimes \mathbf{z} = (z_{d_{\text{in}}+\gamma+1}, \dots, z_{d_{\text{in}}+2\gamma}, 0, \dots, 0), \quad W_2 \otimes \mathbf{y} = (0, \dots, 0, -y_1, \dots, -y_{t_r}, 0, \dots, 0),$$

where the nonzero block of $W_2 \otimes \mathbf{y}$ occupies the coordinates $d_{\text{in}} + \gamma + 1, \dots, d_{\text{in}} + 2\gamma$. Namely, f_2 erases the coordinates $d_{\text{in}} + \gamma + 1, \dots, d_{\text{in}} + 2\gamma$ of each column, which are finite and non-negative by the third property above. Let $g_2 = f_2 \circ f_1$ and $\phi = g_2 \circ g_1 \in \mathcal{B}^{1,t_m,t_r,t_d,n}$. Then, by the definitions of g_1, g_2, ϕ satisfies the desired properties. This completes the proof. $\square$

C.3. Proof of Theorem 6

Since the feed-forward networks are column-wise applications, they are permutation-equivariant, i.e., $\mathcal{G}(\mathcal{T}_n)$ -equivariant. Furthermore, one can observe that for $d', m' \in \mathbb{N}$, $W \in \mathbb{F}^{d' \times m'}$, and $\mathcal{G}(\mathcal{T}_n)$ -equivariant $\psi'_1, \dots, \psi'_h : \mathbb{F}^{d \times n} \rightarrow \mathbb{F}^{m' \times n}$, the maps $Z \mapsto W \hat{\otimes} \psi'_1(Z)$ and $Z \mapsto \bigoplus_{i=1}^h \psi'_i(Z)$ are $\mathcal{G}(\mathcal{T}_n)$ -equivariant, and so is $Z \mapsto Z \oplus \psi'_1(Z)$ when $m' = d$. Hence, it suffices to show that $g(Z) = (W^V \hat{\otimes} Z) \hat{\otimes} \sigma((W^K \hat{\otimes} Z)^\top \hat{\otimes} (W^Q \hat{\otimes} Z))$ is $\mathcal{G}(\mathcal{T}_n)$ -equivariant for all $W^V, W^K, W^Q \in \mathbb{F}^{m \times d}$.

Let $Z \in \mathbb{F}^{d \times n}$ and $Z' = \pi Z$ for some $\pi \in \mathcal{G}(\mathcal{T}_n)$. Let $V = [\mathbf{v}_1, \dots, \mathbf{v}_n] = W^V \hat{\otimes} Z$, $V' = W^V \hat{\otimes} Z'$, $S = \sigma((W^K \hat{\otimes} Z)^\top \hat{\otimes} (W^Q \hat{\otimes} Z))$, and $S' = \sigma((W^K \hat{\otimes} Z')^\top \hat{\otimes} (W^Q \hat{\otimes} Z'))$. Then, $V' = \pi V$. Furthermore, by the definition of the floating-point softmax σ , we have

$$S' = \begin{bmatrix} S_{\pi(1)\pi(1)} & S_{\pi(1)\pi(2)} & S_{\pi(1)\pi(3)} & \cdots & S_{\pi(1)\pi(n)} \\ S_{\pi(2)\pi(1)} & S_{\pi(2)\pi(2)} & S_{\pi(2)\pi(3)} & \cdots & S_{\pi(2)\pi(n)} \\ S_{\pi(3)\pi(1)} & S_{\pi(3)\pi(2)} & S_{\pi(3)\pi(3)} & \cdots & S_{\pi(3)\pi(n)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ S_{\pi(n)\pi(1)} & S_{\pi(n)\pi(2)} & S_{\pi(n)\pi(3)} & \cdots & S_{\pi(n)\pi(n)} \end{bmatrix} \quad \text{where} \quad S = \begin{bmatrix} S_{11} & S_{12} & S_{13} & \cdots & S_{1n} \\ S_{21} & S_{22} & S_{23} & \cdots & S_{2n} \\ S_{31} & S_{32} & S_{33} & \cdots & S_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ S_{n1} & S_{n2} & S_{n3} & \cdots & S_{nn} \end{bmatrix}.$$

Here, the identity $S'_{ij} = S_{\pi(i)\pi(j)}$ holds because the softmax denominator is summed in the $\mathcal{T}_n$ order (Eq. (4)). Let M be the score matrix $(W^K \hat{\otimes} Z)^\top \hat{\otimes} (W^Q \hat{\otimes} Z)$. Then, we have $M' = (W^K \hat{\otimes} Z')^\top \hat{\otimes} (W^Q \hat{\otimes} Z')$ with $M'_{ij} = M_{\pi(i)\pi(j)}$, the shift $x_* = \max_k M_{k\pi(j)}$ is permutation-invariant, and for $\pi \in \mathcal{G}(\mathcal{T}_n)$,

$$\bigoplus_{k \in [n]; \mathbb{F}'}^{\mathcal{T}_n} [\exp]_{\mathbb{F}'} (M_{\pi(k)\pi(j)} \ominus_{\mathbb{F}'} x_*) = \bigoplus_{k \in [n]; \mathbb{F}'}^{\mathcal{T}_n} [\exp]_{\mathbb{F}'} (M_{k\pi(j)} \ominus_{\mathbb{F}'} x_*)$$

by Lemma 1. Hence, the j -th softmax column of M' is equal to the $\pi(j)$ -th softmax column of M up to the row permutation π .

By the definition of g , it holds that $g(Z)_{ij} = \left[\bigoplus_{k \in [n]; \mathbb{F}'}^{\mathcal{T}_n} (V_{ik} \otimes_{\mathbb{F}'} S_{kj}) \right]_{\mathbb{F}}$, and $g(Z')_{ij} = \left[\bigoplus_{k \in [n]; \mathbb{F}'}^{\mathcal{T}_n} (V'_{ik} \otimes_{\mathbb{F}'} S'_{kj}) \right]_{\mathbb{F}}$. Furthermore, by Lemma 1, we have

$$g(Z')_{ij} = \left[\bigoplus_{k \in [n]; \mathbb{F}'}^{\mathcal{T}_n} (V_{i\pi(k)} \otimes_{\mathbb{F}'} S_{\pi(k)\pi(j)}) \right]_{\mathbb{F}} = g(Z)_{i\pi(j)}$$

for all $i \in [m]$ and $j \in [n]$. Hence, $g(Z') = \pi g(Z)$. This completes the proof.

C.4. Proof of Theorem 7

Since feed-forward networks are token-wise applications in floating-point transformers, they preserve equality. Furthermore, for $W \in \mathbb{F}^{m \times d}$, $Z \in \mathbb{F}^{d \times n}$, and equality preserving $\psi' : \mathbb{F}^{d \times n} \rightarrow \mathbb{F}^{d \times n}$, $\phi(Z) = W \hat{\otimes} Z$ and $\psi(Z) = Z \oplus \psi'(Z)$ also preserve equality. Lastly, one can observe that for $W^K, W^Q \in \mathbb{F}^{m \times d}$ and $Z = [\mathbf{z}_1, \dots, \mathbf{z}_n] \in \mathbb{F}^{d \times n}$, $\sigma((W^K \hat{\otimes} Z)^\top \hat{\otimes} (W^Q \hat{\otimes} Z)) = [\mathbf{s}_1, \dots, \mathbf{s}_n]$ satisfies $\mathbf{s}_i = \mathbf{s}_j$ when $\mathbf{z}_i = \mathbf{z}_j$, i.e., $\sigma((W^K \hat{\otimes} Z)^\top \hat{\otimes} (W^Q \hat{\otimes} Z))$ preserves equality. Hence, all floating-point transformers preserve equality. This completes the proof.

C.5. Proof of Lemma 8

Since the proof is trivial if $z \in \{\pm\infty, \text{NaN}\}$, without loss of generality, suppose $z > 0$. If $2^{\epsilon_{\min}+1} \leq z < \Omega$, then $\text{ulp}_{\mathbb{F}}(z) \geq 2\omega$, so the three exact sums $0 + z$, $\omega + z$, and $2\omega + z$ all lie in the closed interval $[z, z^+]$ between z and the next float z^+ above z . Since round-to-nearest maps this interval into the two-element set $\{z, z^+\}$, two of the three distinct floats $0, \omega, 2\omega$ have the same image under $x \mapsto x \oplus z$. If $z = \Omega$, then $\Omega^+ = \infty$, but the three sums $\Omega, \Omega + \omega$, and $\Omega + 2\omega$ still all round to Ω because $2\omega = 2^{\epsilon_{\min}-p+1} < 2^{\epsilon_{\max}-p-1}$ by Condition 1, so again two (indeed all three) of the images coincide. For $z < 2^{\epsilon_{\min}+1}$, let x_0 be the smallest positive float such that $x_0 \oplus z = x_0$; then, $x_0 \leq 2^{\epsilon_{\min}+p+2} \leq \Omega$ by Condition 1, which gives $\epsilon_{\min} + p + 2 \leq \epsilon_{\max}$. Since $0 \oplus z = z > 0$, by the pigeonhole principle, there must be two distinct floats (say a, b) between zero and x_0 such that $a \oplus z = b \oplus z$. Hence, $x \mapsto x \oplus z$ is not injective for all $z \in \mathbb{F} \setminus \{0\}$.

C.6. Proof of Lemma 9

For each $X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \Delta_n$, let $[y_{X,1}, \dots, y_{X,n}] = f^*(X)$ and choose $\tilde{f} : \mathbb{R}^{d_{\text{in}}} \times \mathbb{R}^{d_{\text{in}} \times n} \rightarrow \mathbb{R}^{d_{\text{out}}}$ such that $\tilde{f}(\mathbf{x}_i, X) = y_{X,i}$. Then, $\tilde{f}$ is well-defined since $X \in \Delta_n$, i.e., for each $\mathbf{x} \in \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, there exists a unique i such that $\mathbf{x}_i = \mathbf{x}$. We choose arbitrary values for $\tilde{f}(\mathbf{x}, X)$ for all pairs $(\mathbf{x}, X)$ not covered above, i.e., for $X \notin \Delta_n$ or $\mathbf{x} \notin \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$; these values are not used in the statement. Furthermore, since f^* is $\mathcal{G}(\mathcal{T}_n)$ -equivariant, for $\pi \in \mathcal{G}(\mathcal{T}_n)$, it holds that

$$\begin{aligned} [\tilde{f}(\mathbf{x}_1, X), \dots, \tilde{f}(\mathbf{x}_n, X)] &= f^*(X) = \pi^{-1} f^*(\pi X) = \pi^{-1} [y_{\pi X,1}, \dots, y_{\pi X,n}] \\ &= \pi^{-1} [\tilde{f}(\mathbf{x}_{\pi(1)}, \pi X), \dots, \tilde{f}(\mathbf{x}_{\pi(n)}, \pi X)] \\ &= [\tilde{f}(\mathbf{x}_1, \pi X), \dots, \tilde{f}(\mathbf{x}_n, \pi X)] \end{aligned}$$

for all $X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \Delta_n$. This completes the proof.

C.7. Proof of Lemma 10

Let $f_{\mathcal{T}_n}(x_1, \dots, x_n) = \bigoplus_{i \in [n]; \mathbb{F}'} x_i$. We first show $\mathcal{G}(\mathcal{T}_n)\pi \subset \mathbf{q}^{-1}(\mathbf{q}(\pi))$. Let $g \in \mathcal{G}(\mathcal{T}_n)$. By Lemma 1,

$$f_{\mathcal{T}_n}(x_1, \dots, x_n) = f_{\mathcal{T}_n}(x_{g(1)}, \dots, x_{g(n)}) \quad \text{for all } x \in (\mathbb{F}')^n.$$

Fix an ordered quadruple $\mathcal{I} = (\iota_1, \dots, \iota_4)$ of distinct indices and (j, k) with $j \neq k$, and define $y_i = \delta_{(\pi(\iota_1), \dots, \pi(\iota_4)), (j, k), i}$. Then, it holds that

$$\tau_{(\pi(\iota_1), \dots, \pi(\iota_4)), (j, k)} = f_{\mathcal{T}_n}(y_1, \dots, y_n).$$

In addition, by the definition,

$$\delta_{(g\pi(\iota_1), \dots, g\pi(\iota_4)), (j, k), i} = y_{g^{-1}(i)}.$$

Therefore,

$$\tau_{(g\pi(\iota_1), \dots, g\pi(\iota_4)), (j, k)} = f_{\mathcal{T}_n}(y_{g^{-1}(1)}, \dots, y_{g^{-1}(n)}) = f_{\mathcal{T}_n}(y_1, \dots, y_n),$$

where the last equality uses $g^{-1} \in \mathcal{G}(\mathcal{T}_n)$. Hence $\mathbf{q}(g\pi) = \mathbf{q}(\pi)$, and so $\mathcal{G}(\mathcal{T}_n)\pi \subset \mathbf{q}^{-1}(\mathbf{q}(\pi))$.

Conversely, let $\pi' \in \mathbf{q}^{-1}(\mathbf{q}(\pi))$ and set $\theta = \pi' \pi^{-1}$. Then $\mathbf{q}(\theta\pi) = \mathbf{q}(\pi)$. Hence, for any ordered quadruple $\mathcal{I} = (\iota_1, \dots, \iota_4)$ of distinct elements and for any (j, k) with $j \neq k$,

$$\tau_{(\theta(\pi(\iota_1)), \dots, \theta(\pi(\iota_4))), (j, k)} = \tau_{(\pi(\iota_1), \dots, \pi(\iota_4)), (j, k)}.$$

Let $\mathcal{I}' = (\pi(\iota_1), \dots, \pi(\iota_4))$. Since π is a bijection, $\mathcal{I}'$ ranges over all ordered quadruples of distinct elements. Thus

$$\tau_{(\theta(\iota'_1), \dots, \theta(\iota'_4)), (j, k)} = \tau_{(\iota'_1, \dots, \iota'_4), (j, k)}$$

for all ordered quadruples ι' and all (j, k) with $j \neq k$. This implies that $\mathbf{q}(\theta) = \mathbf{q}(\text{id})$, where $\text{id} \in \mathcal{S}_n$ denotes the identity function.

We claim that θ preserves every induced four-leaf tree. Indeed, for an ordered quadruple $\mathcal{I} = (\iota_1, \dots, \iota_4)$, all coordinates outside $\{\iota_1, \dots, \iota_4\}$ are assigned value 0 in the definition of $\tau_{\mathcal{I}, (j, k)}$. Hence, $\tau_{\mathcal{I}, (j, k)}$ depends only on the induced four-leaf tree $\mathcal{T}_n|_{\mathcal{I}}$ (induced by $\mathcal{I}$). By Lemma 28, the vector $(\tau_{\mathcal{I}, (j, k)})_{j \neq k}$ uniquely determines which of the fifteen rooted full binary trees is realized on the ordered leaves $(\iota_1, \dots, \iota_4)$. This is the only place where the hypothesis $p' \geq p + 2$ of Lemma 10 is used. Here, Lemma 28 is stated at unit scale and is applied to the values $\delta_{\mathcal{I}, (j, k), i}$, which are exactly its values a, b, c, d scaled by the power of two 2^{e^*} . By Condition 1, this rescaling is exact and causes neither overflow nor underflow, so all fifteen identities transfer verbatim. Since $\mathbf{q}(\theta) = \mathbf{q}(\text{id})$, the induced four-leaf tree on $(\iota_1, \dots, \iota_4)$ is isomorphic to that on $(\theta(\iota_1), \dots, \theta(\iota_4))$ for every $\mathcal{I}$.

We now prove that this implies $\theta \in \mathcal{G}(\mathcal{T}_n)$. For distinct leaves a, b, c , write $ab|c$ if the lowest common ancestor of a and b is a strict descendant of the lowest common ancestor of a, b, c . Since $n \geq 4$, for any distinct a, b, c we may choose $d \notin \{a, b, c\}$. The rooted triple relation among a, b, c is determined by the induced four-leaf tree on (a, b, c, d) . Therefore, since θ preserves all induced four-leaf trees, we have

$$ab|c \iff \theta(a)\theta(b)|\theta(c)$$

for all distinct a, b, c .

For each node v of $\mathcal{T}_n$, let $\mathcal{C}_v$ be the set of leaves below v . We use the following characterization: for any subset $\mathcal{C} \subset [n]$ with $|\mathcal{C}| \geq 2$, $\mathcal{C} = \mathcal{C}_v$ for some node v if and only if for every distinct $a, b \in \mathcal{C}$ and every $c \notin \mathcal{C}$, one has $ab|c$. The forward direction is immediate. For the converse, let v be the lowest common ancestor of the leaves in $\mathcal{C}$. If $\mathcal{C} \neq \mathcal{C}_v$, choose $c \in \mathcal{C}_v \setminus \mathcal{C}$. Since v is the lowest common ancestor of $\mathcal{C}$, there exist distinct $a, b \in \mathcal{C}$ lying in different child subtrees of v . Then the rooted triple on a, b, c does not satisfy $ab|c$, a contradiction. Hence $\mathcal{C} = \mathcal{C}_v$.

Now fix a node v with $|\mathcal{C}_v| \geq 2$. Applying the triple-preservation property to the cluster $\mathcal{C}_v$, we obtain that for every distinct $a, b \in \mathcal{C}_v$ and every $c \notin \mathcal{C}_v$,

$$\theta(a)\theta(b)|\theta(c).$$

Thus, the set $\theta(\mathcal{C}_v) = \{\theta(a) : a \in \mathcal{C}_v\}$ satisfies the characterization above, so $\theta(\mathcal{C}_v) = \mathcal{C}_w$ for some node w . Singleton clusters are trivially preserved since θ is a permutation on leaves. Therefore, θ maps clusters of $\mathcal{T}_n$ to clusters of $\mathcal{T}_n$.

Since θ is bijective, the same argument applied to θ^{-1} shows that θ maps the collection of all clusters bijectively onto itself. Moreover, for any nodes u, v ,

$$\mathcal{C}_u \subset \mathcal{C}_v \iff \theta(\mathcal{C}_u) \subset \theta(\mathcal{C}_v).$$

The rooted tree structure of $\mathcal{T}_n$ is completely determined by its cluster family together with the inclusion relations among clusters: for each non-root cluster $\mathcal{C}_u$, its parent is the minimal cluster strictly containing $\mathcal{C}_u$. Therefore, the bijection induced by θ on the cluster family preserves the parent-child relations of the tree. Hence, θ induces an automorphism of the rooted tree $\mathcal{T}_n$.

Finally, the automorphism group of a rooted full binary tree is exactly $\mathcal{G}(\mathcal{T}_n)$, since every automorphism is generated by swaps of isomorphic left and right subtrees at internal nodes. Thus $\theta \in \mathcal{G}(\mathcal{T}_n)$. Since $\pi' = \theta\pi$, we have $\pi' \in \mathcal{G}(\mathcal{T}_n)\pi$, proving

$$\mathfrak{q}^{-1}(\mathfrak{q}(\pi)) \subset \mathcal{G}(\mathcal{T}_n)\pi.$$

Combining the two inclusions gives $\mathfrak{q}^{-1}(\mathfrak{q}(\pi)) = \mathcal{G}(\mathcal{T}_n)\pi$ and completes the proof.

C.8. Proof of Lemma 12

For notational simplicity, we use t_d and t_r to denote t'_d and t'_r . Since all attention layers can represent the identity map for all $m, d, h \in \mathbb{N}$ by choosing zero weights, we only consider feed-forward networks in this proof. By Lemma 23, there exist $d_1 = e_0, e_1, e_2, e_3, e_4 = d_2 \in \mathbb{N}$ and affine transformations $\phi_i : \mathbb{F}^{e_{i-1}} \rightarrow \mathbb{F}^{e_i}$ for all $i \in [4]$, computed under the left-to-right summation order, such that for each $\mathbf{x} \in \mathbb{F}^{e_0}$,

$$\phi_4 \circ \rho \circ \phi_3 \circ \rho \circ \phi_2 \circ \rho \circ \phi_1(\mathbf{x}) = \psi^*(\mathbf{x})$$

where

$$\phi_i(z_1, \dots, z_{e_{i-1}})_j = \left(\bigoplus_{k=1}^{e_{i-1}} w_{jk} \otimes z_k \right) \oplus b_j$$

for some $w_{jk}, b_j \in \mathbb{F}$ for all $k \in [e_{i-1}]$ and $j \in [e_i]$. Here, e_1, e_2, e_3 depend only on e_0 and every product $w_{jk} \otimes z_k$ arising in the evaluation satisfies $w_{jk} \in \{0, \pm 2^{-1}, \pm 1\}$ or $z_k \in \{0, 1\}$.

Choose $t_d = \sum_{i=0}^4 e_i + e_4$ and $t_r = \max\{2 \sum_{i=0}^4 e_i, 4e_4\}$; these depend only on d_1 and d_2 . Then, it suffices to show that there exist $N \in \mathbb{N}$ and $\psi_1, \dots, \psi_N : \mathbb{F}^{t_d} \rightarrow \mathbb{F}^{t_d}$ such that

$$\psi_i(\mathbf{z}) = \mathbf{z} \oplus (g_{i,2} \circ \rho \circ g_{i,1}(\mathbf{z}))$$

for some floating-point affine transformations $g_{i,1} : \mathbb{F}^{t_d} \rightarrow \mathbb{F}^{t_r}$ and $g_{i,2} : \mathbb{F}^{t_r} \rightarrow \mathbb{F}^{t_d}$ for all $i \in [N]$, and such that ψ defined by

$$\psi([\mathbf{x}_1, \dots, \mathbf{x}_n]) = [\psi_N \circ \dots \circ \psi_1(\mathbf{x}_1), \dots, \psi_N \circ \dots \circ \psi_1(\mathbf{x}_n)]$$

satisfies the desired property of ψ . Here, we assume $d = t_d$ and $r = t_r$. For $d > t_d$ or $r > t_r$, we pad unused dimensions/neurons.

Let $\alpha_i = \sum_{j=0}^{i-2} e_j$ for $i \in [6]$ and $\beta = 4 + \sum_{i=0}^3 e_i$. Under this choice, $t_d = \alpha_6 + e_4$; we use the coordinates $\alpha_{i+1} + 1, \dots, \alpha_{i+2}$ to store the output of ϕ_i for each $i \in [4]$, and the coordinates $\alpha_6 + 1, \dots, \alpha_6 + e_4$ as an auxiliary block

(disjoint from the former). We construct ψ_l for all $l \in [\beta]$ as follows, where the parameters $w_{i,jk}, b_{i,j}$ below are those of ϕ_i , $\text{sgn}(w) \in \{1, -1\}$ denotes the sign of w (with $\text{sgn}(0) := 1$). Specifically, let $l = (\alpha_i + i - 1) + k$ for some $i \in [4]$ and $k \in [e_{i-1} + 1]$. If $i = 1$ and $k \leq e_0$, we add the k -th linear term of ϕ_1 , whose input is not rectified:

$$g_{l,1}(z_1, \dots, z_{t_d})_j = \begin{cases} w_{1,jk} \otimes z_k & \text{if } j \leq e_1, \\ (-w_{1,(j-e_1)k}) \otimes z_k & \text{if } e_1 < j \leq 2e_1, \\ 0 & \text{otherwise,} \end{cases}$$

$$g_{l,2}(y_1, \dots, y_{t_r})_j = \begin{cases} y_{j-\alpha_2} - y_{j-\alpha_2+e_1} & \text{if } \alpha_2 < j \leq \alpha_3, \\ 0 & \text{otherwise.} \end{cases}$$

If $i \geq 2$ and $k \leq e_{i-1}$, we add the k -th linear term of ϕ_i , applied to the rectified output of ϕ_{i-1} :

$$g_{l,1}(z_1, \dots, z_{t_d})_j = \begin{cases} z_{\alpha_i+k} & \text{if } j = 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$g_{l,2}(y_1, \dots, y_{t_r})_j = \begin{cases} w_{i,(j-\alpha_{i+1})k} \otimes y_1 & \text{if } \alpha_{i+1} < j \leq \alpha_{i+2}, \\ 0 & \text{otherwise.} \end{cases}$$

If $k = e_{i-1} + 1$, we add the bias of ϕ_i :

$$g_{l,1}(z_1, \dots, z_{t_d}) = \mathbf{0}_{t_r}, \quad g_{l,2}(y_1, \dots, y_{t_r})_j = \begin{cases} b_{i,j-\alpha_{i+1}} & \text{if } \alpha_{i+1} < j \leq \alpha_{i+2}, \\ 0 & \text{otherwise.} \end{cases}$$

For $i = 1$, since one of $\rho(v), \rho(-v)$ is 0 and $(-w) \otimes z = -(w \otimes z)$, the l -th block adds $w_{jk} \otimes z_k$ exactly to the coordinate $\alpha_2 + j$ of $\mathbf{z}$. For $i \geq 2$, the l -th block adds $w_{i,jk} \otimes \rho(z_{\alpha_i+k})$ exactly to the coordinate $\alpha_{i+1} + j$. Since the blocks of phase i write only to the coordinates $\alpha_{i+1} + 1, \dots, \alpha_{i+2}$ and read only the coordinates of the previous phase, the coordinate $\alpha_{i+1} + j$ accumulates in the left-to-right order to output $\phi_i(\rho(\mathbf{z}^{(i)}))_j$ where $\mathbf{z}^{(i)}$ denotes the content of the coordinates $\alpha_i + 1, \dots, \alpha_{i+1}$ (for $i = 1$, $\phi_1(z_1, \dots, z_{e_0})_j$ with no rectification). This coincides with the left-to-right computation of ϕ_i in Lemma 23.

We also define $\psi_{\beta+1}, \psi_{\beta+2}$, and $\psi_{\beta+3}$ so that

$$g_{\beta+1,2} \circ \rho \circ g_{\beta+1,1}(z_1, \dots, z_{t_d})_j = \begin{cases} 1 \otimes \rho(1 \otimes z_{\alpha_5+k}) \oplus (-1) \otimes \rho((-1) \otimes z_{\alpha_5+k}) & \text{if } j = (t_d - e_4) + k \text{ for some } k \in [e_4], \\ 0 & \text{otherwise,} \end{cases}$$

$$g_{\beta+2,2} \circ \rho \circ g_{\beta+2,1}(z_1, \dots, z_{t_d})_j = \begin{cases} (-1) \otimes \rho(1 \otimes z_j) \oplus 1 \otimes \rho((-1) \otimes z_j) & \text{if } j \leq \alpha_6, \\ 0 & \text{otherwise,} \end{cases}$$

$$g_{\beta+3,2} \circ \rho \circ g_{\beta+3,1}(z_1, \dots, z_{t_d})_j = \begin{cases} 1 \otimes \rho(1 \otimes z_{(t_d-e_4)+j}) \oplus (-1) \otimes \rho((-1) \otimes z_{(t_d-e_4)+j}) & \text{if } j \in [e_4], \\ (-1) \otimes \rho(1 \otimes z_j) \oplus 1 \otimes \rho((-1) \otimes z_j) & \text{if } j > t_d - e_4, \\ 0 & \text{otherwise.} \end{cases}$$

That is, $\psi_{\beta+1}$ copies the output of ϕ_4 (i.e., ψ^*) from the coordinates $\alpha_5 + 1, \dots, \alpha_6$ to the auxiliary block, $\psi_{\beta+2}$ erases the coordinates $1, \dots, \alpha_6$, and $\psi_{\beta+3}$ copies the auxiliary block to the coordinates $1, \dots, e_4$ (which are zero after $\psi_{\beta+2}$) and erases the auxiliary block. Here, each transfer is exact since one of $\rho(v), \rho(-v)$ is 0. The required numbers of hidden neurons are $2e_1$ and 1 for the blocks of phases 1 and $i \geq 2$, and $2e_4, 2\alpha_6$, and $4e_4$ for $\psi_{\beta+1}, \psi_{\beta+2}$, and $\psi_{\beta+3}$, respectively, all of which are at most $t_r = \max\{2\alpha_6, 4e_4\}$. Hence, for each $j \in [e_4]$ and each $(z_1, \dots, z_{t_d}) \in \mathbb{F}^{t_d}$ with $z_{e_0+1} = \dots = z_{t_d} = 0$ (which holds for every column of the input of ψ by the hypothesis $Z_{d+1:} = \mathbf{0}_{(d-d_1) \times n}$),

$$\psi \circ \dots \circ \psi_1(z_1, \dots, z_{t_d})_{\alpha_5+j} = \psi^*(z_1, \dots, z_{e_0})_j.$$

Then, for $\psi = \psi_{\beta+3} \circ \dots \circ \psi_1$ and each $(z_1, \dots, z_{t_d}) \in \mathbb{F}^{t_d}$ with $z_{e_0+1} = \dots = z_{t_d} = 0$, it holds that

$$\psi(z_1, \dots, z_{t_d})_j = \begin{cases} \psi^*(z_1, \dots, z_{e_0})_j & \text{if } j \leq e_4, \\ 0 & \text{if } j > e_4. \end{cases}$$

It remains to verify that the arithmetic of these blocks agrees with the left-to-right $\mathbb{F}$ -arithmetic of Lemma 23. First, for every block constructed in this proof, i.e., for all $l \in [\beta + 3]$, each row of $g_{l,1}$ has at most one nonzero entry, and each row of $g_{l,2}$ has at most two nonzero entries, and when it has two, one of the two corresponding inputs is always zero. Hence, every reduction occurring in $g_{l,1}$ and $g_{l,2}$ involves at most one nonzero summand and is therefore exact and independent of the reduction order. The attention layer of each ψ_l uses zero weights, so its output is exactly $\mathbf{0}$ and the residual connection leaves $\mathbf{z}$ unchanged.

Second, for the products $w \otimes z$ arising in Lemma 23, the high-precision product coincides with the $\otimes = \otimes_{\mathbb{F}}$ used there, i.e., $\lceil w \otimes_{\mathbb{F}'} z \rceil_{\mathbb{F}} = w \otimes z$. This is immediate if $w = 0$, $z \in \{0, 1\}$, or $(p', q') = (p, q)$. Otherwise $(p' \geq p \text{ and } q' \geq q \text{ with at least one of them strict})$ $w \in \{\pm 2^{-1}, \pm 1\}$, so $w \times z$ has the mantissa of z with the exponent lowered by at most one and is thus exact in $\mathbb{F}'$. In addition, overflow cannot occur: otherwise $|w \times z| \geq 2^{\epsilon_{\max}+1}$ and the product under $\mathbb{F}$ would round to $\pm\infty$, contradicting the finiteness guarantee of Lemma 23. Hence $\lceil w \otimes_{\mathbb{F}'} z \rceil_{\mathbb{F}} = \lceil w \times z \rceil_{\mathbb{F}} = w \otimes z$.

Finally, every coordinate of $\mathbf{z}$ that the blocks read is finite: it is either 0 (a coordinate whose phase has not been executed yet, the auxiliary block before $\psi_{\beta+1}$, or a coordinate erased by $\psi_{\beta+2}$) or a value arising in the left-to-right evaluation of ψ^* , including its intermediate partial sums, which are finite by Lemma 23. Hence every product formed with a zero weight is exactly zero, and every product whose weight lies in $\{0, \pm 1\}$ is exact; this covers all products of $g_{l,1}$ in the blocks of phase $i \geq 2$, all products in $\psi_{\beta+1}, \psi_{\beta+2}, \psi_{\beta+3}$, and all products of $g_{l,2}$ in the blocks of phase 1. Two families remain: in $g_{l,1}$ of the blocks of phase 1, the weight is $\pm w_{1,jk}$ and the operand is the input coordinate z_k , and in $g_{l,2}$ of the blocks of phase $i \geq 2$ the weight is $w_{i,(j-\alpha_i+1)k}$ and the operand is $\rho(z_{\alpha_i+k})$. In both cases, the operand is the corresponding factor in the evaluation of ψ^* , so the identity above applies. Therefore, every value that the construction feeds forward agrees with the corresponding value in the evaluation of ψ^* . This completes the proof.

C.9. Proof of Lemma 11

Throughout this proof, $\gamma = \binom{|\mathbb{F}|^{d_{\text{in}}}}{4}$ (not the quantity denoted by γ in the proofs of Lemmas 14 and 30), and $\mathcal{Z}_1, \dots, \mathcal{Z}_{\gamma}$, $(\alpha_k, \beta_k)_{k \in [12]}$, ε_j , $\zeta_{i,j,k}$, and e^* are as defined in Lemma 11 and Section 5.1. Let $\mathbb{F} = \mathbb{F}_{p,q}$ and $h = \gamma$. Without loss of generality, we assume that $d = t_d$, $m = t_m$, and $r = t_r$; otherwise, we can add unused dimensions. We set $t_m = 12$, and we will choose the explicit values of t_d, t_r later with $t_d \geq d_{\text{in}} + 25\gamma + 1$. We construct ϕ as a composition of two transformer blocks $g_1, g_2 \in \mathcal{B}^{\gamma, t_m, t_r, t_d, n}$. Here, g_1 maps $Z \in \mathbb{F}^{t_d \times n}$ with $Z_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \Delta_n$ and $Z_{d_{\text{in}}+1:} = \mathbf{0}_{(t_d-d_{\text{in}}) \times n}$ to $Y = [\mathbf{y}_1, \dots, \mathbf{y}_n] \in \mathbb{F}^{t_d \times n}$ such that

$$(\mathbf{y}_i)_l = \begin{cases} (\mathbf{x}_i)_l & \text{if } l \leq d_{\text{in}}, \\ 2^{e^*+2} & \text{if } l = d_{\text{in}} + 12\gamma + 12(j-1) + k \text{ and } \mathbf{x}_i = \mathbf{z}_{j,\alpha_k} \text{ for some } j \in [\gamma], k \in [12], \\ 2^{e^*-p+1} & \text{if } l = d_{\text{in}} + 12\gamma + 12(j-1) + k \text{ and } \mathbf{x}_i = \mathbf{z}_{j,\beta_k} \text{ for some } j \in [\gamma], k \in [12], \\ 2^{e^*-p'+1} & \text{if } l = d_{\text{in}} + 12\gamma + 12(j-1) + k \text{ and } \mathbf{x}_i \in \mathcal{Z}_j \setminus \{\mathbf{z}_{j,\alpha_k}, \mathbf{z}_{j,\beta_k}\} \text{ for some } j \in [\gamma], k \in [12], \\ 0 & \text{if } l = d_{\text{in}} + 24\gamma + j \text{ and } \mathbf{x}_i \in \mathcal{Z}_j \text{ for some } j \in [\gamma], \\ -\Omega/2 & \text{if } l = d_{\text{in}} + 24\gamma + j \text{ and } \mathbf{x}_i \notin \mathcal{Z}_j \text{ for some } j \in [\gamma], \\ 2 & \text{if } l = d_{\text{in}} + 25\gamma + 1, \\ 0 & \text{otherwise,} \end{cases}$$

for all $i \in [n]$ and $l \in [t_d]$ where Ω denotes the largest positive float in $\mathbb{F}$. That is, the values $\zeta_{i,j,k}$ are stored in the rows $d_{\text{in}} + 12\gamma + 1, \dots, d_{\text{in}} + 24\gamma$, the row $d_{\text{in}} + 24\gamma + j$ indicates (for each quadruple $j \in [\gamma]$ separately) whether $\mathbf{x}_i \in \mathcal{Z}_j$, and the row $d_{\text{in}} + 25\gamma + 1$ holds the constant 2. Here, the rows $d_{\text{in}} + 1, \dots, d_{\text{in}} + 12\gamma$ and all rows above $d_{\text{in}} + 25\gamma + 1$ are zero. Note that these row positions do not depend on t_d .

We also note that $\lceil \exp \rceil(-\Omega) = 0$, where $\lceil \exp \rceil$ is computed under $\mathbb{F}'$ as in Eq. (4): since $\Omega = (2 - 2^{-p}) \cdot 2^{2^{q-1}-1} \geq \frac{7}{4} \cdot 2^{2^{q-1}-1}$ and $\log_2 e \geq \frac{10}{7}$, we have $\Omega \log_2 e \geq \frac{5}{2} \cdot 2^{2^{q-1}-1} = 2^{2^{q-1}-1} + \frac{3}{2} \cdot 2^{2^{q-1}-1} \geq 2^{q'-1} + p'$, where the last inequality uses $q' \leq 2^{q-1}$ and $p' \leq p + 2^q - 4 \leq 3 \cdot 2^{q-1} - 7 \leq \frac{3}{2} \cdot 2^{2^{q-1}-1}$ (as $2^{q-1} \geq p + 3 \geq 5$ by Condition 1). Hence, $e^{-\Omega} \leq 2^{-2^{q'-1}-p'}$, which is smaller than $2^{1-2^{q'-1}-p'}$, i.e., half of the smallest positive value of $\mathbb{F}'$, so $e^{-\Omega}$ rounds to 0 under $\mathbb{F}'$. Note that 2^{e^*+2} , 2^{e^*-p+1} , and $2^{e^*-p'+1}$ are in $\mathbb{F}$ under $2^q - 4 \geq p' - p$. By Lemma 12 applied with $d_1 = d_{\text{in}}$, $d_2 = d_{\text{in}} + 25\gamma + 1$, and the present $m = t_m$, $h = \gamma$, and n , there exist $t'_d, t'_r \in \mathbb{N}$ (which do not depend on t_d or t_r) such that the desired $g_1 \in \mathcal{B}^{\gamma, t_m, r', d', n}$ exists for all $d' \geq t'_d$ and $r' \geq t'_r$. Choose $t_d = \max\{t'_d, d_{\text{in}} + 25\gamma + 1\}$, $t_r = \max\{t'_r, 26\gamma + 2\}$, and $g_1 \in \mathcal{B}^{\gamma, t_m, t_r, t_d, n}$.

We now construct g_2 . Define an attention layer $f_1 : \mathbb{R}^{t_d \times n} \rightarrow \mathbb{F}^{t_d \times n}$ as

$$f_1(Z) = Z \oplus \left(\bigoplus_{j=1}^h \left(W_j^O \hat{\otimes} ((W_j^V \hat{\otimes} Z) \hat{\otimes} \sigma((W_j^K \hat{\otimes} Z)^\top \hat{\otimes} (W_j^Q \hat{\otimes} Z))) \right) \right)$$

for some $W_j^K, W_j^Q, W_j^V \in \mathbb{F}^{t_m \times t_d}$ and $W_j^O \in \mathbb{F}^{t_d \times t_m}$. Specifically, for all $j \in [h]$, we choose W_j^K and W_j^Q so that $(W_j^K)_{1, d_{\text{in}}+24\gamma+j} = 1 = (W_j^Q)_{1, d_{\text{in}}+25\gamma+1}$ and all other entries are zero. Since each row of W_j^K and W_j^Q has at most one nonzero entry, the products $W_j^K \hat{\otimes} Y$ and $W_j^Q \hat{\otimes} Y$ are exact for the output Y of g_1 . Hence, the (i, ι) -th entry of $M_j := (W_j^K \hat{\otimes} Y)^\top \hat{\otimes} (W_j^Q \hat{\otimes} Y)$ equals $(\mathbf{y}_i)_{d_{\text{in}}+24\gamma+j} \otimes 2$, which is 0 if $\mathbf{x}_i \in \mathcal{Z}_j$ and $(-\Omega/2) \otimes 2 = -\Omega$ otherwise (exact, as $-\Omega \in \mathbb{F}$). In particular, all columns of M_j are identical.

We now compute $\sigma(M_j)$ column-wise via Eq. (4). Let $k_j = |\{i \in [n] : \mathbf{x}_i \in \mathcal{Z}_j\}|$, and note that $k_j \leq 4$ since $\mathbf{x}_1, \dots, \mathbf{x}_n$ are distinct. If $k_j \geq 1$, the column maximum is $x_* = 0$, and the numerators are $\lceil \exp \rceil (0 \ominus 0) = \exp(0) = 1$ for the k_j indices i with $\mathbf{x}_i \in \mathcal{Z}_j$ and $\lceil \exp \rceil (-\Omega \ominus 0) = \lceil \exp \rceil (-\Omega) = 0$ for the remaining indices, as shown above. Since $v \oplus_{\mathbb{F}'} 0 = v$ for all $v \in \mathbb{F}'$, the denominator equals the exact integer k_j under any reduction order. Hence, the i -th entry of each column of $\sigma(M_j)$ is $\lceil 1 \otimes_{\mathbb{F}'} k_j \rceil_{\mathbb{F}} = \varepsilon_j$ if $\mathbf{x}_i \in \mathcal{Z}_j$ and $\lceil 0 \otimes_{\mathbb{F}'} k_j \rceil_{\mathbb{F}} = 0$ otherwise. If $k_j = 0$, all scores are $-\Omega$, so $x_* = -\Omega$, all numerators are $\lceil \exp \rceil (-\Omega \ominus (-\Omega)) = \exp(0) = 1$, and every entry of $\sigma(M_j)$ is $\lceil 1 \otimes_{\mathbb{F}'} (\bigoplus_{i \in [n]; \mathbb{F}'} 1) \rceil_{\mathbb{F}} = \varepsilon_j$. We will not use this value; it will be multiplied only by $\zeta_{i,j,k} = 0$.

We also choose W_j^V, W_j^O so that

$$\begin{aligned} W_j^V \hat{\otimes} Z &= Z_{d_{\text{in}}+12\gamma+12(j-1)+1:d_{\text{in}}+12\gamma+12(j-1)+12}, \\ (W_j^O \hat{\otimes} Y)_{d_{\text{in}}+12(j-1)+1:d_{\text{in}}+12(j-1)+12} &= Y, \end{aligned}$$

and all other rows of $W_j^O \hat{\otimes} Y$ are zero for all $Y \in \mathbb{F}^{t_m \times n}$. Then, by the definitions of g_1, f_1 , it holds that for each $Z \in \mathbb{F}^{t_d \times n}$ with $Z_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \Delta_n$ and $Z_{d_{\text{in}}+1:} = \mathbf{0}_{(t_d-d_{\text{in}}) \times n}$, for each $j \in [\gamma]$ and $k \in [12]$,

- $(f_1 \circ g_1(Z))_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$,
- $(f_1 \circ g_1(Z))_{d_{\text{in}}+12(j-1)+k} = \left\lceil \bigoplus_{i \in [n]; \mathbb{F}'} \varepsilon_j \otimes_{\mathbb{F}'} \zeta_{i,j,k} \right\rceil_{\mathbb{F}} \times \mathbf{1}_n^\top$,
- $(f_1 \circ g_1(Z))_{d_{\text{in}}+25\gamma+2:} = \mathbf{0}_{(t_d-d_{\text{in}}-25\gamma-1) \times n}$, and the rows $d_{\text{in}} + 12\gamma + 1, \dots, d_{\text{in}} + 25\gamma + 1$ of $f_1 \circ g_1(Z)$ coincide with the corresponding rows of $g_1(Z)$. In particular, all their entries are finite.

Here, the second property holds since the i -th summand of $((W_j^V \hat{\otimes} Y) \hat{\otimes} \sigma(M_j))_{k,\iota} = \left\lceil \bigoplus_{i \in [n]; \mathbb{F}'} \zeta_{i,j,k} \otimes_{\mathbb{F}'} \sigma(M_j)_{i\iota} \right\rceil_{\mathbb{F}}$ equals $\varepsilon_j \otimes_{\mathbb{F}'} \zeta_{i,j,k}$ for every i : if $\mathbf{x}_i \in \mathcal{Z}_j$ then $\sigma(M_j)_{i\iota} = \varepsilon_j$, and otherwise $\zeta_{i,j,k} = 0$ so both are 0. Furthermore, the routing by W_j^O is exact (one nonzero entry per row), the heads write to disjoint rows so their sum $\bigoplus_{j=1}^h$ is exact, and the residual connection adds these values to entries of $g_1(Z)$ that are zero, which is also exact.

Choose a feed-forward network $f_2 : \mathbb{F}^{t_d \times n} \rightarrow \mathbb{F}^{t_d \times n}$ as $f_2(Z) = Z \oplus (W_2 \hat{\otimes} \rho(W_1 \hat{\otimes} Z))$ where $W_1 \in \mathbb{F}^{t_r \times t_d}$ and $W_2 \in \mathbb{F}^{t_d \times t_r}$ are matrices satisfying that for $\mathbf{z} = (z_1, \dots, z_{t_d}) \in \mathbb{F}^{t_d}$ and $\mathbf{y} = (y_1, \dots, y_{t_r}) \in \mathbb{F}^{t_r}$,

$$\begin{aligned} W_1 \otimes \mathbf{z} &= (z_{d_{\text{in}}+12\gamma+1}, \dots, z_{d_{\text{in}}+25\gamma+1}, -z_{d_{\text{in}}+12\gamma+1}, \dots, -z_{d_{\text{in}}+25\gamma+1}, 0, \dots, 0), \\ W_2 \otimes \mathbf{y} &= (0, \dots, 0, y_{13\gamma+2} - y_1, \dots, y_{26\gamma+2} - y_{13\gamma+1}, 0, \dots, 0), \end{aligned}$$

where the nonzero block of $W_2 \otimes \mathbf{y}$ occupies the coordinates $d_{\text{in}} + 12\gamma + 1, \dots, d_{\text{in}} + 25\gamma + 1$. Each row of W_1 has at most one nonzero entry (which is ± 1), so every reduction in $W_1 \hat{\otimes} Z$ has at most one nonzero summand and is therefore exact under any reduction order. For each $l \in \{d_{\text{in}} + 12\gamma + 1, \dots, d_{\text{in}} + 25\gamma + 1\}$, the l -th coordinate of each column of $W_2 \hat{\otimes} \rho(W_1 \hat{\otimes} Z)$ is $\lceil (-\rho(v)) \oplus_{\mathbb{F}'} \rho(-v) \rceil_{\mathbb{F}}$ where v is the l -th coordinate of the corresponding column of Z : since one of $\rho(v), \rho(-v)$ is 0, this equals $-v$ exactly for every finite v of either sign, and the residual connection yields $v \oplus (-v) = 0$. Namely, f_2 erases the coordinates $d_{\text{in}} + 12\gamma + 1, \dots, d_{\text{in}} + 25\gamma + 1$ of each column, whose entries are finite by the third property above; this requires $t_r \geq 26\gamma + 2$, consistent with our choice of t_r . Let $g_2 = f_2 \circ f_1$ and $\phi = g_2 \circ g_1 \in \mathcal{B}^{\gamma, t_m, t_r, t_d, n}$. Then, by the definitions of g_1 and g_2 , ϕ satisfies the desired properties. This completes the proof.

C.10. Proof of Lemma 13

For each $X = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{F}^{d_{\text{in}} \times n}$ and $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, define $f_X : \mathcal{X} \rightarrow \mathbb{F}^{d_{\text{out}}}$ as $f_X(\mathbf{x}) = \mathbf{y}_i$ for $i \in [n]$ satisfying $\mathbf{x}_i = \mathbf{x}$ where $[\mathbf{y}_1, \dots, \mathbf{y}_n] = f^*(X)$. Since f^* is permutation-equivariant, f_X is well-defined.

Since f^* is permutation-equivariant, for any $\pi \in \mathcal{S}_n$, $f^*([\mathbf{x}_{\pi(1)}, \dots, \mathbf{x}_{\pi(n)}]) = [\mathbf{y}_{\pi(1)}, \dots, \mathbf{y}_{\pi(n)}]$, i.e., $f_X = f_{\pi X}$ for all $\pi \in \mathcal{S}_n$. Furthermore, since $\mathbf{c}_X = \mathbf{c}_Z$ if and only if $X = \pi Z$ for some $\pi \in \mathcal{S}_n$, we can define $\hat{f}(\mathbf{x}_i, \mathbf{c}_X) := f_X(\mathbf{x}_i)$ for all $X \in \mathbb{F}^{d_{\text{in}} \times n}$; note that this is well-defined as $f_X = f_{\pi X}$ for all $\pi \in \mathcal{S}_n$. Choose arbitrary values to $\hat{f}(\mathbf{x}, \mathbf{c})$ for all $\mathbf{x}, \mathbf{c}$ that we did not consider yet. This completes the proof.

C.11. Proof of Lemma 14

Let $\gamma = |\mathbb{F}|^{d_{\text{in}}}, 1 \otimes (\bigoplus_{i=1}^n 1) = \alpha \geq 2^{-p-1}$, and $\beta \in \mathbb{F}$ such that $\beta \otimes \alpha = 1^+$. Such β always exists by Lemma 19: write $\alpha = a \times 2^{-s}$ with $a \in (1, 2]_{\mathbb{F}}$ and $s \in \{1, \dots, p+2\}$ (this scaling is exact since it is by a power of two, and $s \leq p+2$ since $\alpha \geq 2^{-p-1}$ and $a \leq 2$), take $y \in (2^{-1}, 1]_{\mathbb{F}}$ from Lemma 19 with $a \otimes y = 1^+$, and set $\beta := y \times 2^s$, which lies in $\mathbb{F}$ since $\beta \leq 2^{p+2} \leq \Omega$ by Condition 1. Then $\beta \otimes \alpha = a \otimes y$, and hence, $\beta \otimes \alpha = a \otimes y = 1^+$. Without loss of generality, we assume that $d = t_d \geq d_{\text{in}} + 2\gamma$, $m = t_m$, and $r = t_r$; otherwise, we can add unused dimensions. We set $t_m = \gamma$, and we will choose the explicit values of t_d, t_r later. We construct ϕ as a composition of two transformer blocks $g_1, g_2 \in \mathcal{B}^{1, t_m, t_r, t_d, n}$. Here, g_1 maps $Z \in \mathbb{F}^{t_d \times n}$ with $Z_{:, d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{F}^{d_{\text{in}} \times n}$ and $Z_{d_{\text{in}}+1:} = \mathbf{0}_{(t_d - d_{\text{in}}) \times n}$ to $Y = [\mathbf{y}_1, \dots, \mathbf{y}_n] \in \mathbb{F}^{t_d \times n}$ such that

$$(\mathbf{y}_i)_l = \begin{cases} (\mathbf{x}_i)_l & \text{if } l \leq d_{\text{in}}, \\ \beta & \text{if } l = d_{\text{in}} + \gamma + j \text{ and } \mathbf{x}_i = \mathbf{z}_j \text{ for some } j \in [\gamma], \\ 0 & \text{otherwise,} \end{cases}$$

for all $i \in [n]$ and $l \in [t_d]$. Note that these row positions do not depend on t_d . By Lemma 12 applied with $d_1 = d_{\text{in}}$, $d_2 = d_{\text{in}} + 2\gamma$, and the present $m = t_m$, $h = 1$, and n , there exist $t'_d, t'_r \in \mathbb{N}$ (which do not depend on t_d or t_r) such that the desired $g_1 \in \mathcal{B}^{1, t_m, r', d', n}$ exists for all $d' \geq t'_d$ and $r' \geq t'_r$. Choose $t_d = \max\{t'_d, d_{\text{in}} + 2\gamma\}$, $t_r = \max\{t'_r, \gamma\}$, and the desired $g_1 \in \mathcal{B}^{1, t_m, t_r, t_d, n}$.

We now construct g_2 . Define an attention layer $f_1 : \mathbb{F}^{t_d \times n} \rightarrow \mathbb{F}^{t_d \times n}$ as

$$\begin{aligned} f_1(Z) &= Z \oplus (W^O \otimes ((W^V \otimes Z) \otimes \sigma((\mathbf{0}_{t_m \times t_d} \otimes Z)^\top \otimes (\mathbf{0}_{t_m \times t_d} \otimes Z)))) \\ &= Z \oplus (W^O \otimes ((W^V \otimes Z) \otimes (\alpha \mathbf{1}_{n \times n}))) \end{aligned}$$

for some $W^V \in \mathbb{F}^{t_m \times t_d}$ and $W^O \in \mathbb{F}^{t_d \times t_m}$. Specifically, we choose W^V, W^O so that $W^V \otimes Z = Z_{d_{\text{in}} + \gamma + 1: d_{\text{in}} + 2\gamma}$, $(W^O \otimes Y)_{d_{\text{in}} + 1: d_{\text{in}} + \gamma} = Y$, and all other rows of $W^O \otimes Y$ are zero for all $Y \in \mathbb{F}^{t_m \times n}$. Then, by the definitions of g_1, f_1 , it holds that for each $Z \in \mathbb{F}^{t_d \times n}$ with $Z_{:, d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{F}^{d_{\text{in}} \times n}$ and $Z_{d_{\text{in}}+1:} = \mathbf{0}_{(t_d - d_{\text{in}}) \times n}$, for each $j \in [\gamma]$,

- $(f_1 \circ g_1(Z))_{:, d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$,
- $(f_1 \circ g_1(Z))_{d_{\text{in}}+j} = (\bigoplus_{i=1}^{k_j} 1^+) \times \mathbf{1}_n^\top$ where k_j denotes the number of $i \in [n]$ such that $\mathbf{x}_i = \mathbf{z}_j$,
- $(f_1 \circ g_1(Z))_{d_{\text{in}}+2\gamma+1:} = \mathbf{0}_{(t_d - 2\gamma - d_{\text{in}}) \times n}$, and the rows $d_{\text{in}} + \gamma + 1, \dots, d_{\text{in}} + 2\gamma$ of $f_1 \circ g_1(Z)$ coincide with the corresponding rows of $g_1(Z)$. In particular, they are non-negative.

Choose a feed-forward network $f_2 : \mathbb{F}^{t_d \times n} \rightarrow \mathbb{F}^{t_d \times n}$ as $f_2(Z) = Z \oplus (W_2 \otimes \rho(W_1 \otimes Z))$ where $W_1 \in \mathbb{F}^{t_r \times t_d}$ and $W_2 \in \mathbb{F}^{t_d \times t_r}$ are matrices satisfying that for $\mathbf{z} = (z_1, \dots, z_{t_d}) \in \mathbb{F}^{t_d}$ and $\mathbf{y} = (y_1, \dots, y_{t_r}) \in \mathbb{F}^{t_r}$,

$$W_1 \otimes \mathbf{z} = (z_{d_{\text{in}} + \gamma + 1}, \dots, z_{d_{\text{in}} + 2\gamma}, 0, \dots, 0), \quad W_2 \otimes \mathbf{y} = (0, \dots, 0, -y_1, \dots, -y_\gamma, 0, \dots, 0),$$

where the nonzero block of $W_2 \otimes \mathbf{y}$ occupies the coordinates $d_{\text{in}} + \gamma + 1, \dots, d_{\text{in}} + 2\gamma$. Namely, f_2 erases the coordinates $d_{\text{in}} + \gamma + 1, \dots, d_{\text{in}} + 2\gamma$ of each column, which are finite and non-negative by the third property above. Let $g_2 = f_2 \circ f_1$ and $\phi = g_2 \circ g_1 \in \mathcal{B}^{1, t_m, t_r, t_d, n}$. Then, by the definitions of g_1, g_2, ϕ satisfies the desired properties. This completes the proof.

C.12. Proof of Lemma 15

Let $x_k = \bigoplus_{i=1}^k 1^+$ for all $k \in [3 \times 2^p - 1] \cup \{0\}$. By direct computation, we have $x_0 = 0$, $x_1 = 1^+$, $x_2 = 2^+$, $x_3 = 3^{++}$, $x_4 = 4^+$. Furthermore, for $k \in \{4, \dots, 2^p\}$, $x_k = k^+$. This can be shown via mathematical induction. The base case $k = 4$ holds by our direct computation. Suppose that $x_{k-1} = (k-1)^+$ for some $k \in \{5, \dots, 2^p\}$ and a natural number e with $2^e < k \leq 2^{e+1}$; here, $e \geq 2$. If $k \neq 2^{e+1}$, then $x_{k-1} + 1^+ = k + 2^{e-p} + 2^{-p}$, i.e., $x_k = x_{k-1} \oplus 1^+ = k^+$. If $k = 2^{e+1}$, then $x_{k-1} + 1^+ = 2^{e+1} + 2^{e-p} + 2^{-p}$, i.e., $x_k = x_{k-1} \oplus 1^+ = (2^{e+1})^+ = k^+$.

Since $x_{2^p} = 2^p + 1$, for $k \in \{2^p + 1, \dots, 2^{p+1} - 1\}$, one can observe that $x_k = k + 1$. Furthermore, for $k \in \{2^{p+1}, \dots, 3 \times 2^p - 1\}$, we have $x_k = 2^{p+1} + (k - 2^{p+1} + 1) \times 2$ due to the rounding. Hence, x_k can be written as follows:

$$x_k = \begin{cases} 0 & \text{if } k = 0, \\ 1^+ & \text{if } k = 1, \\ 2^+ & \text{if } k = 2, \\ 3^{++} & \text{if } k = 3, \\ k^+ & \text{if } k \in \{4, \dots, 2^p\}, \\ k + 1 & \text{if } k \in \{2^p + 1, \dots, 2^{p+1} - 1\}, \\ 2^{p+1} + (k - 2^{p+1} + 1) \times 2 & \text{if } k \in \{2^{p+1}, \dots, 3 \times 2^p - 1\}. \end{cases}$$

This implies that the values x_k , $k \in [3 \times 2^p - 1] \cup \{0\}$, are all distinct, and $x_{3 \times 2^p - 1} = 2^{p+2}$. Finally, for $v = 2^{p+2}$, the spacing of $\mathbb{F}$ on $[2^{p+2}, 2^{p+3})$ is 2^2 , and $1^+ = 1 + 2^{-p} < 2$ is smaller than half of this spacing; hence $v \oplus 1^+ = v$. Therefore, $\bigoplus_{i=1}^{3 \times 2^p - 1 + n} 1^+ = 2^{p+2}$ for all $n \in \mathbb{N} \cup \{0\}$, i.e., the sum is constant from this point.

C.13. Proof of Lemma 16

The proof is trivial when $x \in \{\pm\infty, \text{NaN}\}$. For $x \in \mathbb{F}$, we may assume $x \geq 0$ since $\lceil \cdot \rceil_{\mathbb{F}}$ is odd and hence $\bigoplus_{i=1}^k (-x) = -\bigoplus_{i=1}^k x$ for all k . The result then follows from Lemma 18 with $z = 0$.

C.14. Proof of Lemma 17

Without loss of generality, we assume that z, x are finite and $x > 0$; if $x \in \{0, \pm\infty, \text{NaN}\}$ or $z \in \{\pm\infty, \text{NaN}\}$, then the result directly follows. We further assume that $z < 0$ since if $z \geq 0$, the result reduces to Lemma 18. Choose $e_z \in \mathbb{Z}$ such that $z = -2^{e_z}$. We now consider all possible cases.

Case $e_z \leq \epsilon_{\min} + 1$. There exists $k \leq 2^{p+1}$ such that $z \oplus \bigoplus_{i=1}^k x \geq 0$ since $x \geq 2^{\epsilon_{\min} - p} = \omega$. By Lemma 18, the result follows.

Case $e_z > \epsilon_{\min} + 1$ and $x \leq 2^{e_z - p - 2}$. In this case, $x \leq 2^{e_z - p - 2} = \frac{1}{2}(z^+ - z)$. Since $z = -2^{e_z}$, this implies that $z \oplus \bigoplus_{i=1}^k x = z$ for all $k \in \mathbb{N}$ and the result follows.

Case $e_z > \epsilon_{\min} + 1$ and $x > 2^{e_z - p - 2}$. Since $x > 2^{e_z - p - 2} = \frac{1}{2}(z^+ - z)$, it holds that $z \oplus \bigoplus_{i=1}^{2^p} x \geq -2^{e_z - 1}$. Furthermore, it holds that

$$z \oplus \bigoplus_{i=1}^{3 \times 2^p} x \geq -2^{e_z - 1} \oplus \bigoplus_{i=1}^{2^{p+1}} x \geq -2^{e_z - 1} \oplus \bigoplus_{i=1}^{2^{p+1}} 2^{e_z - p - 2} = 0.$$

Hence, by Lemma 18, the result follows. This completes the proof.

D. Proof of Theorem 2 when $p' \leq p + 1$

The proof outline is identical to that for the $p' \geq p + 2$ case; we choose the same W_{in} , W_{out} , and construct similar ϕ , ψ . One major difference is that we identify the reduction order of each triple of inputs here, instead of quadruples. For each triple of distinct indices $\mathcal{I} = (\iota_1, \iota_2, \iota_3)$ in $[n]$ and each triple of distinct elements (j, k, l) in $[3]$, we define (throughout this section, the symbols τ and δ are redefined for triples)

$$\tau_{\mathcal{I},(j,k,l)} = \bigoplus_{i \in [n]; \mathbb{F}'}^{\mathcal{T}_n} \delta_{\mathcal{I},(j,k,l),i}, \quad \delta_{\mathcal{I},(j,k,l),i} = \begin{cases} a_{p,p'} & \text{if } i = \iota_j, \\ b_{p,p'} & \text{if } i = \iota_k, \\ c_{p,p'} & \text{if } i = \iota_l, \\ 0 & \text{otherwise,} \end{cases}$$

$$a_{p,p'} = \begin{cases} 1 & \text{if } p' = p + 1, \\ 1 + 2^{-p} & \text{if } p' = p, p \geq 3, \\ 1 & \text{if } p' = p, p = 2, \end{cases} \quad b_{p,p'} = \begin{cases} 2^{-p-1} & \text{if } p' = p + 1, \\ 1 + 2^{-p+1} & \text{if } p' = p, p \geq 3, \\ 1 + 2^{-p} & \text{if } p' = p, p = 2, \end{cases} \quad c_{p,p'} = \begin{cases} 2^{-p'-1} & \text{if } p' = p + 1, \\ 1 + 2^{-p+1} & \text{if } p' = p, p \geq 3, \\ 1 + 2^{-p} & \text{if } p' = p, p = 2. \end{cases}$$

We then define

$$\mathbf{t}(\pi) = (\tau_{(\pi(\iota_1), \pi(\iota_2), \pi(\iota_3)), (j,k,l)})_{\mathcal{I},(j,k,l)}$$

where $\mathcal{I} = (\iota_1, \iota_2, \iota_3)$ ranges over all triples of distinct indices in $[n]$ and (j, k, l) ranges over all ordered triples of distinct elements in $[3]$. We then introduce the following lemma. The proof of Lemma 31 is in Appendix D.1.

Lemma 31. *Suppose $p' \leq p + 1$. Let $n \geq 4$ and $\mathcal{T}_n$ be a full binary tree with n leaves indexed by $[n]$. Then, for each $\pi \in \mathcal{S}_n$,*

$$\mathbf{t}^{-1}(\mathbf{t}(\pi)) = \mathcal{G}(\mathcal{T}_n)\pi.$$

Lemma 31 implies that given $\mathbf{t}(\pi)$, we can recover the permutation π up to its $\mathcal{G}(\mathcal{T}_n)$ -equivalent counterpart. Using this, we introduce Lemma 32, similar to Lemma 11 for the $p' \geq p + 2$ case. Here, we let $\gamma_0 = |\mathbb{F}|^{d_{\text{in}}}$, $\bar{\mathbf{z}}_1, \dots, \bar{\mathbf{z}}_{\gamma_0}$ denote all elements of $\mathbb{F}^{d_{\text{in}}}$, $\gamma = 24 \binom{|\mathbb{F}|^{d_{\text{in}}}}{4}$, $\mathcal{Z}_1 = (\mathbf{z}_{1,1}, \mathbf{z}_{1,2}, \mathbf{z}_{1,3}, \mathbf{z}_{1,4}), \dots, \mathcal{Z}_\gamma = (\mathbf{z}_{\gamma,1}, \mathbf{z}_{\gamma,2}, \mathbf{z}_{\gamma,3}, \mathbf{z}_{\gamma,4})$ denote all ordered quadruples of distinct elements in $\mathbb{F}^{d_{\text{in}}}$, and write $\mathbf{x} \in \mathcal{Z}_j$ if $\mathbf{x} \in \{\mathbf{z}_{j,1}, \mathbf{z}_{j,2}, \mathbf{z}_{j,3}, \mathbf{z}_{j,4}\}$. The proof of Lemma 32 is in Appendix D.2.

Lemma 32. *Suppose $p' \leq p + 1$ and $q' \leq 2^{q-1}$. Let $d_{\text{in}}, n \in \mathbb{N}$ with $n \geq 4$, and $e = d_{\text{in}} + \gamma_0 + \gamma$. There exist $h, t_d, t_m, t_r \in \mathbb{N}$ such that for any $d \geq t_d$, $m \geq t_m$, and $r \geq t_r$, there exists $\phi \in \mathcal{B}^{h,m,r,d,n}$ satisfying the following properties. For each $Z \in \mathbb{F}^{d \times n}$ with $Z_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \Delta_n$ and $Z_{d_{\text{in}}+1:} = \mathbf{0}_{(d-d_{\text{in}}) \times n}$, $j_0 \in [\gamma_0]$, and $j \in [\gamma]$,*

- $\phi(Z)_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ and $\phi(Z)_{e+1:} = \mathbf{0}_{(d-e) \times n}$,
- $\phi(Z)_{d_{\text{in}}+j_0} = \mathbf{1}_n^\top$ if $\bar{\mathbf{z}}_{j_0} \in \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ and $\phi(Z)_{d_{\text{in}}+j_0} = \mathbf{0}_n^\top$ otherwise,
- $\phi(Z)_{d_{\text{in}}+\gamma_0+j} = \left[\bigoplus_{i \in [n]; \mathbb{F}'}^{\mathcal{T}_n} \varepsilon_j \otimes_{\mathbb{F}'} \zeta_{i,j} \right]_{\mathbb{F}} \times \mathbf{1}_n^\top$ where

$$\varepsilon_j = \begin{cases} \left[1 \otimes_{\mathbb{F}'} \left(\bigoplus_{i \in [n]; \mathbb{F}'}^{\mathcal{T}_n} 1 \right) \right]_{\mathbb{F}} & \text{if } \mathbf{x}_i \notin \mathcal{Z}_j \forall i, \\ \left[1 \otimes_{\mathbb{F}'} |\{i \in [n] : \mathbf{x}_i \in \mathcal{Z}_j\}| \right]_{\mathbb{F}} & \text{otherwise,} \end{cases} \quad \zeta_{i,j} = \begin{cases} 4a_{p,p'} & \text{if } \mathbf{x}_i = \mathbf{z}_{j,1}, \\ 4b_{p,p'} & \text{if } \mathbf{x}_i = \mathbf{z}_{j,2}, \\ 4c_{p,p'} & \text{if } \mathbf{x}_i = \mathbf{z}_{j,3}, \\ 0 & \text{otherwise.} \end{cases}$$

We now explain how the outputs of Lemma 32 identify the input up to the action of $\mathcal{G}(\mathcal{T}_n)$. By the second property, the coordinates $d_{\text{in}} + 1, \dots, d_{\text{in}} + \gamma_0$ of each column identify the token set $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, and hence the ordered quadruples $\mathcal{Z}_j$ whose four entries all lie in $\mathcal{X}$. For each such $\mathcal{Z}_j$, the tokens being distinct gives $\varepsilon_j = \lceil 1 \otimes_{\mathbb{F}'} 4 \rceil_{\mathbb{F}} = 2^{-2}$ and $\varepsilon_j \otimes_{\mathbb{F}'} \zeta_{i,j} \in \{a_{p,p'}, b_{p,p'}, c_{p,p'}, 0\}$, both exact as they scale by a power of two. Hence the coordinate $d_{\text{in}} + \gamma_0 + j$ stores $\lceil \tau_{\mathcal{I},(1,2,3)} \rceil_{\mathbb{F}}$, where $\mathcal{I}$ is the triple of positions of $\mathbf{z}_{j,1}, \mathbf{z}_{j,2}, \mathbf{z}_{j,3}$ and all remaining positions contribute exact zeros. Since $n \geq 4$, every ordered triple of distinct tokens in $\mathcal{X}$ arises this way under every role assignment (j, k, l) , so the outputs determine $\lceil \tau_{\mathcal{I},(j,k,l)} \rceil_{\mathbb{F}}$ for all triples $\mathcal{I}$ of distinct positions and all (j, k, l) . By Lemmas 24, 25, and 27, these values determine the induced three-leaf topology of each triple; this identifies the input up to $\mathcal{G}(\mathcal{T}_n)$ as in the proof of Lemma 31. As in the case $p' \geq p + 2$, combining Lemmas 9, 12, 31, and 32 completes the proof.

D.1. Proof of Lemma 31

The proof of Lemma 31 is almost identical to that of Lemma 10. The only difference arises in the converse part (i.e., $t^{-1}(t(\pi)) \subset \mathcal{G}(\mathcal{T}_n)\pi$), where we use Lemmas 24, 25, and 27 to identify the reduction order of each triple instead of Lemma 28. In particular, the hypothesis $p' \geq p + 2$ of Lemma 10 is used in its proof only through Lemma 28, which is exactly the step replaced here. Hence, there is no conflict with the hypothesis $p' \leq p + 1$ of Lemma 31. Here, while Lemmas 24 and 25 are stated for $\oplus$ under $\mathbb{F}$, the same identities hold for $\oplus_{\mathbb{F}'}$ followed by rounding to $\mathbb{F}$ when $p' = p$: the two formats have the same mantissa precision and $\mathbb{F}'$ has a wider exponent range ($q' \geq q$), so every intermediate value and every rounding in these computations, which involve only values in $[1, 4]$, coincides in the two formats.

D.2. Proof of Lemma 32

Throughout this proof, $\gamma_0, \gamma, \bar{\mathbf{z}}_{j_0}, \mathcal{Z}_j, \varepsilon_j$, and $\zeta_{i,j}$ are defined before Lemma 32; in particular, γ is not the quantity denoted by γ in the proofs of Lemmas 14 and 30. Let $\mathbb{F} = \mathbb{F}_{p,q}$ and $h = \gamma_0 + \gamma$. Without loss of generality, we assume that $d = t_d, m = t_m$, and $r = t_r$; otherwise, we can add unused dimensions. We set $t_m = 1$, and we will choose the explicit values of t_d, t_r later with $t_d \geq d_{\text{in}} + 3\gamma_0 + 3\gamma + 1$. We construct ϕ as a composition of two transformer blocks $g_1, g_2 \in \mathcal{B}^{h, t_m, t_r, t_d, n}$. Here, g_1 maps $Z \in \mathbb{F}^{t_d \times n}$ with $Z_{:, d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \Delta_n$ and $Z_{d_{\text{in}}+1:} = \mathbf{0}_{(t_d - d_{\text{in}}) \times n}$ to $Y = [\mathbf{y}_1, \dots, \mathbf{y}_n] \in \mathbb{F}^{t_d \times n}$ such that

$$(\mathbf{y}_i)_l = \begin{cases} (\mathbf{x}_i)_l & \text{if } l \leq d_{\text{in}}, \\ 1 & \text{if } l = d_{\text{in}} + \gamma_0 + \gamma + j_0 \text{ and } \mathbf{x}_i = \bar{\mathbf{z}}_{j_0} \text{ for some } j_0 \in [\gamma_0], \\ 0 & \text{if } l = d_{\text{in}} + 2\gamma_0 + \gamma + j_0 \text{ and } \mathbf{x}_i = \bar{\mathbf{z}}_{j_0} \text{ for some } j_0 \in [\gamma_0], \\ -\Omega/2 & \text{if } l = d_{\text{in}} + 2\gamma_0 + \gamma + j_0 \text{ and } \mathbf{x}_i \neq \bar{\mathbf{z}}_{j_0} \text{ for some } j_0 \in [\gamma_0], \\ 4a_{p,p'} & \text{if } l = d_{\text{in}} + 3\gamma_0 + \gamma + j \text{ and } \mathbf{x}_i = \mathbf{z}_{j,1} \text{ for some } j \in [\gamma], \\ 4b_{p,p'} & \text{if } l = d_{\text{in}} + 3\gamma_0 + \gamma + j \text{ and } \mathbf{x}_i = \mathbf{z}_{j,2} \text{ for some } j \in [\gamma], \\ 4c_{p,p'} & \text{if } l = d_{\text{in}} + 3\gamma_0 + \gamma + j \text{ and } \mathbf{x}_i = \mathbf{z}_{j,3} \text{ for some } j \in [\gamma], \\ 0 & \text{if } l = d_{\text{in}} + 3\gamma_0 + 2\gamma + j \text{ and } \mathbf{x}_i \in \mathcal{Z}_j \text{ for some } j \in [\gamma], \\ -\Omega/2 & \text{if } l = d_{\text{in}} + 3\gamma_0 + 2\gamma + j \text{ and } \mathbf{x}_i \notin \mathcal{Z}_j \text{ for some } j \in [\gamma], \\ 2 & \text{if } l = d_{\text{in}} + 3\gamma_0 + 3\gamma + 1, \\ 0 & \text{otherwise,} \end{cases}$$

for all $i \in [n]$ and $l \in [t_d]$ where Ω denotes the largest positive float in $\mathbb{F}$. That is, the row $d_{\text{in}} + \gamma_0 + \gamma + j_0$ indicates whether $\mathbf{x}_i = \bar{\mathbf{z}}_{j_0}$, the values $\zeta_{i,j}$ are stored in the rows $d_{\text{in}} + 3\gamma_0 + \gamma + 1, \dots, d_{\text{in}} + 3\gamma_0 + 2\gamma$ ($4a_{p,p'}, 4b_{p,p'}, 4c_{p,p'} \in \mathbb{F}$ since each of them is either a power of two with exponent at least $1 - p' \geq -p \geq \epsilon_{\text{min}}$ or a power of two times a value of the form $1 + 2^{-t}$ with $t \leq p$). The rows $d_{\text{in}} + 2\gamma_0 + \gamma + j_0$ and $d_{\text{in}} + 3\gamma_0 + 2\gamma + j$ indicate (for each $j_0 \in [\gamma_0]$ and each $j \in [\gamma]$ separately) whether $\mathbf{x}_i = \bar{\mathbf{z}}_{j_0}$ and whether $\mathbf{x}_i \in \mathcal{Z}_j$, respectively, and the row $d_{\text{in}} + 3\gamma_0 + 3\gamma + 1$ holds the constant 2. Here, the rows $d_{\text{in}} + 1, \dots, d_{\text{in}} + \gamma_0 + \gamma$ and all rows above $d_{\text{in}} + 3\gamma_0 + 3\gamma + 1$ are zero. Note that these row positions do not depend on t_d . As in the proof of Lemma 11, Condition 1 and $q' \leq 2^{q-1}$ imply $\lceil \exp \rceil(-\Omega) = 0$ under $\mathbb{F}'$ (here, $p' \leq p + 1 \leq 3 \cdot 2^{q-1} - 7$). By Lemma 12 applied with $d_1 = d_{\text{in}}, d_2 = d_{\text{in}} + 3\gamma_0 + 3\gamma + 1$, and the present $m = t_m, h = \gamma_0 + \gamma$, and n , there exist $t'_d, t'_r \in \mathbb{N}$ (which do not depend on t_d or t_r) such that the desired $g_1 \in \mathcal{B}^{h, t_m, r', d', n}$ exists for all $d' \geq t'_d$ and $r' \geq t'_r$. Choose $t_d = \max\{t'_d, d_{\text{in}} + 3\gamma_0 + 3\gamma + 1\}$, $t_r = \max\{t'_r, 4\gamma_0 + 4\gamma + 2\}$, and $g_1 \in \mathcal{B}^{h, t_m, t_r, t_d, n}$.

We now construct g_2 . Define an attention layer $f_1 : \mathbb{F}^{t_d \times n} \rightarrow \mathbb{F}^{t_d \times n}$ as

$$f_1(Z) = Z \oplus \left(\bigoplus_{j=1}^h \left(W_j^O \hat{\otimes} ((W_j^V \hat{\otimes} Z) \hat{\otimes} \sigma((W_j^K \hat{\otimes} Z)^\top \hat{\otimes} (W_j^Q \hat{\otimes} Z))) \right) \right)$$

for some $W_j^K, W_j^Q, W_j^V \in \mathbb{F}^{t_m \times t_d}$ and $W_j^O \in \mathbb{F}^{t_d \times t_m}$. Specifically, for the heads $j_0 \in [\gamma_0]$, we choose $W_{j_0}^K$ and $W_{j_0}^Q$ so that $(W_{j_0}^K)_{1, d_{\text{in}} + 2\gamma_0 + \gamma + j_0} = 1 = (W_{j_0}^Q)_{1, d_{\text{in}} + 3\gamma_0 + 3\gamma + 1}$, and for the heads $\gamma_0 + j$ with $j \in [\gamma]$, we choose $W_{\gamma_0 + j}^K$ and $W_{\gamma_0 + j}^Q$ so that $(W_{\gamma_0 + j}^K)_{1, d_{\text{in}} + 3\gamma_0 + 2\gamma + j} = 1 = (W_{\gamma_0 + j}^Q)_{1, d_{\text{in}} + 3\gamma_0 + 3\gamma + 1}$, with all other entries zero. Since each row of these matrices has at most one nonzero entry, their products with the output Y of g_1 are exact. Hence, for the token set $\mathcal{K}$ marked by each head ($\mathcal{K} = \{\bar{\mathbf{z}}_{j_0}\}$ for head $j_0 \in [\gamma_0]$ and $\mathcal{K} = \{\mathbf{z}_{j,1}, \dots, \mathbf{z}_{j,4}\}$ for head $\gamma_0 + j$ with $j \in [\gamma]$), the (i, ι) -th entry of the score

matrix $M_K := (W^K \hat{\otimes} Y)^\top \hat{\otimes} (W^Q \hat{\otimes} Y)$ of the head equals the corresponding indicator entry times 2, which is 0 if $\mathbf{x}_i \in \mathcal{K}$ and $(-\Omega/2) \otimes 2 = -\Omega$ otherwise (exact, as $-\Omega \in \mathbb{F}$). In particular, all columns of M_K are identical.

We now compute $\sigma(M_K)$ column-wise via Eq. (4). Let $k_K = |\{i \in [n] : \mathbf{x}_i \in \mathcal{K}\}|$, and note that $k_K \leq |\mathcal{K}| \leq 4$ since $\mathbf{x}_1, \dots, \mathbf{x}_n$ are distinct. In particular, $k_K \leq 1$ for the heads $j_0 \in [\gamma_0]$. If $k_K \geq 1$, the column maximum is $x_* = 0$, and the numerators are $\lceil \exp \rceil (0 \ominus 0) = \exp(0) = 1$ for the k_K indices i with $\mathbf{x}_i \in \mathcal{K}$ and $\lceil \exp \rceil (-\Omega \ominus 0) = \lceil \exp \rceil (-\Omega) = 0$ for the remaining indices, as shown above. Since $v \oplus_{\mathbb{F}'} 0 = v$ for all $v \in \mathbb{F}'$, the denominator equals the exact integer k_K under any reduction order. Hence, the i -th entry of each column of $\sigma(M_K)$ is $\lceil 1 \oslash_{\mathbb{F}'} k_K \rceil_{\mathbb{F}}$ if $\mathbf{x}_i \in \mathcal{K}$ and $\lceil 0 \oslash_{\mathbb{F}'} k_K \rceil_{\mathbb{F}} = 0$ otherwise. For the heads $j_0 \in [\gamma_0]$ with $k_K = 1$, the former value is $\lceil 1 \oslash_{\mathbb{F}'} 1 \rceil_{\mathbb{F}} = 1$, and for the heads $\gamma_0 + j$, it is ε_j . If $k_K = 0$, all scores are $-\Omega$, so $x_* = -\Omega$, all numerators are $\lceil \exp \rceil (-\Omega \ominus (-\Omega)) = \exp(0) = 1$, and every entry of $\sigma(M_K)$ is $\lceil 1 \oslash_{\mathbb{F}'} (\bigoplus_{i \in [n]; \mathbb{F}'}^{\tau_n} 1) \rceil_{\mathbb{F}}$; we do not use this value since it will be multiplied only by zero entries.

Given $m = t_m = 1$, we also choose $W_{j_0}^V \hat{\otimes} Z = Z_{d_{\text{in}} + \gamma_0 + \gamma + j_0}$ and $(W_{j_0}^O \hat{\otimes} Y)_{d_{\text{in}} + j_0} = Y$ for the heads $j_0 \in [\gamma_0]$, and $W_{\gamma_0 + j}^V \hat{\otimes} Z = Z_{d_{\text{in}} + 3\gamma_0 + \gamma + j}$ and $(W_{\gamma_0 + j}^O \hat{\otimes} Y)_{d_{\text{in}} + \gamma_0 + j} = Y$ for the heads $\gamma_0 + j$ with $j \in [\gamma]$, with all other rows of $W^O \hat{\otimes} Y$ zero for all $Y \in \mathbb{F}^{t_m \times n}$. Then, by the definitions of g_1, f_1 , it holds that for each $Z \in \mathbb{F}^{t_d \times n}$ with $Z_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \Delta_n$ and $Z_{d_{\text{in}}+1:} = \mathbf{0}_{(t_d - d_{\text{in}}) \times n}$, for each $j_0 \in [\gamma_0]$ and $j \in [\gamma]$,

- $(f_1 \circ g_1(Z))_{:d_{\text{in}}} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$,
- $(f_1 \circ g_1(Z))_{d_{\text{in}} + j_0} = \mathbf{1}_n^\top$ if $\bar{\mathbf{z}}_{j_0} \in \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ and $(f_1 \circ g_1(Z))_{d_{\text{in}} + j_0} = \mathbf{0}_{1 \times n}$ otherwise,
- $(f_1 \circ g_1(Z))_{d_{\text{in}} + \gamma_0 + j} = \left[\bigoplus_{i \in [n]; \mathbb{F}'}^{\tau_n} \varepsilon_j \otimes_{\mathbb{F}'} \zeta_{i,j} \right]_{\mathbb{F}} \times \mathbf{1}_n^\top$,
- $(f_1 \circ g_1(Z))_{d_{\text{in}} + 3\gamma_0 + 3\gamma + 2:} = \mathbf{0}_{(t_d - d_{\text{in}} - 3\gamma_0 - 3\gamma - 1) \times n}$, and the rows $d_{\text{in}} + \gamma_0 + \gamma + 1, \dots, d_{\text{in}} + 3\gamma_0 + 3\gamma + 1$ of $f_1 \circ g_1(Z)$ coincide with the corresponding rows of $g_1(Z)$. In particular, all their entries are finite.

Here, the second property holds since the value row of head j_0 contains 1 at the (unique) index i with $\mathbf{x}_i = \bar{\mathbf{z}}_{j_0}$ (if it exists) and 0 elsewhere, so the summand at that index is $1 \otimes_{\mathbb{F}'} 1 = 1$, and the summands elsewhere are $0 \otimes_{\mathbb{F}'} \sigma(M_K)_{i\iota} = 0$, yielding the exact value 1 if $\bar{\mathbf{z}}_{j_0} \in \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ and 0 otherwise. The third property holds since the i -th summand of $((W_{\gamma_0 + j}^V \hat{\otimes} Y) \hat{\otimes} \sigma(M_K))_{1,\iota} = \left[\bigoplus_{i \in [n]; \mathbb{F}'}^{\tau_n} \zeta_{i,j} \otimes_{\mathbb{F}'} \sigma(M_K)_{i\iota} \right]_{\mathbb{F}}$ equals $\varepsilon_j \otimes_{\mathbb{F}'} \zeta_{i,j}$ for all i : if $\mathbf{x}_i \in \mathcal{Z}_j$ then $\sigma(M_K)_{i\iota} = \varepsilon_j$, and otherwise $\zeta_{i,j} = 0$ so both are 0. Furthermore, the routing by W^O is exact (one nonzero entry per row), the heads write to disjoint rows so their sum $\bigoplus$ over heads is exact, and the residual connection adds these values to entries of $g_1(Z)$ that are zero, which is also exact.

Choose a feed-forward network $f_2 : \mathbb{F}^{t_d \times n} \rightarrow \mathbb{F}^{t_d \times n}$ as $f_2(Z) = Z \oplus (W_2 \hat{\otimes} \rho(W_1 \hat{\otimes} Z))$ where $W_1 \in \mathbb{F}^{t_r \times t_d}$ and $W_2 \in \mathbb{F}^{t_d \times t_r}$ are matrices satisfying that for $\mathbf{z} = (z_1, \dots, z_{t_d}) \in \mathbb{F}^{t_d}$ and $\mathbf{y} = (y_1, \dots, y_{t_r}) \in \mathbb{F}^{t_r}$,

$$\begin{aligned} W_1 \otimes \mathbf{z} &= (z_{d_{\text{in}} + \gamma_0 + \gamma + 1}, \dots, z_{d_{\text{in}} + 3\gamma_0 + 3\gamma + 1}, -z_{d_{\text{in}} + \gamma_0 + \gamma + 1}, \dots, -z_{d_{\text{in}} + 3\gamma_0 + 3\gamma + 1}, 0, \dots, 0), \\ W_2 \otimes \mathbf{y} &= (0, \dots, 0, y_{2\gamma_0 + 2\gamma + 2} - y_1, \dots, y_{4\gamma_0 + 4\gamma + 2} - y_{2\gamma_0 + 2\gamma + 1}, 0, \dots, 0), \end{aligned}$$

where the nonzero block of $W_2 \otimes \mathbf{y}$ occupies the coordinates $d_{\text{in}} + \gamma_0 + \gamma + 1, \dots, d_{\text{in}} + 3\gamma_0 + 3\gamma + 1$. Exactly as in the proof of Lemma 11, f_2 erases these coordinates of each column, whose entries are finite by the last property above; this requires $t_r \geq 4\gamma_0 + 4\gamma + 2$, consistent with our choice of t_r . Let $g_2 = f_2 \circ f_1$ and $\phi = g_2 \circ g_1 \in \mathcal{B}^{h, t_m, t_r, t_d, n}$. Then, by the definitions of g_1 and g_2 , ϕ satisfies the desired properties. This completes the proof.